

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250282889

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SHIKU; Hiroshi et al.

BISPECIFIC ANTIBODY

Abstract

An antibody that can be used as an anti-tumor agent and an anti-tumor agent comprises a molecule containing such an antibody. Such an antibody or a binding fragment thereof comprises: a heavy chain CDRH1 consisting of SEQ ID NO: 54, a heavy chain CDRH2 consisting of SEQ ID NO: 55, a heavy chain CDRH3 consisting of SEQ ID NO: 56, a light chain CDRL1 consisting of SEQ ID NO: 57 or a light chain CDRL1 consisting of an amino acid sequence derived from SEQ ID NO: 57 in which amino acid 7 is W and/or amino acid 8 is K, a light chain CDRL2 consisting of SEQ ID NO: 58, and a light chain CDRL3 consisting of SEQ ID NO: 59 or a light chain CDRL3 consisting of an amino acid sequence derived from SEQ ID NO: 59 in which amino acid 2 is A or S.

Inventors: SHIKU; Hiroshi (Tsu-shi, JP), AKAHORI; Yasushi (Tsu-shi, JP), TANAKA; Kento (Tokyo, JP), YATSU; Ayaka (Tokyo, JP), ICHIKAWA; Junya (Tokyo, JP), OHTSUKA; Toshiaki (Tokyo, JP), KOZUMA; Shiho (Tokyo, JP), HASHIMOTO; Ryuji (Tokyo, JP), NAKAYAMA; Makiko (Tokyo, JP), SHINOZAKI; Naoya (Tokyo, JP), NAKAMURA; Kensuke (Tokyo, JP), WATANABE; Ichiro (Tokyo, JP), FURUZONO; Shinji (Tokyo, JP)

Applicant: MIE UNIVERSITY (Tsu-shi, JP); DAIICHI SANKYO COMPANY, LIMITED (Tokyo, JP)

Family ID: 1000008620933

Assignee: MIE UNIVERSITY (Tsu-shi, JP); DAIICHI SANKYO COMPANY, LIMITED (Tokyo, JP)

Appl. No.: 19/218008

Filed: May 23, 2025

Foreign Application Priority Data

JP	2020-061476	Mar. 30, 2020
----	-------------	---------------

Related U.S. Application Data

parent US division 17799648 20220812 parent-grant-document US 12344674 WO division PCT/JP2021/013378 20210329 child US 19218008

Publication Classification

Int. Cl.: C07K16/30 (20060101); A61K39/00 (20060101); A61P35/00 (20060101); C07K16/28 (20060101); C07K16/32 (20060101); C07K16/46 (20060101); C12N5/0783 (20100101)

U.S. Cl.:

CPC **C07K16/30** (20130101); **A61P35/00** (20180101); **C07K16/2809** (20130101); **C07K16/2833** (20130101); **C07K16/32** (20130101); **C07K16/468** (20130101); **C12N5/0636** (20130101); **A61K2039/505** (20130101); **C07K2317/31** (20130101); **C07K2317/52** (20130101); **C07K2317/522** (20130101); **C07K2317/524** (20130101); **C07K2317/526** (20130101); **C07K2317/528** (20130101); **C07K2317/53** (20130101); **C07K2317/55** (20130101); **C07K2317/56** (20130101); **C07K2317/565** (20130101); **C07K2317/622** (20130101); **C07K2317/73** (20130101); **C07K2317/92** (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a Divisional of U.S. patent application Ser. No. 17/799,648, filed on Aug. 12, 2022, which claims priority under 35 U.S.C. § 371 to International Patent Application No. PCT/JP2021/013378, filed Mar. 29, 2021, which claims priority to and the benefit of Japanese Patent Application No. 2020-061476, filed on Mar. 30, 2020. The contents of these applications are hereby incorporated by reference in their entirety.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted via USPTO Patent Center is hereby incorporated by reference in its entirety. Said electronic copy, is named 127565-0109_SL.xml and is 372 kb in size.

TECHNICAL FIELD

[0003] The present invention relates to an antibody, a bispecific antibody, and the like that are useful for the treatment of tumors.

BACKGROUND ART

[0004] NY-ESO-1 is a molecule that has been identified in esophageal cancer by a method of serological analysis of recombinant cDNA expression libraries (SEREX) (Non-Patent Literature 1), and LAGE-1 is also referred to as “NY-ESO-2,” which is a molecule identified via representational difference analysis of the tumor cDNA library (Non-Patent Literature 2). While expression of such molecules is known to be localized in the testis in the case of normal tissue, the mechanism thereof remains unknown. Since expression of NY-ESO-1 and LAGE-1 has been reported to occur in a wide variety of cancer species, such as melanoma, lung cancer, bladder cancer, ovarian cancer, soft-tissue sarcoma, and myeloma (Non-Patent Literature 3), association thereof with cancer is also suggested. In addition, reports have been made concerning the correlation with the degree of malignancy. For example, the NY-ESO-1 expression level is higher in the metastatic focus of melanoma than in the primary focus thereof (Non-Patent Literature 4), the NY-ESO-1 and LAGE-1 expression levels are higher in the advanced urothelial carcinoma than in the early stage urothelial carcinoma (Non-Patent Literature 5), and the NY-ESO-1 expression level is higher in high-risk myeloma with chromosomal abnormalities than in myeloma without chromosomal abnormalities (Non-Patent Literature 6). On the basis of such information, NY-ESO-1 and LAGE-1 have drawn attention as molecules with high cancer-specificity, and a great deal of research and development has been made aimed at drug discovery in cancer vaccine therapy. To date, however, there have been no pharmaceutical products approved in this respect.

[0005] The 9-mer NY-ESO peptides of NY-ESO-1 and of LAGE-1 each comprising residues 157 to 165 (i.e., SLLMWITQC) are known to form a complex with HLA (histocompatibility leukocyte antigen)-A2 (i.e., the HLA/NY-ESO peptide complex) and present the complex extracellularly (Non-Patent Literature 7). As described above, expression of NY-ESO-1 and LAGE-1 is cancer-specific. This indicates that the HLA/NY-ESO peptide complex is a therapeutic target specific to cancer that is selectively present on HLA-A2-positive and NY-ESO-1- or LAGE-1-positive cancer cells (Non-Patent Literature 8). In addition, molecules that bind to the HLA/NY-ESO peptide complex, such as TCR (Patent Literature 1 and Patent Literature 2) and antibodies (Patent Literature 3 and Patent Literature 8) have been reported.

[0006] An application of a molecule binding to a cancer-targeting molecule is a CD3 bispecific antibody that functions on the basis of the T cell redirecting mechanism whereby it recruits T cells to cancer cells and thereby exerts antitumor effects by cytotoxicity (Non-Patent Literature 9 and Non-Patent Literature 10). An example of a currently available CD3 bispecific antibody medicine is Blinatumomab, which is a CD19 bispecific T-cell engager (BiTE). Such antibody medicine is approved for acute lymphoblastic leukemia (ALL), and clinical trials targeting other blood cancers have been in progress. However, the bispecific antibody format thereof is tandem scFv (taFv) without the Fc region, and the blood half-life upon administration thereof to a patient is significantly shorter than that attained with the use of the IgG type antibody that is commonly used as a therapeutic antibody (Non-Patent Literature 11).

[0007] Concerning bispecific antibodies having the heterodimer Fc region exhibiting the blood half-life equivalent to that of the IgG type antibody, research and clinical trials on CD3 bispecific antibodies with a variety of antibody formats, such as knobs-into-holes, CrossMab, and DuoBody® (Patent Literatures 4, 5, 6, and 7), have been in

progress, and a CD3 bispecific antibody utilizing an antibody against the HLA/NY-ESO peptide complex is reported (Non-Patent Literature 12).

CITATION LIST

Patent Literature

[0008] Patent Literature 1: WO 2005/113595 [0009] Patent Literature 2: WO 2017/109496 [0010] Patent Literature 3: WO 2010/106431 [0011] Patent Literature 4: WO 1998/050431 [0012] Patent Literature 5: WO 2006/106905 [0013] Patent Literature 6: WO 2011/028952 [0014] Patent Literature 7: WO 2011/131746 [0015] Patent Literature 8: WO 2021/003357

Non-Patent Literature

[0016] Non-Patent Literature 1: Proc. Natl. Acad. Sci., U.S.A., 94 (5), 1914-8, 1997 [0017] Non-Patent Literature 2: Int. J. Cancer, 76 (6), 903-8, 1998 [0018] Non-Patent Literature 3: Immunol. Cell Biol., 84 (3), 303-17, 2006 [0019] Non-Patent Literature 4: J. Surg. Res., 98 (2), 76-80, 2001 [0020] Non-Patent Literature 5: Cancer Res., 61 (12), 4671-4, 2001 [0021] Non-Patent Literature 6: Blood, 105 (10), 3939-44, 2005 [0022] Non-Patent Literature 7: J. Exp. Med., 187 (2), 265-70, 1998 [0023] Non-Patent Literature 8: J. Immunol., 176 (12), 7308-16, 2006 [0024] Non-Patent Literature 9: Nature, 314 (6012), 628-31, 1985 [0025] Non-Patent Literature 10: Int. Rev. Immunol., 4 (2), 159-73, 1989 [0026] Non-Patent Literature 11: Drug Des. Devel. Ther., 10, 757-765, 2016 [0027] Non-Patent Literature 12: Abstracts of the 21st Annual Meeting of the Japanese Association of Cancer Immunology, 013-4

SUMMARY OF INVENTION

Technical Problem

[0028] The present invention provides a novel anti-HLA-A2/NY-ESO antibody that can be used as an anti-tumor agent and an anti-tumor agent comprising, as an active ingredient, a molecule that binds to HLA-A2/NY-ESO comprising such antibody and the like.

Solution to Problem

[0029] The present inventors have conducted concentrated studies in order to solve the above problem. As a result, they discovered a novel anti-HLA-A2/NY-ESO antibody and a molecule that binds to HLA-A2/NY-ESO comprising such an antibody and the like. This has led to the completion of the present invention.

[0030] Specifically, the present invention includes the following. [0031] [1] An antibody that binds specifically to human HLA/NY-ESO or a binding fragment thereof comprising: [0032] a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54, [0033] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55, [0034] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56, [0035] a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57 or a light chain CDRL1 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 in which amino acid 7 is W and/or amino acid 8 is K, [0036] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and [0037] a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59 or a light chain CDRL3 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 in which amino acid 2 is A or S. [0038] [2] The antibody according to [1] comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 of one or more groups selected from the group consisting of the groups (i) to (v) below or a binding fragment thereof: [0039] (i) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54, [0040] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55, [0041] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56, [0042] a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57, [0043] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and [0044] a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59; [0045] (ii) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54, [0046] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55, [0047] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56, [0048] a light chain CDRL1 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 in which amino acid 7 is W, [0049] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and [0050] a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59; [0051] (iii) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54, [0052] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55, [0053] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56, [0054] a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57, [0055] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and [0056] a light chain CDRL3 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 in which amino acid 2 is A; [0057] (iv) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54, [0058] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55, [0059] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56, [0060] a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57, [0061] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and [0062]

a light chain CDRL3 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 in which amino acid 2 is S; and [0063] (v) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54, [0064] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55, [0065] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56, [0066] a light chain CDRL1 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 in which amino acid 7 is W and amino acid 8 is K, [0067] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and [0068] a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59. [0069] [3] The antibody or the binding fragment thereof according to [1] comprising a heavy chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27 or the amino acid sequence as shown in SEQ ID NO: 38 or 39 and a light chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27 or SEQ ID NO: 52 or the amino acid sequence as shown in SEQ ID NO: 40. [0070] [4] The antibody or the binding fragment thereof according to [1] comprising: [0071] (H1) a heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 6, [0072] (H2) a heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 18, [0073] (H3) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 29, [0074] (H4) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 26, [0075] (H5) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27, [0076] (H6) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 28, [0077] (H7) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 36, [0078] (H8) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 47, [0079] (H9) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 48, [0080] (H10) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 50; [0081] (H11) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 51, [0082] (H12) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 52, [0083] (H13) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 53, [0084] (H14) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 30; or [0085] (H15) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156; and [0086] (L1) a light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 8, [0087] (L2) a light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 20, [0088] (L3) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 29, [0089] (L4) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 26, [0090] (L5) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27, [0091] (L6) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 28, [0092] (L7) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 36, [0093] (L8) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 47, [0094] (L9) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 48, [0095] (L10) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 50, [0096] (L11) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 51, [0097] (L12) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 52, [0098] (L13) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53, [0099] (L14) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 30; or [0100] (L15) a light chain variable region consisting of an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156. [0101] [5] The antibody or the binding fragment thereof according to [4] comprising: [0102] (H1L1) a heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 6 and a light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 8, [0103] (H2L2) a heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 18 and a light chain variable region consisting of the amino acid

[illegible]

cell comprising the polynucleotide according to [8] or the vector according to [9]. [0137] [11] A method for producing an antibody that binds specifically to human HLA/NY-ESO or a binding fragment thereof comprising: (i) a step of culturing the host cell according to [10]; and (ii) a step of purifying an antibody or a binding fragment thereof from the culture product obtained in step (i). [0138] [12] An antibody that binds specifically to human HLA/NY-ESO or a binding fragment thereof, which is obtained by the method according to [11]. [0139] [13] An antibody that has the properties (i) or (ii) below and binds to HLA-A2/NY-ESO or a binding fragment thereof: [0140] (i) binding to a site on HLA-A2/NY-ESO recognized by the antibody or the binding fragment thereof according to [7]; or [0141] (ii) binding to human HLA-A2/NY-ESO competitively with the antibody or the binding fragment thereof according to [7]. [0142] [14] A pharmaceutical composition comprising, as an active ingredient, the antibody or the binding fragment thereof according to any of [1] to [7], [12], and [13], the polynucleotide according to [8], the vector according to [9], or the cell according to [10]. [0143] [15] A molecule that binds specifically to human HLA/NY-ESO comprising the antibody or the binding fragment thereof according to any of [1] to [7], [12], and [13]. [0144] [16] The molecule according to [15], which is a multispecific antibody. [0145] [17] The molecule according to [15], which is a bispecific antibody. [0146] [18] The molecule according to any of [15] to [17], which comprises an antibody that binds specifically to CD3 or a binding fragment thereof. [0147] [19] The molecule according to [18], wherein the antibody that binds specifically to CD3 or the binding fragment thereof comprises: [0148] (CCH1) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 141; [0149] (CCH2) a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 142 or a heavy chain CDRH2 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 142 in which amino acid 3 is N or S; [0150] (CCH3) a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 143; [0151] (CCL1) a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 144; [0152] (CCL2) a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 145 or a light chain CDRL2 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 145 in which amino acid 2 is N; and [0153] (CCL3) a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 146. [0154] [20] The molecule according to [19], wherein the antibody that binds specifically to CD3 or the binding fragment thereof comprises: [0155] (CH1) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136, [0156] (CH2) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137, [0157] (CH3) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 147, [0158] (CH4) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 138, [0159] (CH5) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139, [0160] (CH6) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140, [0161] (CH7) a heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155, [0162] (CH8) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156; or [0163] (CH9) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157; and [0164] (CL1) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136, [0165] (CL2) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137, [0166] (CL3) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 147, [0167] (CL4) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138, [0168] (CL5) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139, [0169] (CL6) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140, [0170] (CL7) a light chain variable region consisting of an amino acid sequence of amino acids 405 to 511 of the amino acid sequence as shown in SEQ ID NO: 155, [0171] (CL8) a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156, or [0172] (CL9) a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157. [0173] [21] The molecule according to [20], wherein the antibody that binds specifically to CD3 or the binding fragment thereof comprises: [0174] (CH1CL1) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136; [0175] (CH2CL2) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137; [0176] (CH3CL3) a heavy chain variable

region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 147 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 147; [0177] (CH4CL4) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 138 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138; [0178] (CH5CL5) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139; [0179] (CH6CL6) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140; [0180] (CH7CL7) a heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155 and a light chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155; [0181] (CH8CL8) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156 and a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156; or [0182] (CH9CL9) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157 and a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157. [0183] [22] The molecule according to any of [18] to [21], wherein the antibody that binds specifically to CD3 or the binding fragment thereof is scFv. [0184] [23] The molecule according to [22], wherein scFv comprises: [0185] (CS1) scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 136; [0186] (CS2) scFv consisting of an amino acid sequence of amino acids 2 to 241 of the amino acid sequence as shown in SEQ ID NO: 137; [0187] (CS3) scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 147; [0188] (CS4) scFv consisting of an amino acid sequence of amino acids 2 to 241 of the amino acid sequence as shown in SEQ ID NO: 138; [0189] (CS5) scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 139; [0190] (CS6) scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 140; [0191] (CS7) scFv consisting of an amino acid sequence of amino acids 272 to 511 of the amino acid sequence as shown in SEQ ID NO: 155; [0192] (CS8) scFv consisting of an amino acid sequence of amino acids 277 to 516 of the amino acid sequence as shown in SEQ ID NO: 156; or [0193] (CS9) scFv consisting of an amino acid sequence of amino acids 277 to 516 of the amino acid sequence as shown in SEQ ID NO: 157. [0194] [24] The molecule according to any of [18] to [23] comprising: a first polypeptide comprising scFv that binds specifically to human HLA/NY-ESO, scFv that binds specifically to CD3, and an Fc region (i) in that order from the N terminus toward the C terminus; and a second polypeptide comprising an Fc region (ii), wherein the first polypeptide is preferably associated with the second polypeptide at the Fc region (i) and the Fc region (ii). [0195] [25] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [1], which is scFv. [0196] [26] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [2], which is scFv. [0197] [27] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [3], which is scFv. [0198] [28] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [4], which is scFv. [0199] [29] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [5], which is scFv. [0200] [30] The molecule according to any of [19] to [24], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [18], which is scFv. [0201] [31] The molecule according to any of [19] to [24], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [19], which is scFv. [0202] [32] The molecule according to any of [19] to [24], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [20] or [21], which is scFv. [0203] [33] The molecule according to any of [19] to [24], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [23], which is scFv. [0204] [34] The molecule according to any of [24] to [33] comprising an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino

acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 150. [0205] [35] The molecule according to any of [24] to [33], which comprises an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 155, an amino acid sequence of amino acids 21 to 516 of the amino acid sequence as shown in SEQ ID NO: 156, and an amino acid sequence of amino acids 21 to 516 of the amino acid sequence as shown in SEQ ID NO: 157. [0206] [36] The molecule according to any of [24] to [34], wherein the first polypeptide comprises an amino acid sequence of amino acids 529 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 86, 149, or 150. [0207] [37] The molecule according to any of [24] to [33] and [35], wherein the first polypeptide comprises an amino acid sequence of amino acids 529 to 745 of the amino acid sequence as shown in SEQ ID NO: 155, an amino acid sequence of amino acids 534 to 750 of the amino acid sequence as shown in SEQ ID NO: 156, or an amino acid sequence of amino acids 534 to 750 of the amino acid sequence as shown in SEQ ID NO: 157. [0208] [38] The molecule according to [34] or [36], wherein the first polypeptide consists of an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 150. [0209] [39] The molecule according to [35] or [37], wherein the first polypeptide consists of an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 155, an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 156, and an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 157. [0210] [40] The molecule according to any of [24] to [39], wherein the second polypeptide comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84. [0211] [41] The molecule according to any of [18] to [23] comprising a first polypeptide, a second polypeptide, and a third polypeptide, wherein [0212] the first polypeptide comprises scFv that binds specifically to human HLA/NY-ESO, scFv that binds specifically to CD3, and an Fc region (i) in that order from the N terminus toward the C terminus, [0213] the second polypeptide comprises an immunoglobulin heavy chain comprising an Fc region (ii), and [0214] the third polypeptide comprises an immunoglobulin light chain; and [0215] the second polypeptide is preferably associated with the third polypeptide, and [0216] the first polypeptide is preferably associated with the second polypeptide in the Fc regions. [0217] [42] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [1], which is scFv. [0218] [43] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [2], which is scFv. [0219] [44] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [3], which is scFv. [0220] [45] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [4], which is scFv. [0221] [46] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [5], which is scFv. [0222] [47] The molecule according to any of [42] to [46], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof

according to [18], which is scFv. [0223] [48] The molecule according to any of [42] to [46], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [19], which is scFv. [0224] [49] The molecule according to any of [42] to [46], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [20] or [21], which is scFv. [0225] [50] The molecule according to any of [42] to [46], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [23], which is scFv. [0226] [51] The molecule according to any of [41] to [50], wherein the second polypeptide comprises an amino acid sequence of amino acids 20 to 242 of the amino acid sequence as shown in SEQ ID NO: 99. [0227] [52] The molecule according to any of [41] to [51], wherein the third polypeptide comprises the amino acid sequence as shown in SEQ ID NO: 100. [0228] [53] The molecule according to any of [41] to [52], wherein the first polypeptide comprises an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 150. [0229] [54] The molecule according to any of [41] to [53], wherein the first polypeptide consists of an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 150. [0230] [55] The molecule according to any one of [18] to [23] comprising a first polypeptide that comprises scFv that binds specifically to human HLA/NY-ESO, a heavy chain variable region and constant region CH1 of the antibody that binds specifically to CD3, and an immunoglobulin Fc region (i) in that order from the N terminus toward the C terminus, a second polypeptide that comprises a hinge region and an Fc region (ii) of the immunoglobulin, and a third polypeptide that comprises an antibody light chain consisting of a variable region and a constant region, wherein the first polypeptide is preferably associated with the second polypeptide at the Fc region (i) and the Fc region (ii), and the first polypeptide is preferably associated with the third polypeptide in the heavy chain variable region and the constant region CH1 of the antibody. [0231] [56] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [1], which is scFv. [0232] [57] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [2], which is scFv. [0233] [58] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [3], which is scFv. [0234] [59] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [4], which is scFv. [0235] [60] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [5], which is scFv. [0236] [61] The molecule according to any of [18] to [21], wherein the antibody that binds specifically to CD3 or the binding fragment thereof is Fab. [0237] [62] The molecule according to any of [56] to [61], wherein the Fab that binds specifically to CD3 is the antibody that binds specifically to CD3 or

the binding fragment thereof according to [18], which is Fab. [0238] [63] The molecule according to any of [56] to [61], wherein the Fab that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [19], which is Fab. [0239] [64] The molecule according to any of [56] to [61], wherein the Fab that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [20] or [21], which is Fab. [0240] [65] The molecule according to any of [55] to [64], wherein the first polypeptide comprises an amino acid sequence of amino acids 21 to 394 of the amino acid sequence as shown in SEQ ID NO: 160. [0241] [66] The molecule according to any of [55] to [65], wherein the first polypeptide comprises the amino acid sequence according to any one of (i) to (iii) below: [0242] (i) an amino acid sequence of amino acids 20 to 724 of the amino acid sequence as shown in SEQ ID NO: 160; [0243] (ii) an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 197; and [0244] (iii) an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 198. [0245] [67] The molecule according to any of [55] to [66], wherein the second polypeptide comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84. [0246] [68] The molecule according to any of [56] to [67], wherein the third polypeptide comprises an amino acid sequence of amino acids 21 to 127 of the amino acid sequence as shown in SEQ ID NO: 161. [0247] [69] The molecule according to any of [56] to [68], wherein the third polypeptide comprises an amino acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161. [0248] [70] The molecule according to any of [15] to [69], wherein 1 or 2 amino acids are deleted from the carboxyl terminus of the amino acid sequence of at least one polypeptide included in the molecule. [0249] [71] A polynucleotide comprising a nucleotide sequence encoding an amino acid sequence included in the molecule according to any of [15] to [70]. [0250] [72] A vector comprising the polynucleotide according to [71]. [0251] [73] A host cell comprising the polynucleotide according to [71] or the vector according to [72]. [0252] [74] A method for producing a molecule that binds specifically to human HLA/NY-ESO and human CD3 comprising: (i) a step of culturing the host cell according to [73]; and (ii) a step of purifying an antibody or a binding fragment thereof from the culture product obtained in the step (i). [0253] [75] A molecule that binds specifically to human HLA/NY-ESO and human CD3, which is obtained by the method according to [74]. [0254] [76] A pharmaceutical composition comprising, as an active ingredient, the molecule according to any of [15] to [70] and [75], the polynucleotide according to [71], the vector according to [72], or the host cell according to [73]. [0255] [77] The pharmaceutical composition according to [14] or [76], which is an anti-cancer agent. [0256] [78] The pharmaceutical composition according to [77], wherein cancer is one or more cancers selected from the group consisting of renal cancer, melanoma, squamous cell carcinoma, basal cell cancer, conjunctival cancer, oral cavity cancer, laryngeal cancer, pharyngeal cancer, thyroid gland cancer, lung cancer (non-small cell lung cancer (adenocarcinoma, epidermoid cancer, large cell cancer), and small cell lung cancer), breast cancer, esophageal cancer, gastric cancer, duodenal carcinoma, small bowel cancer, large bowel cancer, rectal cancer, appendiceal cancer, anal cancer, liver cancer, gallbladder cancer, bile duct cancer, pancreatic cancer, adrenal cancer, bladder cancer, prostate cancer, uterine cancer, vaginal cancer, liposarcoma, angiosarcoma, chondrosarcoma, rhabdomyosarcoma, Ewing's sarcoma, osteosarcoma, undifferentiated pleomorphic sarcoma, myxofibrosarcoma, malignant peripheral nerve sheath tumor, retroperitoneal sarcoma, synovial sarcoma, uterine sarcoma, gastrointestinal stromal tumor, leiomyosarcoma, epithelioid sarcoma, B cell lymphoma, T/NK-cell lymphoma, Hodgkin's lymphoma, myelogenic leukemia, lymphatic leukemia, myeloproliferative disorder, myelodysplastic syndrome, multiple myeloma, testicular carcinoma, and ovarian cancer. [0257] [79] The pharmaceutical composition according to any of [76] to [78], which is used in combination with another agent.

Advantageous Effects of Invention

[0258] According to the present invention, an antibody that binds to HLA-A2/NY-ESO and a novel bispecific antibody (bispecific molecule) that binds to HLA-A2/NY-ESO and to CD3 is obtained. Also, a novel pharmaceutical composition comprising, as an active ingredient, such an antibody (molecule) is obtained. Such antibody or molecule has cytotoxicity and is thus useful as an agent for treatment or prevention of cancer and the like.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0259] FIG. 1 shows a table demonstrating standardized gMFI of anti-HLA/NY-ESO scFv: NYA-0001, NYA-1143, NYA-1154, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, NYA-2061, NYA-2143, NYC-0003, and NYC-0004, relative to T2 cells supplemented with various point-mutant peptides. * Each underlined value indicates a half or lower than the standardized gMFI of each scFv relative to that of T2 cells supplemented with the NY-ESO peptide.

[0260] FIG. 2A shows a table demonstrating information concerning the selected homologous peptides. Binding affinity to HLA-A0201 predicted using NetMHCPan2.8 is indicated in terms of the 50% inhibitory concentration (IC_{sub}50).

[0261] FIG. 2B shows a table demonstrating standardized gMFI of anti-HLA/NY-ESO scFv: NYA-0001, NYA-1143, NYA-1154, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, NYA-2061, NYA-2143, NYC-0003, and NYC-0004, relative to T2 cells supplemented with various homologous peptides. * Each underlined value indicates a value larger than the standardized gMFI of each scFv relative to that of T2 cells supplemented with DMSO.

[0262] FIG. 3 shows antibody formats demonstrated in the examples. (a) shows an scFv format in which the antibody H chain variable region (VH, described below) is ligated to the antibody L chain variable region (VL, described below) (both blank) with a linker. In the Examples, an anti-HLA-A2/NY-ESO scFv and an anti-CD3 scFv were used for evaluation. (b) shows an Fab format that comprises VH (blank), the antibody H chain constant region (CH1: checkered pattern), VL (blank), and the antibody L chain constant region (border lines). In the Examples, anti-HLA-A2/NY-ESO Fab and the like were used for evaluation. (c) shows a taFv format in which two types of scFv constructs (blank and positive slopes) are ligated to each other with a linker. In the Examples, taFv including anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv were used for evaluation. (d) shows a taFv-heterodimer Fc-type format in which Fc (negative slopes) comprising a heterodimer-forming mutation introduced into a C-terminal site of taFv (the first polypeptide) is heterologously associated with another Fc (solid: the second polypeptide). In the example, the taFv-heterodimer Fc comprising anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv was used for evaluation. (e) shows a taFv-Fab-heterodimer Fc-type format in which Fab is added to the taFv-heterodimer Fc-type format. In the example, taFv including anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv and taFv-Fab-heterodimer-Fc including HLA-A2/NY-ESO Fab were used for evaluation. (f) shows the first polypeptide that is common between the taFv-heterodimer Fc-type and the taFv-Fab-heterodimer Fc-type. The first polypeptide comprises scFv that binds specifically to human HLA/NY-ESO, scFv that binds specifically to CD3, and the Fc region (i) in that order from the N terminus toward the C terminus. (g) shows the second polypeptide of the taFv-heterodimer Fc-type. The second polypeptide comprises a hinge region and the Fc region (ii). (h) shows the second polypeptide of the taFv-Fab-heterodimer Fc-type. The second polypeptide comprises an immunoglobulin heavy chain comprising a hinge region and the Fc region (ii). (i) shows the third polypeptide of the taFv-Fab-heterodimer Fc-type. The third polypeptide comprises an immunoglobulin light chain.

[0263] FIG. 4A demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, and NYF-0047, exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0264] FIG. 4B demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016 and NYF-0058, exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0265] FIG. 4C demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0023, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, exert cytotoxicity on the endogenous human NY-ESO-expressing U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0266] FIG. 4D demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, and NYF-0047, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0267] FIG. 4E demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016 and NYF-0058, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0268] FIG. 4F demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0023, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0269] FIG. 4G demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, and NYF-0047, do not exert cytotoxicity on human AGS cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0270] FIG. 4H demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016 and NYF-0058, do not exert cytotoxicity on human AGS cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0271] FIG. 4I demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0023, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, do not exert cytotoxicity on human AGS cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0272] FIG. 4J demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, and NYF-0047, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0273] FIG. 4K demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016 and NYF-0058, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0274] FIG. 4L demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0023, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0275] FIG. 5A demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0044, and NYF-0047, exert antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5, but only n=4 on Day 32 in the NYF-0044 group).

[0276] FIG. 5B demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0022, NYF-0023, NYF-0027, NYF-0035, and NYF-0058, exert antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5, n=6 only in the Vehicle control group).

[0277] FIG. 5C demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, exert antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5).

[0278] FIG. 6A shows antibody formats described in the examples. (a): Hybrid-type format: Fc (slopes) comprising a heterodimer-forming mutation and Fc (solid) comprising a heterodimer-forming mutation are added to the C terminal region of Fab and scFv, respectively, and the two Fc constructs are associated. In the example, a hybrid-type format comprising anti-HLA-A2/NY-ESO Fab and anti-CD3 scFv was used for evaluation. (b): Dual-type format: Fc (negative slopes) comprising a heterodimer-forming mutation and Fc (solid) comprising a heterodimer-forming mutation are added to the C terminal regions of two different types of scFv constructs and the two Fc constructs are heterologously associated. In the example, a dual-type format comprising anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv was used for evaluation. (c): scFv-Fab-heterodimer Fc-type format: Fc (negative slopes) comprising a heterodimer-forming mutation is added to the C terminal region where scFv is ligated to Fab with a linker and associated with another Fc (solid). In the example, an scFv-Fab-heterodimer Fc-type format comprising anti-CD3 scFv (positive slopes) and anti-HLA-A2/NY-ESO Fab was used for evaluation. Also, the scFv-Fab-heterodimer Fc-type comprising anti-HLA-A2/NY-ESO scFv (positive slopes) and anti-CD3 Fab was used for evaluation.

[0279] FIG. 6B shows antibody formats described in the examples. (a): taFv-heterodimer Fc-type format: the same as FIG. 3 (d); Fe (negative slopes) comprising a heterodimer-forming mutation is added to the C terminal region of taFv and heterologously associated with another Fe (solid). In the example, the taFv-heterodimer Fc-type comprising anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv was used for evaluation. (b): taFv (inversed)-heterodimer Fc-type format: the order of the two scFv constructs is inverted in the taFv-heterodimer Fc-type format. In the example, taFv (inversed)-heterodimer Fc-type comprising anti-CD3 scFv and anti-HLA-A2/NY-ESO scFv was used for evaluation. (c) shows the first polypeptide of the taFv (inversed)-heterodimer Fc-type. The first polypeptide comprises scFv that binds specifically to CD3, scFv that binds specifically to human HLA/NY-ESO, and the Fc region (i) in that order from the N terminus toward the C terminus. (d) shows the second polypeptide of the taFv (inversed)-heterodimer Fc-type. The second polypeptide comprises a hinge region and the Fc region (ii).

[0280] FIG. 7A demonstrates that various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; Hybrid-type (NYG-3143), Dual-type (NYG-2143), and taFv-heterodimer Fc-type (NYF-0011), exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0281] FIG. 7B demonstrates that various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; scFv-Fab-heterodimer Fc-type (NYF-0003), taFv-heterodimer Fc-type (NYF-0010), taFv (inversed)-heterodimer Fc-type (NYF-0004), exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0282] FIG. 8 The amino acid sequence of the peptide in NY-ESO (SEQ ID NO: 1)

[0283] FIG. 9 The amino acid sequence of the peptide in MAGEC-1 (SEQ ID NO: 2)

[0284] FIG. 10 The scFv sequence analysis primer 1 (SEQ ID NO: 3)

[0285] FIG. 11 The scFv sequence analysis primer 2 (SEQ ID NO: 4)

[0286] FIG. 12 The nucleotide sequence of the heavy chain variable region of NYA-0001 (SEQ ID NO: 5)

[0287] FIG. 13 The amino acid sequence of the heavy chain variable region of NYA-0001 (SEQ ID NO: 6)

[0288] FIG. 14 The nucleotide sequence of the light chain variable region of NYA-0001 (SEQ ID NO: 7)

[0289] FIG. 15 The amino acid sequence of the light chain variable region of NYA-0001 (SEQ ID NO: 8)

[0290] FIG. **16** The nucleotide sequence of the heavy chain variable region of NYA-0060 (SEQ ID NO: 9)
[0291] FIG. **17** The amino acid sequence of the heavy chain variable region of NYA-0060 (SEQ ID NO: 10)
[0292] FIG. **18** The nucleotide sequence of the light chain variable region of NYA-0060 (SEQ ID NO: 11)
[0293] FIG. **19** The amino acid sequence of the light chain variable region of NYA-0060 (SEQ ID NO: 12)
[0294] FIG. **20** The nucleotide sequence of the heavy chain variable region of NYA-0068 (SEQ ID NO: 13)
[0295] FIG. **21** The amino acid sequence of the heavy chain variable region of NYA-0068 (SEQ ID NO: 14)
[0296] FIG. **22** The nucleotide sequence of the light chain variable region of NYA-0068 (SEQ ID NO: 15)
[0297] FIG. **23** The amino acid sequence of the light chain variable region of NYA-0068 (SEQ ID NO: 16)
[0298] FIG. **24** The nucleotide sequence of the heavy chain variable region of NYA-0082 (SEQ ID NO: 17)
[0299] FIG. **25** The amino acid sequence of the heavy chain variable region of NYA-0082 (SEQ ID NO: 18)
[0300] FIG. **26** The nucleotide sequence of the light chain variable region of NYA-0082 (SEQ ID NO: 19)
[0301] FIG. **27** The amino acid sequence of the light chain variable region of NYA-0082 (SEQ ID NO: 20)
[0302] FIG. **28** The nucleotide sequence of the NYA-1163 tag adduct (SEQ ID NO: 21)
[0303] FIG. **29** The nucleotide sequence of the NYA-2023 tag adduct (SEQ ID NO: 22)
[0304] FIG. **30** The nucleotide sequence of the NYA-2027 tag adduct (SEQ ID NO: 23)
[0305] FIG. **31** The nucleotide sequence of the NYA-1143 tag adduct (SEQ ID NO: 24)
[0306] FIG. **32** The nucleotide sequence of the NYA-2143 tag adduct (SEQ ID NO: 25)
[0307] FIG. **33** The amino acid sequence of the NYA-1163 tag adduct (SEQ ID NO: 26); NYA-1163: amino acids 21 to 266
[0308] FIG. **34** The amino acid sequence of the NYA-2023 tag adduct (SEQ ID NO: 27); NYA-2023: amino acids 21 to 266
[0309] FIG. **35** The amino acid sequence of the NYA-2027 tag adduct (SEQ ID NO: 28); NYA-2027: amino acids 21 to 266
[0310] FIG. **36** The amino acid sequence of the NYA-1143 tag adduct (SEQ ID NO: 29); NYA-1143: amino acids 21 to 266
[0311] FIG. **37** The amino acid sequence of the NYA-2143 tag adduct (SEQ ID NO: 30); NYA-2143: amino acids 21 to 266
[0312] FIG. **38** The nucleotide sequence of the NYA-1154 tag adduct (SEQ ID NO: 31)
[0313] FIG. **39** The amino acid sequence of the NYA-1154 tag adduct (SEQ ID NO: 32); NYA-1154: amino acids 21 to 266
[0314] FIG. **40** The amino acid sequence of HLA-A*0201 (GenBank: ASA47534.1) truncate (SEQ ID NO: 33)
[0315] FIG. **41** The amino acid sequence of 02-microglobulin (SEQ ID NO: 34)
[0316] FIG. **42** The nucleotide sequence of the NYA-2035 tag adduct (SEQ ID NO: 35)
[0317] FIG. **43** The amino acid sequence of the NYA-2035 tag adduct (SEQ ID NO: 36); NYA-2035: amino acids 21 to 266
[0318] FIG. **44** The amino acid sequence of NYA-1143-VH01 (SEQ ID NO: 37)
[0319] FIG. **45** The amino acid sequence of NYA-1143-VH02 (SEQ ID NO: 38)
[0320] FIG. **46** The amino acid sequence of NYA-1143-VH03 (SEQ ID NO: 39)
[0321] FIG. **47** The amino acid sequence of NYA-1143-VL01 (SEQ ID NO: 40)
[0322] FIG. **48** The nucleotide sequence of the NYA-2044 tag adduct (SEQ ID NO: 41)
[0323] FIG. **49** The nucleotide sequence of the NYA-2045 tag adduct (SEQ ID NO: 42)
[0324] FIG. **50** The nucleotide sequence of the NYA-2047 tag adduct (SEQ ID NO: 43)
[0325] FIG. **51** The nucleotide sequence of the NYA-2048 tag adduct (SEQ ID NO: 44)
[0326] FIG. **52** The nucleotide sequence of the NYA-2060 tag adduct (SEQ ID NO: 45)
[0327] FIG. **53** The nucleotide sequence of the NYA-2061 tag adduct (SEQ ID NO: 46)
[0328] FIG. **54** The amino acid sequence of the NYA-2044 tag adduct (SEQ ID NO: 47); NYA-2044: amino acids 21 to 266
[0329] FIG. **55** The amino acid sequence of the NYA-2045 tag adduct (SEQ ID NO: 48); NYA-2045: amino acids 21 to 266
[0330] FIG. **56** The amino acid sequence of the NYA-0082 (SEQ ID NO: 49)
[0331] FIG. **57** The amino acid sequence of the NYA-2047 tag adduct (SEQ ID NO: 50); NYA-2047: amino acids 21 to 266
[0332] FIG. **58** The amino acid sequence of the NYA-2048 tag adduct (SEQ ID NO: 51); NYA-2048: amino acids 21 to 266
[0333] FIG. **59** The amino acid sequence of the NYA-2060 tag adduct (SEQ ID NO: 52); NYA-2060: amino acids 21 to 266
[0334] FIG. **60** The amino acid sequence of the NYA-2061 tag adduct (SEQ ID NO: 53); NYA-2061: amino acids 21 to 266
[0335] FIG. **61** The amino acid sequences of the CDRH1 to CDRH3 of the NYA-0001 heavy chain and CDRL1 to

CDRL3 of the NYA-0001 light chain (SEQ ID NOs: 54 to 59)

[0336] FIG. **62** The amino acid sequence of CDRL1 of NYA-2023 (SEQ ID NO: 60)

[0337] FIG. **63** The amino acid sequence of CDRL3 of NYA-2027 (SEQ ID NO: 61)

[0338] FIG. **64** The amino acid sequences of CDRH3 and CDRL3 of NYA-1154 (SEQ ID NOs: 62 and 63)

[0339] FIG. **65** The amino acid sequence of CDRL1 of NYA-0035 (SEQ ID NO: 64)

[0340] FIG. **66** The nucleotide sequence of the NYC-0003 tag adduct (SEQ ID NO: 65)

[0341] FIG. **67** The nucleotide sequence of the NYC-0004 tag adduct (SEQ ID NO: 66)

[0342] FIG. **68** The amino acid sequence of the NYC-0003 tag adduct (SEQ ID NO: 67); NYC-0003: amino acids 21 to 263

[0343] FIG. **69** The amino acid sequence of the NYC-0004 tag adduct (SEQ ID NO: 68); NYC-0004: amino acids 21 to 263

[0344] FIG. **70** The nucleotide sequence of the NYA-0001 tag adduct (SEQ ID NO: 69)

[0345] FIG. **71** The amino acid sequence of the NYA-0001 tag adduct (SEQ ID NO: 70); NYA-0001: amino acids 21 to 266

[0346] FIG. **72** The nucleotide sequence of HC1 (SEQ ID NO: 71)

[0347] FIG. **73** The nucleotide sequence of NYF-0016-HC2 (SEQ ID NO: 72)

[0348] FIG. **74** The nucleotide sequence of NYF-0019-HC2 (SEQ ID NO: 73)

[0349] FIG. **75** The nucleotide sequence of NYF-0022-HC2 (SEQ ID NO: 74)

[0350] FIG. **76** The nucleotide sequence of NYF-0023-HC2 (SEQ ID NO: 75)

[0351] FIG. **77** The nucleotide sequence of NYF-0027-HC2 (SEQ ID NO: 76)

[0352] FIG. **78** The nucleotide sequence of NYF-0035-HC2 (SEQ ID NO: 77)

[0353] FIG. **79** The nucleotide sequence of NYF-0044-HC2 (SEQ ID NO: 78)

[0354] FIG. **80** The nucleotide sequence of NYF-0045-HC2 (SEQ ID NO: 79)

[0355] FIG. **81** The nucleotide sequence of NYF-0047-HC2 (SEQ ID NO: 80)

[0356] FIG. **82** The nucleotide sequence of NYF-0048-HC2 (SEQ ID NO: 81)

[0357] FIG. **83** The nucleotide sequence of NYF-0060-HC2 (SEQ ID NO: 82)

[0358] FIG. **84** The nucleotide sequence of NYF-0061-HC2 (SEQ ID NO: 83)

[0359] FIG. **85** The amino acid sequence of HC1 (SEQ ID NO: 84)

[0360] FIG. **86** The amino acid sequence of NYF-0016-HC2 (SEQ ID NO: 85); NYA-1143: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0361] FIG. **87** The amino acid sequence of NYF-0019-HC2 (SEQ ID NO: 86); NYA-2143: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0362] FIG. **88** The amino acid sequence of NYF-0022-HC2 (SEQ ID NO: 87); NYA-1163: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0363] FIG. **89** The amino acid sequence of NYF-0023-HC2 (SEQ ID NO: 88); NYA-2023: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0364] FIG. **90** The amino acid sequence of NYF-0027-HC2 (SEQ ID NO: 89); NYA-2027: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0365] FIG. **91** The amino acid sequence of NYF-0035-HC2 (SEQ ID NO: 90); NYA-2035: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0366] FIG. **92** The amino acid sequence of NYF-0044-HC2 (SEQ ID NO: 91); NYA-2044: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0367] FIG. **93** The amino acid sequence of NYF-0045-HC2 (SEQ ID NO: 92); NYA-2045: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0368] FIG. **94** The amino acid sequence of NYF-0047-HC2 (SEQ ID NO: 93); NYA-2047: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0369] FIG. **95** The amino acid sequence of NYF-0048-HC2 (SEQ ID NO: 94); NYA-2048: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0370] FIG. **96** The amino acid sequence of NYF-0060-HC2 (SEQ ID NO: 95); NYA-2060: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0371] FIG. **97** The amino acid sequence of NYF-0061-HC2 (SEQ ID NO: 96); NYA-2061: amino acids 21 to 266; C3E-7085: amino acids 272 to 511

[0372] FIG. **98** The nucleotide sequence of NYA-0001-Fab-HC1-k delete (SEQ ID NO: 97)

[0373] FIG. **99** The nucleotide sequence of NYA-0001-LC (SEQ ID NO: 98)

[0374] FIG. **100** The amino acid sequence of NYA-0001-Fab-HC1-k delete (SEQ ID NO: 99); NYA-0001 heavy chain variable region: amino acids 20 to 139

[0375] FIG. **101** The amino acid sequence of NYA-0001-LC (SEQ ID NO: 100); NYA-0001 light chain variable region: amino acids 21 to 131

[0376] FIG. **102** The nucleotide sequence of NYA-1143-Fab-HC1-k delete (SEQ ID NO: 101)

[0377] FIG. **103** The nucleotide sequence of NYA-1143-LC (SEQ ID NO: 102)
[0378] FIG. **104** The nucleotide sequence of C3E-7085-HC2-k deleteC (SEQ ID NO: 103)
[0379] FIG. **105** The amino acid sequence of NYA-1143-Fab-HC1-k delete (SEQ ID NO: 104); the NYA-1143 heavy chain variable region: amino acids 20 to 139
[0380] FIG. **106** The amino acid sequence of NYA-1143-LC (SEQ ID NO: 105); the NYA-1143 light chain variable region: amino acids 21 to 131
[0381] FIG. **107** The amino acid sequence of C3E-7085-HC2-k delete (SEQ ID NO: 106); C3E-7085: amino acids 21 to 260
[0382] FIG. **108** The nucleotide sequence of NYA-1143-HC1-k delete (SEQ ID NO: 107)
[0383] FIG. **109** The amino acid sequence of NYA-1143-HC1-k delete (SEQ ID NO: 108); NYA-1143: amino acids 21 to 266
[0384] FIG. **110** The nucleotide sequence of C3E-7085-NYA-1154-Fab-HC2-k delete (SEQ ID NO: 109)
[0385] FIG. **111** The nucleotide sequence of NYA-1154-LC (SEQ ID NO: 110)
[0386] FIG. **112** The nucleotide sequence of OAA-HC1-k delete (SEQ ID NO: 111)
[0387] FIG. **113** The amino acid sequence of C3E-7085-NYA-1154-Fab-HC2-k delete (SEQ ID NO: 112); C3E-7085: amino acids 21 to 260; the NYA-1154 heavy chain variable region: amino acids 266 to 285
[0388] FIG. **114** The amino acid sequence of NYA-1154-LC (SEQ ID NO: 113); the NYA-1154 light chain variable region: amino acids 21 to 131
[0389] FIG. **115** The amino acid sequence of OAA-HC1-k delete (SEQ ID NO: 114)
[0390] FIG. **116** The nucleotide sequence of NYF-0010-HC2-k delete (SEQ ID NO: 115)
[0391] FIG. **117** The nucleotide sequence of NYF-0004-HC2-k delete (SEQ ID NO: 116)
[0392] FIG. **118** The nucleotide sequence of NYF-0011-HC2-k delete (SEQ ID NO: 117)
[0393] FIG. **119** The amino acid sequence of NYF-0010-HC2-k delete (SEQ ID NO: 118); NYA-1154: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
[0394] FIG. **120** The amino acid sequence of NYF-0004-HC2-k delete (SEQ ID NO: 119); C3E-7085: amino acids 21 to 260; NYA-1154: amino acids 272 to 511
[0395] FIG. **121** The amino acid sequence of NYF-0011-HC2-k delete (SEQ ID NO: 120); NYA-1143: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
[0396] FIG. **122** The amino acid sequence of the point-mutant NY-ESO peptide 1F (SEQ ID NO: 121)
[0397] FIG. **123** The amino acid sequence of the point-mutant NY-ESO peptide 2M (SEQ ID NO: 122)
[0398] FIG. **124** The amino acid sequence of the point-mutant NY-ESO peptide 3A (SEQ ID NO: 123)
[0399] FIG. **125** The amino acid sequence of the point-mutant NY-ESO peptide 4A (SEQ ID NO: 124)
[0400] FIG. **126** The amino acid sequence of the point-mutant NY-ESO peptide 5A (SEQ ID NO: 125)
[0401] FIG. **127** The amino acid sequence of the point-mutant NY-ESO peptide 6L (SEQ ID NO: 126)
[0402] FIG. **128** The amino acid sequence of the point-mutant NY-ESO peptide 7F (SEQ ID NO: 127)
[0403] FIG. **129** The amino acid sequence of the point-mutant NY-ESO peptide 8A (SEQ ID NO: 128)
[0404] FIG. **130** The amino acid sequence of the point-mutant NY-ESO peptide 9A (SEQ ID NO: 129)
[0405] FIG. **131** The amino acid sequence of the gp100 peptide (SEQ ID NO: 130)
[0406] FIG. **132** The amino acid sequence of the homologous peptide DOLPP1 (SEQ ID NO: 131)
[0407] FIG. **133** The amino acid sequence of the homologous peptide IL20RB (SEQ ID NO: 132)
[0408] FIG. **134** The amino acid sequence of the homologous peptide PRKD2 (SEQ ID NO: 133)
[0409] FIG. **135** The amino acid sequence of the homologous peptide CD163 (SEQ ID NO: 134)
[0410] FIG. **136** The amino acid sequence of the homologous peptide P2RY8 (SEQ ID NO: 135)
[0411] FIG. **137** The amino acid sequence of C3E-7034 (SEQ ID NO: 136)
[0412] FIG. **138** The amino acid sequence of C3E-7036 (SEQ ID NO: 137)
[0413] FIG. **139** The amino acid sequence of C3E-7085 (SEQ ID NO: 138)
[0414] FIG. **140** The amino acid sequence of C3E-7088 (SEQ ID NO: 139)
[0415] FIG. **141** The amino acid sequence of C3E-7093 (SEQ ID NO: 140)
[0416] FIG. **142** The amino acid sequences of CDRH1 to CDRH3 of the C3E-7085 heavy chain and CDRL1 to CDRL3 of the C3E-7085 light chain (SEQ ID NOs: 141 to 146)
[0417] FIG. **143** The amino acid sequence of C3E-7078 (SEQ ID NO: 147)
[0418] FIG. **144** The nucleotide sequence of NYF-0014-HC2 (SEQ ID NO: 148)
[0419] FIG. **145** The amino acid sequence of NYF-0014-HC2 (SEQ ID NO: 149); NYA-0001: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
[0420] FIG. **146** The amino acid sequence of NYF-0082-HC2 (SEQ ID NO: 150); NYA-0082: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
[0421] FIG. **147** The amino acid sequence of human CD3 P (SEQ ID NO: 151)
[0422] FIG. **148** The nucleotide sequence of the entire NYZ-0038-HC2 (SEQ ID NO: 152)
[0423] FIG. **149** The nucleotide sequence of the entire NYZ-0082-HC2 (SEQ ID NO: 153)

[0424] FIG. **150** The nucleotide sequence of the entire NYZ-0083-HC2 (SEQ ID NO: 154)

[0425] FIG. **151** The amino acid sequence of the entire NYZ-0038-HC2 (SEQ ID NO: 155); NYA-2061: amino acids 21 to 266; C3E-7096: amino acids 272 to 511

[0426] FIG. **152** The amino acid sequence of the entire NYZ-0082-HC2 (SEQ ID NO: 156); NYA-3061: amino acids 21 to 271; C3E-7096: amino acids 277 to 516.

[0427] FIG. **153** The amino acid sequence of the entire NYZ-0083-HC2 (SEQ ID NO: 157); NYA-3061: amino acids 21 to 271; C3E-7097: amino acids 277 to 516.

[0428] FIG. **154** The nucleotide sequence of the full-length NYZ-1010-HC2 (SEQ ID NO: 158)

[0429] FIG. **155** The nucleotide sequence of the full-length C3E-7085-LC (SEQ ID NO: 159)

[0430] FIG. **156** The amino acid sequence of the full-length NYZ-1010-HC2 (SEQ ID NO: 160); NYA-3061: amino acids 21 to 271; the C3E-7085 heavy chain variable region: amino acids 277 to 394

[0431] FIG. **157** The amino acid sequence of the full-length C3E-7085-LC (SEQ ID NO: 161)

[0432] FIG. **158A** FIG. **158A** shows a table demonstrating standardized gMFI of the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010, relative to CD3e knockout T2 cells supplemented with various point-mutant peptides. * Each underlined value indicates a half or lower than the standardized gMFI of each Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule, relative to CD3e knockout T2 cells supplemented with the NY-ESO peptide, and * each bold value indicates a quarter or lower of the standardized gMFI.

[0433] FIG. **158B** shows a table demonstrating standardized gMFI of the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010, relative to CD3e knockout T2 cells supplemented with various homologous peptides. * Each underlined value indicates a value of each Fc-added anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule larger than the standardized gMFI of CD3e knockout T2 cells supplemented with DMSO.

[0434] FIG. **159A** FIG. **159A** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038, NYZ-0082, and NYZ-0083, exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0435] FIG. **159B** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-1010, exerts cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0436] FIG. **159C** FIG. **159C** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0082 and NYZ-1010, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0437] FIG. **159D** FIG. **159D** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038 and NYZ-0083, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0438] FIG. **159E** FIG. **159E** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038, NYZ-0082, and NYZ-0083, do not exert cytotoxicity on human AGS cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0439] FIG. **159F** FIG. **159F** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-1010, does not exert cytotoxicity on human AGS cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0440] FIG. **159G** FIG. **159G** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0082 and NYZ-1010, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0441] FIG. **159H** FIG. **159H** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038 and NYZ-0083, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0442] FIG. **160A** FIG. **160A** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-0038, exerts antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5).

[0443] FIG. **160B** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-0082, exerts antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5).

[0444] FIG. **160C** FIG. **160C** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-0083, exerts antitumor activity on human PBMC-transfected models. An error bar in the figure

indicates the standard deviation (n=5).

[0445] FIG. **160D** FIG. **160D** demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-1010, exerts antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5).

[0446] FIG. **161** The amino acid sequence of the peptide linker (SEQ ID NO: 162)

[0447] FIG. **162** shows the polymer content (%) upon acid treatment of various Fc-conjugated anti-HLA/NY-ESO-anti-CD3 bispecific molecules.

[0448] FIG. **163** shows the evaluation results of solution stability of anti-HLA/NY-ESO scFv-heterodimer Fc.

[0449] FIG. **164** The nucleotide sequence of the full-length NYA-3061 (SEQ ID NO: 163)

[0450] FIG. **165** The amino acid sequence of the full-length NYA-3061 (SEQ ID NO: 164)

[0451] FIG. **166** The nucleotide sequence of the full-length NYC-0005 (SEQ ID NO: 165)

[0452] FIG. **167** The amino acid sequence of the full-length NYC-0005 (SEQ ID NO: 166)

[0453] FIG. **168** The nucleotide sequence of the full-length NYC-0006 (SEQ ID NO: 167)

[0454] FIG. **169** The amino acid sequence of the full-length NYC-0006 (SEQ ID NO: 168)

[0455] FIG. **170** The nucleotide sequence of the full-length NYC-0007 (SEQ ID NO: 169)

[0456] FIG. **171** The amino acid sequence of the full-length NYC-0007 (SEQ ID NO: 170)

[0457] FIG. **172** The nucleotide sequence of the full-length NYC-0008 (SEQ ID NO: 171)

[0458] FIG. **173** The amino acid sequence of the full-length NYC-0008 (SEQ ID NO: 172)

[0459] FIG. **174** The nucleotide sequence of the full-length NYC-0009 (SEQ ID NO: 173)

[0460] FIG. **175** The amino acid sequence of the full-length NYC-0009 (SEQ ID NO: 174)

[0461] FIG. **176** The nucleotide sequence of the full-length NYC-0010 (SEQ ID NO: 175)

[0462] FIG. **177** The amino acid sequence of the full-length NYC-0010 (SEQ ID NO: 176)

[0463] FIG. **178** The nucleotide sequence of the full-length HC-h (SEQ ID NO: 177)

[0464] FIG. **179** The amino acid sequence of the full-length HC-h (SEQ ID NO: 178)

[0465] FIG. **180** The nucleotide sequence of the full-length NYD-2047-HC-k (SEQ ID NO: 179)

[0466] FIG. **181** The amino acid sequence of the full-length NYD-2047-HC-k (SEQ ID NO: 180)

[0467] FIG. **182** The nucleotide sequence of the full-length NYD-2061-HC-k (SEQ ID NO: 181)

[0468] FIG. **183** The amino acid sequence of the full-length NYD-2061-HC-k (SEQ ID NO: 182)

[0469] FIG. **184** The nucleotide sequence of the full-length NYD-3061-HC-k (SEQ ID NO: 183)

[0470] FIG. **185** The amino acid sequence of the full-length NYD-3061-HC-k (SEQ ID NO: 184)

[0471] FIG. **186** The nucleotide sequence of the full-length NYC-0011-HC-k (SEQ ID NO: 185)

[0472] FIG. **187** The amino acid sequence of the full-length NYC-0011-HC-k (SEQ ID NO: 186)

[0473] FIG. **188** The nucleotide sequence of the full-length NYC-0012-HC-k (SEQ ID NO: 187)

[0474] FIG. **189** The amino acid sequence of the full-length NYC-0012-HC-k (SEQ ID NO: 188)

[0475] FIG. **190** The nucleotide sequence of the full-length NYC-0013-HC-k (SEQ ID NO: 189)

[0476] FIG. **191** The amino acid sequence of the full-length NYC-0013-HC-k (SEQ ID NO: 190)

[0477] FIG. **192** The nucleotide sequence of the full-length NYC-0014-HC-k (SEQ ID NO: 191)

[0478] FIG. **193** The amino acid sequence of the full-length NYC-0014-HC-k (SEQ ID NO: 192)

[0479] FIG. **194** The nucleotide sequence of the full-length NYC-0015-HC-k (SEQ ID NO: 193)

[0480] FIG. **195** The amino acid sequence of the full-length NYC-0015-HC-k (SEQ ID NO: 194)

[0481] FIG. **196** The nucleotide sequence of the full-length NYC-0016-HC-k (SEQ ID NO: 195)

[0482] FIG. **197** The amino acid sequence of the full-length NYC-0016-HC-k (SEQ ID NO: 196)

[0483] FIG. **198** The amino acid sequence of the full-length NYZ-1007-HC2 (SEQ ID NO: 197); NYA-2061: amino acids 21 to 266; the C3E-7085 heavy chain variable region: amino acids 272 to 389

[0484] FIG. **199** The amino acid sequence of the full-length NYZ-1017-HC2 (SEQ ID NO: 198); NYA-2047: amino acids 21 to 266; the C3E-7085 heavy chain variable region: amino acids 277 to 389

DESCRIPTION OF EMBODIMENTS

[0485] Hereafter, the present invention is described in detail.

1. Definition

[0486] In the present invention, the term “gene” refers to a nucleotide chain including a nucleotide sequence encoding an amino acid of a protein or a complementary chain thereof. For example, a polynucleotide, oligonucleotide, DNA, mRNA, cDNA, cRNA, or the like, which is a nucleotide chain including a nucleotide sequence encoding an amino acid of a protein or a complementary chain thereof, is within the scope of the “gene.” Such gene is a single-stranded, double-stranded, or triple or more-stranded nucleotide, and an association of a DNA chain and an RNA chain, a single nucleotide chain comprising both ribonucleotide (RNA) and deoxyribonucleotide (DNA), and a double-stranded or triple-stranded nucleotide comprising such a nucleotide chain are included within the scope of the “gene.” In the present invention, the term “base sequence” is synonymous with the term “nucleotide sequence.”

[0487] In the present invention, the terms “polynucleotide,” “nucleotide chain,” “nucleic acid,” and “nucleic acid

molecule” are synonymous. DNA, RNA, probe, oligonucleotide, and primer are within the scope of “polynucleotide.” Such a polynucleotide consists of 1, 2, 3, or more chains, and an association of a DNA chain and an RNA chain, a single polynucleotide chain comprising both ribonucleotide (RNA) and deoxyribonucleotide (DNA), and a double-stranded or triple-stranded nucleotide comprising such a polynucleotide chain are included within the scope of the “polynucleotide.”

[0488] In the present invention, the terms “polypeptide,” “peptide,” and “protein” are synonymous.

[0489] In the present invention, the term “antigen” may refer to the term “immunogen,” according to need.

[0490] In the present invention, the term “cell” encompasses various cells derived from animals, subculture cells, primary culture cells, cell lines, recombinant cells, and microorganisms.

[0491] In the present invention, the term “antibody” is synonymous with the term “immunoglobulin.” However, the term “antibody” in the case of the anti-HLA/NY-ESO antibody according to the present invention refers to immunoglobulin comprising a constant region and a variable region. An antibody is not particularly limited, and it may be a naturally occurring or partially or completely synthesized immunoglobulin. The anti-HLA/NY-ESO antibody according to the present invention is included in the term “molecule” described below.

[0492] In the present invention, “NY-ESO peptide” indicates a peptide consisting of the 9 amino acids 157 to 165 of NY-ESO-1 and LAGE-1 (SLLMWITQC: SEQ ID NO: 1).

[0493] In the present invention, “HLA-A2/NY-ESO” indicates a complex of NY-ESO peptide and Histocompatibility Leukocyte Antigen-A2 (HLA-A2), and it is also denoted as “HLA/NY-ESO.”

[0494] In the present invention, the term “anti-HLA-A2/NY-ESO antibody” refers to an antibody that binds to HLA-A2/NY-ESO. In other words, the term refers to an antibody that recognizes HLA-A2/NY-ESO. Also, the term “anti-HLA-A2/NY-ESO scFv” refers to scFv that binds to HLA/NY-ESO. In other words, the term refers to scFv that recognizes HLA-A2/NY-ESO. The terms “anti-HLA-A2/NY-ESO antibody” and “anti-HLA-A2/NY-ESO scFv” are also denoted as the “anti-HLA/NY-ESO antibody” and the “anti-HLA/NY-ESO scFv,” respectively.

[0495] A basic four-chain antibody structure is composed of two identical light chains (L chains) and two identical heavy chains (H chains). A light chain binds to a heavy chain via a single covalent disulfide bond. Two heavy chains are bound to each other via one or more disulfide bonds in accordance with heavy chain isotypes. A light chain and a heavy chain each have an intra-chain disulfide bond at regular intervals. In a light chain and a heavy chain, there are a constant region exhibiting very high amino acid sequence similarity and a variable region exhibiting low amino acid sequence similarity. A light chain comprises, at its amino terminus, a variable region (VL) adjacent to a constant region (CL). A heavy chain comprises, at its amino terminus, a variable region (VH) adjacent to 3 constant regions (CH1/CH2/CH3). VL is paired with VH, and CL is aligned with a first constant region of a heavy chain (CH1). VL is paired with VH to form a single antigen-binding site.

[0496] A constant region of the antibody of the present invention is not particularly limited. The antibody of the present invention to be used for treatment or prevention of human diseases preferably comprises a constant region of a human antibody. Examples of heavy chain constant regions of a human antibody include C γ 1, C γ 2, C γ 3, C γ 4, C μ , C δ , C α 1, C α 2, and C ϵ . Examples of light chain constant regions of a human antibody include C κ and C λ .

[0497] Fab comprises a heavy chain VH, CH1 adjacent thereto, a light chain VL, and CL adjacent thereto. VH and VL each comprise a complementarity determining region (CDR).

[0498] Fc (also referred to as an “Fc region”) is a carboxyl terminal region of a heavy chain constant region, it comprises CH2 and CH3, and it is a dimer. Fc of the present invention may comprise a naturally occurring sequence or it may comprise a sequence derived from the naturally occurring sequence by mutation (referred to as “mutant Fc”). In the multispecific molecule and the bispecific molecule of the present invention, a Fc region is preferably mutant Fc, and more preferably a combination of Fc regions capable of forming a heterodimer. An example of a combination of Fc regions is a combination of Fc (i) in the first polypeptide and Fc (ii) in the second polypeptide described below. A combination is not limited thereto, provided that such combination of Fc regions is capable of association (formation of a heterodimer).

[0499] Examples of mutant Fc include, but are not limited to, a modified Fc region included in a heteropolymer with improved stability (including a heterodimer Fc region) disclosed in WO 2013/063702, Fc including an immunoglobulin CD3 region induced from the IgG antibody with a “protrusion” and a “gap” included in a heteropolymer disclosed in WO 1996/27011, Fc including a CH3 domain included in a heterodimer that becomes electrostatically advantageous via substitution of one or more amino acids with charged amino acids disclosed in WO 2009/089004, a heterodimer Fc region included in a heterodimer involving steric mutation and/or pI (isoelectric point) mutation disclosed in WO 2014/110601, and a heterodimer Fc including a CH3 domain with a modification to eliminate or reduce the binding to protein A disclosed in WO 2010/151792.

[0500] A variable region is composed of a region with an extreme variability referred to as a hypervariable region (HVR) and relatively invariable regions referred to as framework regions (FRs) divided by the HVR. Naturally occurring heavy chain and light chain variable regions comprise 4 FRs connected by 3 hypervariable regions, a hypervariable region of each chain and a hypervariable region of other chains being maintained very close thereto, and such regions contribute to formation of an antigen-binding site of an antibody.

[0501] A heavy chain and a light chain of an antibody molecule are known to comprise 3 complementarity determining regions (CDRs). A complementarity determining region is also referred to as a hypervariable region, it is present within variable regions of a heavy chain and a light chain of the antibody where variability of a primary structure is particularly high, and, in general, it is separated in 3 positions in the primary structure of a polypeptide chain of a heavy chain and a light chain. In the present invention, complementarity determining regions of a heavy chain of an antibody are denoted as CDRH1, CDRH2, and CDRH3 from the amino terminus of the heavy chain amino acid sequence, and complementarity determining regions of a light chain are denoted as CDRL1, CDRL2, and CDRL3 from the amino terminus of the light chain amino acid sequence. These regions are adjacent to each other on a steric structure and determine specificity to the antigens to which they bind.

[0502] In the present invention, the position and the length of CDR were determined in accordance with the definition of IMGT (Developmental and Comparative Immunology 27, 2003, 55-77).

[0503] FR is a variable region other than CDR. In general, a variable region comprises 4 FRs; i.e., FR1, FR2, FR3, and FR4.

[0504] CDRs and FRs included in the heavy chain and in the light chain are provided in the orders of FRH1-CDRH1-FRH2-CDRH2-FRH3-CDRH3-FRH4 and FRL1-CDRL1-FRL2-CDRL2-FRL3-CDRL3-FRL4, respectively, from the amino terminus toward the carboxyl terminus.

[0505] CDR and FR positions can be determined in accordance with various definitions well known in the art, such as the definitions of Kabat, Chothia, AbM, contact, in addition to IMGT.

[0506] In the present invention, the term “an antigen-binding fragment of the antibody” refers to a partial fragment of an antibody having the activity of binding to an antigen, which is composed of a heavy chain variable region and a light chain variable region. Examples of “an antigen-binding fragment of the antibody” include, but are not limited to, antigen-binding fragments, such as Fab, F(ab').sub.2, scFv, Fab', Fv, and single-domain antibody (sdAb). Such antigen-binding fragment of the antibody may be obtained by treating a full-length molecule of an antibody protein with an enzyme such as papain or pepsin or it may be a recombinant protein produced in an adequate host cell with the use of a recombinant gene. In the present invention, the term “a binding fragment of the antibody” is synonymous with the term “an antigen-binding fragment of the antibody.”

[0507] In the present invention, a “site” to which an antibody binds; i.e., a “site” that is recognized by an antibody, is a partial peptide or a partial higher-order structure of an antigen to which an antibody binds or which is recognized by the antibody.

[0508] In the present invention, such a site is referred to as an epitope or an antibody binding site. In the present invention, a “mutant antibody” refers to a polypeptide having an amino acid sequence derived from the amino acid sequence of the original antibody by substitution, deletion, or addition (“addition” encompasses “insertion”) (hereafter, collectively referred to as “mutation”) of amino acids and binding to HLA/NY-ESO of the present invention. The number of mutant amino acids in such a mutant antibody is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20, 25, 30, 40, or 50. Such a mutant antibody is within the scope of the “antibody” of the present invention.

[0509] In the present invention, the term “several” in “one or several” indicates 2 to 10.

[0510] The term “molecule” used herein indicates a molecule comprising the antibody or the antigen-binding fragment of the antibody described above. In addition, the term “molecule” encompasses a multispecific molecule formed of an antibody or a plurality of antigen-binding fragments derived therefrom.

[0511] The term “multispecific molecule” used herein is not particularly limited, provided that such a molecule is capable of binding to a plurality of different epitopes on a molecule and/or different epitopes on two or more molecules. A multispecific molecule encompasses an antibody comprising a heavy chain variable region (VH) and a light chain variable region (VL). Examples of such a multispecific molecule include, but are not limited to, a full-length antibody molecule comprising two or more different types of heavy chains and light chains; i.e., an IgG-type multispecific molecule, and a molecule comprising an antigen-binding fragment consisting of two or more types of VLs and VHs; i.e., a molecule derived from Fab, Fab', Fv, scFv, sdAb, or the like in combination, such as tandem scFv, a diabody, a single chain diabody, or a triabody. In addition, a molecule generated by genetically or chemically ligating a protein without an immunoglobulin skeleton and capable of binding to an antigen to an antigen-binding fragment is within the scope of the multispecific molecule. In the present invention, such a multispecific molecule may be referred to as a “multispecific antibody” except for the case in which the molecule does not comprise an antibody or an antigen-binding fragment thereof, according to need.

[0512] The anti-HLA/NY-ESO antibody according to the present invention, an antigen-binding fragment of such an antibody, or the molecule according to the present invention have, for example, biological activity and physicochemical properties (which may also be referred to as physical properties). Specific examples thereof include various biological activity, such as cytotoxicity, ADCC activity, or antitumor activity (described below), and physical properties, such as antigen- or epitope-binding activity, stability during production or storage, and heat stability.

[0513] In the present invention, “hybridization under stringent conditions” is performed in a solution containing 5×SSC at 65° C., in an aqueous solution containing 2×SSC and 0.1% SDS at 65° C. for 20 minutes, in an aqueous solution containing 0.5×SSC and 0.1% SDS at 65° C. for 20 minutes, or in an aqueous solution containing 0.2×SSC

and 0.1% SDS at 65° C. for 20 minutes under washing conditions or conditions equivalent thereto. SSC is an aqueous solution of 150 mM NaCl and 15 mM sodium citrate, and nx SSC is SSC of n-fold concentration.

[0514] In the present invention, “cell damage” is any form of a pathological change that has occurred in a cell. In addition to an external injury, the term “cell damage” refers to any damage caused on a cell structure or functions, such as DNA cleavage, formation of nucleotide dimers, chromosome cleavage, mitotic apparatus damage, or lowering in various enzyme activity. In the present invention, “cytotoxicity” is induction of the cell damage.

[0515] In the present invention, the term “antibody-dependent cellular cytotoxicity (ADCC)” refers to activity of NK cells to damage target cells such as tumor cells mediated by an antibody.

[0516] In the present invention, the term “T-cell-redirecting cytotoxicity” refers to induction of cytotoxicity mediated by a multispecific molecule comprising an anti-tumor antigen antibody and the anti-HLA/NY-ESO antibody.

Specifically, an anti-tumor antigen antibody binds to a target tumor cell, the anti-HLA/NY-ESO antibody binds to a T cell, the distance between the target tumor cell and the T cell is shortened, and cytotoxicity is induced by T cell activation. Such a molecule can be incorporated into a pharmaceutical composition.

2. Antigen

2-1. HLA/NY-ESO Antigen

[0517] In the present invention, “HLA/NY-ESO” is used in the same sense as “HLA/NY-ESO protein.”

[0518] HLA/NY-ESO is a triple complex of HLA-A2, β 2-microglobulin, and the NY-ESO peptide. HLA-A2 is an HLA allele, which is expressed at the highest frequency in Caucasians. HLA forms a triple complex with β 2-microglobulin and a peptide fragment of an autologous protein in the cell endoplasmic reticulum; the complex is presented extracellularly, and it is recognized by the T cell receptor (TCR) of the T cell. The NY-ESO peptide (SLLMWITQC; SEQ ID NO: 1, FIG. 8) is reported to be a peptide consisting of 9 amino acids at positions 157 to 165 of NY-ESO-1 and LAGE-1, which is presented to HLA-A2.

2-2. CD3 Antigen

[0519] In the present invention, “CD3” is used in the same sense as the “CD3 protein.”

[0520] CD3 is expressed on a T cell as a part of a multimolecular T cell receptor complex, and it is a complex of 5 types of polypeptide chains γ , δ , ϵ , ζ , and η (the molecular weights thereof are 25000-28000, 21000, 20000, 16000, and 22000, respectively).

[0521] Examples of CD3 complexes include γ , δ , ϵ , ζ , and η chains, which are also referred to as subunits. When the anti-CD3 antibody binds to a T cell, cell damage is induced by T cell activation. Many anti-CD3 antibodies bind to CD3 ϵ .

[0522] The nucleotide sequence of cDNA encoding human CD3 ϵ is registered under Accession Number: NM_000733 (NM_000733.3) at NCBI/GenBank, and the amino acid sequence of human CD3 ϵ is registered under Accession Number: NP_000724 (NP_000724.1) at NCBI/GenPept. The nucleotide sequence of cDNA encoding cynomolgus monkey CD3 is registered under Accession Number: NM_001283615.1 at GenBank. The amino acid sequence of human CD3 ϵ is shown in SEQ ID NO: 151 in the Sequence Listing (FIG. 147).

2-3. Preparation of Antigen

[0523] The antigen proteins used in the present invention described above; i.e., HLA/NY-ESO and CD3 (hereafter, HLA/NY-ESO and CD3 are collectively referred to as the antigen proteins) can be prepared from animal tissue (including body fluid), cells derived from the animal tissue, or a culture product of the cells via, for example, purification, isolation, gene recombination, in vitro translation, or chemical synthesis.

[0524] cDNA of the antigen protein can be obtained by a so-called PCR method in which a polymerase chain reaction (hereafter referred to as “PCR”) using, as a template, the cDNA library of organs expressing mRNA of the antigen protein and primers specifically amplifying cDNA of the antigen protein (Saiki, R. K., et al., Science, 1988, 239, 487-49).

[0525] The cDNA of the antigen protein encompasses a polynucleotide hybridizing, under stringent conditions, to a polynucleotide consisting of a nucleotide sequence complementary to the nucleotide sequence encoding the antigen protein expressed in a human or rat and encoding a protein having biological activity equivalent to that of the antigen protein.

[0526] In addition, the cDNA of the antigen protein encompasses a splicing variant transcribed from the antigen protein locus expressed in a human or rat and a polynucleotide hybridizing, under stringent conditions, thereto, and encoding a protein having biological activity equivalent to that of the antigen protein.

[0527] A nucleotide sequence encoding a protein comprising the amino acid sequence of the human or rat antigen protein or an amino acid sequence derived from such amino acid sequence by substitution, deletion, or addition of one to several amino acids from which the signal sequence has been deleted and having biological activity equivalent to that of the antigen protein is included in the nucleotide sequence of the antigen protein gene.

[0528] A protein comprising an amino acid sequence encoded by the splicing variant transcribed from the human or rat antigen protein gene locus or an amino acid sequence derived from such an amino acid sequence by substitution, deletion, or addition of one or several amino acids and having biological activity equivalent to that of the antigen protein is included in the antigen protein.

2-4. Binding Specificity to Antigen Protein

[0529] The anti-HLA/NY-ESO antibody according to the present invention, an antigen-binding fragment thereof, and the like recognize HLA/NY-ESO. Specifically, they bind to the HLA/NY-ESO antigen. The presence of HLA/NY-ESO is not known in non-human animals, such as mice, rats, and cynomolgus monkeys.

[0530] The anti-CD3 antibody included in the multispecific molecule of the present invention, the binding fragments thereof, and the like recognize; i.e., bind to the CD3 antigen. Such anti-CD3 antibody and the like preferably bind to, for example, human CD3 and monkey CD3 and more preferably to human CD3 and cynomolgus monkey CD3. In contrast, such a preferable anti-CD3 antibody does not bind to rat and/or mouse CD3.

[0531] The anti-tumor activity of the multispecific molecule of the present invention can be evaluated by, for example, (i) transplanting human cancer cells or human cancer tissue into nonhuman animals into which human peripheral blood lymphocytes have been transplanted, preferably into rats or mice, and more preferably to rats or mice with deficient endogenous effector functions (e.g., immunodeficient rats or mice) or (ii) transplanting mouse cancer cells into which HLA and NY-ESO genes have been transduced into human CD3 gene knock-in nonhuman animals and preferably into rats or mice. By performing evaluation with the use of such immunodeficient animals or knock-in animals, various assays, immunohistochemical assays, and the like can be performed using mouse and/or rat bodies. This is preferable for medicines containing the multispecific molecule of the present invention, nonclinical development, and other purposes.

[0532] In the present invention, “recognition,” i.e., “binding” is binding that is not non-specific adsorption. Whether or not the antibody recognizes; i.e., binds, can be evaluated on the basis of, for example, the dissociation constant (KD). A preferable KD value of the antibody and the like according to the present invention to HLA/NY-ESO or CD3 is 1×10^{-5} M or lower, 5×10^{-6} M or lower, 2×10^{-6} M or lower, or 1×10^{-6} M or lower, a KD value thereof to HLA/NY-ESO is preferably 5×10^{-7} M or lower, 2×10^{-7} M or lower, 1×10^{-7} M or lower, 5×10^{-8} M or lower, 2×10^{-8} M or lower, 1×10^{-8} M or lower, 5×10^{-9} M or lower, or 2×10^{-9} M or lower and more preferably 1×10^{-9} M or lower. Examples of the anti-HLA/NY-ESO scFv of the present invention having excellent antigen-binding activity include NYA-1143, NYA-2023, NYA-2143, NYA-2044, NYA-2045, NYA-2060, NYA-2061, and NYA-3061, and the KD value thereof to HLA/NY-ESO, such as NYA-1143, NYA-2044, NYA-2045, and NYA-2143, is 1×10^{-9} M or lower (e.g., Example 4).

[0533] In the present invention, antigen-antibody binding can be assayed or evaluated by, for example, the system of biomolecular interaction analysis, such as SPR or BLI, ELISA, or RIA. Binding between an antigen and an antibody expressed on a cell surface can be assayed by, for example, flow cytometry.

[0534] The method of surface plasmon resonance (SPR) analysis performs reaction kinetic analysis to determine the association rate constant (Ka value) and the dissociation rate constant (Kd value) and determine the dissociation constant (KD value) serving as an affinity index. Examples of apparatuses used for SPR analysis include Biacore™ (GE Healthcare), ProteOn™ (BioRad), SPR-Navi™ (BioNavis), Spreeta™ (Texas Instruments), SPRi-PlexII™ (Horiba Ltd.), and Autolab SPR™ (Metrohm).

[0535] The method of BioLayer Interferometry (BLI) performs assaying the interaction between biomolecules using biolayer interference. An example of an apparatus used for interaction analysis by the BLI method is the Octet system (Pall ForteBio).

[0536] The method of enzyme-linked immunosorbent assay (ELISA) performs detection and quantification of the target antigen or antibody contained in a sample solution by capturing the target antigen or antibody using a specific antibody or antigen, respectively, with the utilization of an enzyme reaction. The enzyme-labeled antigen or antibody is integrated into the reaction system to detect enzymatic activity. Enzymatic activity is detected using a substrate exhibiting the absorption spectra varying depending on reactions and the enzymatic activity is quantified based on absorption assay.

[0537] Cell-ELISA performs detection and quantification of the target analyte on a cell surface by capturing the target analyte together with the cell with the utilization of an enzyme reaction.

[0538] The method of radio immunoassay (RIA) performs labeling an antibody with a radioactive substance and assaying the radioactivity of the antibody. Thus, the antibody can be quantified.

[0539] In the method of flow cytometry, cells are dispersed in fluid, the fluid is allowed to flow narrowly, and each cell is optically analyzed. The antibody labeled with a fluorescent dye is allowed to bind to a cell surface antigen by the antigen-antibody reaction, and the intensity of fluorescence emitted by the labeled antibody bound to the cell is assayed to quantify antigen-binding capability of the antibody.

[0540] Examples of the anti-HLA/NY-ESO antibodies according to the present invention exhibiting excellent antigen-binding specificity include anti-HLA/NY-ESO scFv, such as NYA-0001, NYA-1143, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, NYA-2061, NYA-2143, and NYA-3061 (e.g., Example 6).

3. Antibody Binding Specifically to HLA/NY-ESO or the Binding Fragment Thereof

3-1. Anti-HLA/NY-ESO or the Binding Fragment Thereof

[0541] The present invention provides an antibody that recognizes and binds to HLA/NY-ESO or a binding fragment

thereof.

[0542] As described above, HLA/NY-ESO is a complex comprising HLA-A2 and a 9-mer NY-ESO peptide (SLLMWITQC; SEQ ID NO: 1). The NY-ESO peptide is a peptide derived from NY-ESO-1 or LAGE-1, which is an intracellular protein and a cancer testis antigen. HLA/NY-ESO is expressed on a cancer cell surface.

[0543] The anti-HLA/NY-ESO antibody according to the present invention and an antigen-binding fragment of the antibody (which hereafter, may be referred to as “the antibody and the like of the present invention”) may be monoclonal or polyclonal antibodies. In the present invention, an isotype of a monoclonal antibody isotype is not particularly limited, and examples include IgG such as IgG1, IgG2, IgG3 and IgG4, IgM, IgA such as IgA1 and IgA2, IgD, and Ig. An isotype and a subclass of a monoclonal antibody can be determined by, for example, the Ouchterlony method, ELISA, or RIA. Examples of the monoclonal antibodies of the present invention include an antibody derived from a non-human animal (a non-human animal antibody), a human antibody, a chimerized antibody (also referred to as a “chimeric antibody”), and a humanized antibody, with the human antibody being preferable. The antibody of the present invention encompasses a mutant of an antibody (the “mutant antibody” described below), and, for example, the human antibody encompasses a human mutant antibody.

[0544] Examples of non-human animal antibodies include antibodies derived from vertebrates, such as mammals and birds. Examples of mammalian-derived antibodies include antibodies derived from rodents, such as mouse antibody and rat antibody. An example of a bird-derived antibody is a chicken antibody.

[0545] Examples of chimerized antibodies include, but are not limited to, antibodies comprising a variable region derived from a non-human animal antibody bound to a constant region derived from a human antibody (human immunoglobulin).

[0546] Examples of humanized antibodies include, but are not limited to, a humanized antibody prepared by transplanting CDR in a variable region of a non-human animal antibody into a human antibody (a variable region of human immunoglobulin), a humanized antibody prepared by transplanting, in addition to CDR, a part of a sequence of a framework region of a non-human animal antibody into a human antibody, and a humanized antibody prepared by substitution of 1 or more amino acids derived from a non-human animal antibody with amino acids derived from a human antibody.

[0547] An antibody can be prepared by a variety of known techniques. For example, an antibody can be prepared by a method involving the use of a hybridoma, cell-mediated immunity, or genetic recombination. Also, a phage-display-derived human antibody selected from a human antibody library can be obtained. In a phage display method, for example, a human antibody variable region may be expressed as scFv on a phage surface, and an antigen-binding phage may then be selected. The gene of the phage selected upon its binding to the antigen may be analyzed, so that a DNA sequence encoding a human antigen variable region binding to the antigen can be determined. If a DNA sequence of the antigen-binding scFv is elucidated, an expression vector comprising such sequence may be prepared, introduced into an adequate host cell, and expressed therein. Thus, a human antibody can be obtained (WO 92/01047, WO 92/20791, WO 93/06213, WO 93/11236, WO 93/19172, WO 95/01438, WO 95/15388, Annu. Rev. Immunol., 1994, 12, 433-455).

[0548] The thus-obtained antibody with high activity may be used as a lead antibody, and a gene encoding such a lead antibody may be mutated, so that a mutant with higher activity can be prepared (“the mutant antibody” described below).

[0549] A preferable combination of CDRH1 to CDRH3 included in the heavy chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof is CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54 (FIG. 61), CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55 (FIG. 61), and CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 56 (FIG. 61) in which amino acid 6 is N (Asn). A more preferable combination of CDRH1 to CDRH3 is included in the NYA-0001 heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 6 (FIG. 13), the NYA-0082 heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 18 (FIG. 25), the NYA-2023 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34), the NYA-2027 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35), the NYA-1143 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 29 (FIG. 36), the NYA-1163 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 26 (FIG. 33), the NYA-2023 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34), the NYA-2027 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35), the NYA-2035 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 36 (FIG. 43), the NYA-2044 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 47 (FIG. 54), the

NYA-2045 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 48 (FIG. 55), the NYA-2047 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 50 (FIG. 57), the NYA-2048 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 51 (FIG. 58), the NYA-2060 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 52 (FIG. 59), the NYA-2061 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 53 (FIG. 60), or the NYA-2143 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 30 (FIG. 37). [0550] A further example of a combination of CDRH1 to CDRH3 is included in the NYA-3061 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152).

[0551] A preferable combination of CDRL1 to CDRL3 included in the light chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof is CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 (FIG. 61) in which amino acid 7 is W (Trp) or amino acid 8 is K (Lys), CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58 (FIG. 61), and CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 (FIG. 61) in which amino acid 2 is A (Ala) or S (Ser).

[0552] A more preferable combination of CDRL1 to CDRL3 is included in the NYA-0001 light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 8 (FIG. 15), the NYA-0082 light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 20 (FIG. 27), the NYA-1143 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 29 (FIG. 36), the NYA-1163 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 26 (FIG. 33), the NYA-2023 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34), the NYA-2027 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35), the NYA-2035 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 36 (FIG. 43), the NYA-2044 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 47 (FIG. 54), the NYA-2045 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 48 (FIG. 55), the NYA-2047 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 50 (FIG. 57), the NYA-2048 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 51 (FIG. 58), the NYA-2060 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 52 (FIG. 59), the NYA-2061 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53 (FIG. 60), or the NYA-2143 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 30 (FIG. 37).

[0553] A further example of a combination of CDRL1 to CDRL3 is included in the NYA-3061 light chain variable region consisting of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152).

[0554] A preferable combination of CDRH1 to CDRH3 in the heavy chain and CDRL1 to CDRL3 in the light chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof is a combination of CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54 (FIG. 61), CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55 (FIG. 61), and CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 56 (FIG. 61) in which amino acid 6 is N (Asn) and a combination of CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 (FIG. 61) in which amino acid 7 is N (Asn) and/or amino acid 8 is K (Lys), CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58 (FIG. 61), and CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 in which amino acid 2 is A (Ala) or S (Ser). A more preferable combination of CDRH1 to CDRH3 and CDRL1 to CDRL3 is included in the NYA-0001 heavy chain variable region and light chain variable region consisting of the amino acid sequences as shown in SEQ ID NO: 6 and SEQ ID NO: 8, the NYA-1143 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 29 (FIG. 36), the NYA-1163 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 26 (FIG. 33), the NYA-2023 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34), the NYA-2027 heavy chain variable region and

light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35), the NYA-2035 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 36 (FIG. 43), the NYA-2044 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 47 (FIG. 54), the NYA-2045 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 48 (FIG. 55), the NYA-2047 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 50 (FIG. 57), the NYA-2048 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 51 (FIG. 58), the NYA-2060 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 52 (FIG. 59), the NYA-2061 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53 (FIG. 60), or the NYA-2143 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 30 (FIG. 37).

[0555] A further example of a combination of CDRH1 to CDRH3 and CDRL1 to CDRL3 is included in the NYA-3061 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152).

[0556] Preferable examples of a heavy chain variable region of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include the heavy chain CDRs described above and variable regions including such heavy chain CDRs in adequate combination. More preferable examples include the NYA-0001 heavy chain variable region, the NYA-0082 heavy chain variable region, the NYA-1143 heavy chain variable region, the NYA-1163 heavy chain variable region, the NYA-2023 heavy chain variable region, the NYA-2027 heavy chain variable region, the NYA-2035 heavy chain variable region, the NYA-2044 heavy chain variable region, the NYA-2045 heavy chain variable region, the NYA-2047 heavy chain variable region, the NYA-2048 heavy chain variable region, the NYA-2060 heavy chain variable region, the NYA-2061 heavy chain variable region, the NYA-2143 heavy chain variable region, and the NYA-3061 heavy chain variable region. The amino acid sequence of each heavy chain variable region is as described above.

[0557] Preferable examples of a light chain variable region of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include the light chain CDRs described above and variable regions including such light chain CDRs in adequate combination. More preferable examples include the NYA-0001 light chain variable region, the NYA-0082 light chain variable region, the NYA-1143 light chain variable region, the NYA-1163 light chain variable region, the NYA-2023 light chain variable region, the NYA-2027 light chain variable region, the NYA-2035 light chain variable region, the NYA-2044 light chain variable region, the NYA-2045 light chain variable region, the NYA-2047 light chain variable region, the NYA-2048 light chain variable region, the NYA-2060 light chain variable region, the NYA-2061 light chain variable region, the NYA-2143 light chain variable region, and the NYA-3061 light chain variable region. The amino acid sequence of each light chain variable region is as described above.

[0558] Preferable examples of a heavy chain variable region and a light chain variable region of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include the heavy chain and light chain CDRs described above or those including such CDRs in adequate combination. More preferable examples include the NYA-0001 heavy chain variable region and light chain variable region, the NYA-0082 heavy chain variable region and light chain variable region, the NYA-1143 heavy chain variable region and light chain variable region, the NYA-1163 heavy chain variable region and light chain variable region, the NYA-2023 heavy chain variable region and light chain variable region, the NYA-2027 heavy chain variable region and light chain variable region, the NYA-2035 heavy chain variable region and light chain variable region, the NYA-2044 heavy chain variable region and light chain variable region, the NYA-2045 heavy chain variable region and light chain variable region, the NYA-2047 heavy chain variable region and light chain variable region, the NYA-2048 heavy chain variable region and light chain variable region, the NYA-2060 heavy chain variable region and light chain variable region, the NYA-2061 heavy chain variable region and light chain variable region, the NYA-2143 heavy chain variable region and light chain variable region, and the NYA-3061 heavy chain variable region and light chain variable region.

[0559] Preferable examples of a heavy chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include heavy chains including the preferable or more preferable heavy chain variable regions described above. More preferable examples include the NYA-0001 heavy chain, the NYA-0082 heavy chain, the NYA-1143 heavy chain, the NYA-1163 heavy chain, the NYA-2023 heavy chain, the NYA-2027 heavy chain, the NYA-2035 heavy chain, the NYA-2044 heavy chain, the NYA-2045 heavy chain, the NYA-2047 heavy chain, the NYA-2048 heavy chain, the NYA-2060 heavy chain, the NYA-2061 heavy chain, the NYA-2143

heavy chain, and the NYA-3061 heavy chain.

[0560] Preferable examples of a light chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include light chains including the preferable or more preferable light chain variable regions described above. More preferable examples include the NYA-0001 light chain, the NYA-0082 light chain, the NYA-1143 light chain, the NYA-1163 light chain, the NYA-2023 light chain, the NYA-2027 light chain, the NYA-2035 light chain, the NYA-2044 light chain, the NYA-2045 light chain, the NYA-2047 light chain, the NYA-2048 light chain, the NYA-2060 light chain, the NYA-2061 light chain, the NYA-2143 light chain, and the NYA-3061 light chain.

[0561] Preferable examples of a heavy chain and a light chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include heavy chains and light chains including the preferable or more preferable heavy chain variable regions and light chain variable regions described above. More preferable examples include the NYA-0001 heavy chain and light chain, the NYA-1143 heavy chain and light chain, the NYA-1163 heavy chain and light chain, the NYA-2023 heavy chain and light chain, the NYA-2027 heavy chain and light chain, the NYA-2035 heavy chain and light chain, the NYA-2044 heavy chain and light chain, the NYA-2045 heavy chain and light chain, the NYA-2047 heavy chain and light chain, the NYA-2048 heavy chain and light chain, the NYA-2060 heavy chain and light chain, the NYA-2061 heavy chain and light chain, the NYA-2143 heavy chain and light chain, and the NYA-3061 heavy chain and light chain.

[0562] An antigen-binding fragment of the antibody is a fragment that retains at least the antigen-binding ability out of the functions of the original antibody or a modified product thereof. Examples of the functions of the antibody include, in general, antigen-binding activity, the activity of regulating antigen activity, antibody-dependent cellular cytotoxicity, and complement-dependent cytotoxicity. Examples of functions of the antibody and the like of the present invention and multispecific molecules comprising the antibody and the like of the present invention include T cell redirection, T cell activation, and cancer cytotoxicity caused by T cell activation.

[0563] An antigen-binding fragment of the antibody is not particularly limited, provided that such a fragment is derived from the antibody that retains at least antigen-binding ability out of the functions of the original antibody. Examples thereof include, but are not limited to, Fab, Fab', F(ab').sub.2, Fv, single chain Fv (scFv) comprising heavy chain Fv ligated to light chain Fv via an adequate linker, and the single domain antibody (sdAb). A molecule comprising a region other than the antigen-binding fragment of the antibody according to the present invention as in the case of scFv comprising a linker portion is within the scope of the antigen-binding fragment of the antibody of the present invention.

[0564] A molecule of the antibody protein lacking at least one or several amino acids from the amino terminus and/or carboxyl terminus and retaining some functions of the antibody is within the scope of the antigen-binding fragment of the antibody. A modified antigen-binding fragment of the antibody is within the scope of the antibody according to the present invention, an antigen-binding fragment thereof, or a modified antibody or fragment (described below).

[0565] An embodiment of the antibody of the present invention or an antigen-binding fragment thereof is scFv. scFv is obtained by ligating the heavy chain variable region of the antibody to the light chain variable region thereof with a polypeptide linker (Pluckthun A., *The Pharmacology of Monoclonal Antibodies* 113, Rosenberg and Moore (ed.), Springer Verlag, New York, 269-315, 1994, *Nature Biotechnology*, 2005, 23, 1126-1136). Also, tandem scFv prepared by connecting two scFv constructs with a polypeptide linker can be used as a bispecific molecule. In addition, a triabody or the like comprising 3 or more scFv constructs can be used as a multispecific molecule.

[0566] Preferable examples of HLA/NY-ESO-specific scFv (also referred to as "anti-HLA/NY-ESO scFv") include scFv comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 described above, more preferable examples thereof include scFv comprising the heavy chain variable region and the light chain variable region, and further preferable examples thereof include NYA-0001 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 70 (FIG. 71)), NYA-0082 (comprising the amino acid sequence as shown in SEQ ID NO: 18 (FIG. 25) and the amino acid sequence as shown in SEQ ID NO: 20 (FIG. 27)), NYA-1143 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 29 (FIG. 36)), NYA-1163 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 26 (FIG. 33)), NYA-2023 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34)), NYA-2027 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35)), NYA-2035 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 36 (FIG. 43)), NYA-2044 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 47 (FIG. 54)), NYA-2045 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 48 (FIG. 55)), NYA-2047 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 50 (FIG. 57)), NYA-2048 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 51 (FIG. 58)), NYA-2060 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 52 (FIG. 59)), NYA-2061 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 53 (FIG. 60)), and NYA-2143 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 30 (FIG. 37)). Another example is NYA-3061 (amino acids 21 to 271 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152)).

[0567] A preferable embodiment of anti-HLA/NY-ESO scFv is scFv comprising a FLAG-His tag fused to its

carboxyl terminus (may be simply referred to as "tag adduct"). Examples of preferable tag adducts include NYA-0001 tag adduct (amino acids 20 to 292 of SEQ ID NO: 70 (FIG. 71)), NYA-1143 tag adduct (amino acids 20 to 292 of SEQ ID NO: 29 (FIG. 36)), NYA-1163 tag adduct (amino acids 20 to 292 of SEQ ID NO: 26 (FIG. 33)), NYA-2023 tag adduct (amino acids 20 to 292 of SEQ ID NO: 27 (FIG. 34)), NYA-2027 tag adduct (amino acids 20 to 292 of SEQ ID NO: 28 (FIG. 35)), NYA-2035 tag adduct (amino acids 20 to 292 of SEQ ID NO: 36 (FIG. 43)), NYA-2044 tag adduct (amino acids 20 to 292 of SEQ ID NO: 47 (FIG. 54)), NYA-2045 tag adduct (amino acids 20 to 292 of SEQ ID NO: 48 (FIG. 55)), NYA-2047 tag adduct (amino acids 20 to 292 of SEQ ID NO: 50 (FIG. 57)), NYA-2048 tag adduct (amino acids 20 to 292 of SEQ ID NO: 51 (FIG. 58)), NYA-2060 tag adduct (amino acids 20 to 292 of SEQ ID NO: 52 (FIG. 59)), NYA-2061 tag adduct (amino acids 20 to 292 of SEQ ID NO: 53 (FIG. 60)), and NYA-2143 tag adduct (amino acids 20 to 292 of SEQ ID NO: 30 (FIG. 37)).

[0568] Of these, NYA-2023 and its tag adduct, NYA-2047 and its tag adduct, NYA-2048 and its tag adduct, NYA-2060 and its tag adduct, and NYA-2061 and its tag adduct exert excellent biological activity, physical properties, and the like in Fc-added bispecific molecules and thus are more preferable.

[0569] When anti-HLA/NY-ESO scFv and its tag adduct are to be expressed in a host cell, a signal peptide can be added to its amino terminus. Examples of amino acid sequences of signal-peptide-added anti-HLA/NY-ESO scFv tag adducts include amino acid sequences as shown in SEQ ID NOs: 70, 29, 26 to 28, 36, 47, 48, 50 to 53, and 30 (FIGS. 71, 36, 33 to 35, 43, 54, 55, 57 to 60, and 37).

[0570] scFv can be obtained by the phage display method in which an antibody variable region is expressed as a single chain antibody (scFv) on a phage surface and an antigen-binding phage is then selected (Nature Biotechnology, 2005, 23, (9), pp. 1105-1116). The gene of the phage selected upon its binding to the antigen may be analyzed, so that a DNA sequence encoding a human antigen variable region binding to the antigen can be determined. If a DNA sequence of the antigen-binding scFv is elucidated, an expression vector comprising such sequence may be prepared, introduced into an adequate host cell, and expressed therein. Thus, a human antibody can be obtained (WO 92/01047, WO 92/20791, WO 93/06213, WO 93/11236, WO 93/19172, WO 95/01438, WO 95/15388, Annu. Rev. Immunol., 1994, 12, pp. 433-455, Nature Biotechnology, 2005, 23 (9), pp. 1105-1116).

[0571] The antibody of the present invention may comprise a single heavy chain variable region and may not comprise a light chain sequence. Such an antibody is referred to as a single domain antibody (sdAb) or nanobody and retains antigen-binding ability (Muyldemans S. et al., Protein Eng., 1994, 7 (9), 1129-35, Hamers-Casterman C. et al., Nature, 1993, 363 (6428), 446-448). Such antibodies are within the scope of the antigen-binding fragment of the antibody according to the present invention.

[0572] The present invention also includes a single chain immunoglobulin comprising the full-length sequence of the heavy chain ligated to that of the light chain of the antibody with an adequate linker (Lee, H-S, et al., Molecular Immunology, 1999, 36, 61-71; Shirmann, T. et al., mAbs, 2010, 2 (1), 1-4). Such single chain immunoglobulin may be dimerized, so that it can retain the structure and activity similar to those of an antibody that is inherently a tetramer. The anti-HLA/NY-ESO antibody according to the present invention may be single chain immunoglobulin.

[0573] In scFv of the present invention, a heavy chain variable region may form a disulfide bond with a light chain variable region.

[0574] The anti-HLA/NY-ESO antibody according to the present invention may be composed of components derived from a plurality of different antibodies, provided that it binds to HLA/NY-ESO. Examples thereof include those resulting from replacement of heavy chains and/or light chains among a plurality of different antibodies, those resulting from replacement of full-length sequences of heavy chains and/or light chains, those resulting from selective replacement of either a variable region or a constant region, and those resulting from selective replacement of a part of or the entire CDR. In a chimerized antibody, a heavy chain variable region and a light chain variable region may be derived from a different anti-HLA/NY-ESO antibody according to the present invention. In the heavy chain and light chain variable regions of a humanized antibody, heavy chain CDR1 to heavy chain CDR3 and light chain CDR1 to light chain CDR3 may be derived from two or more types of the anti-HLA/NY-ESO antibodies according to the present invention. In the heavy chain and light chain variable regions of a human antibody, a combination of heavy chain CDR1 to heavy chain CDR3 and light chain CDR1 to light chain CDR3 may be derived from two or more types of the anti-HLA/NY-ESO antibodies according to the present invention.

[0575] The anti-HLA/NY-ESO antibody according to the present invention encompasses an antibody that comprises an amino acid sequence encoded by a nucleotide sequence included in a polynucleotide hybridizing, under stringent conditions, to a complementary strand of a polynucleotide comprising a nucleotide sequence encoding the amino acid sequence included in the anti-HLA/NY-ESO antibody according to the present invention and binding to HLA/NY-ESO.

[0576] It may be an antibody comprising an amino acid sequence exhibiting at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to the amino acid sequence included in the heavy chain variable region of the anti-HLA/NY-ESO antibody according to the present invention (preferably, an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27, the amino acid sequence as shown in SEQ ID NO: 38, the amino acid sequence as shown in SEQ ID NO: 39, or an amino acid sequence of amino acids 21 to 140

of the amino acid sequence as shown in SEQ ID NO: 160, SEQ ID NO: 197, or SEQ ID NO: 198) and/or an amino acid sequence included in a light chain variable region thereof (preferably, an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27, an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 52, the amino acid sequence as shown in SEQ ID NO: 40, an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 160, or an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 197 or SEQ ID NO: 198) or an antigen-binding fragment thereof.

[0577] When the position and the length of the light chain variable region are determined in accordance with a definition that is different from IMGT (e.g., Kabat, Chothia, AbM, contact), one or more amino acids, such as arginine or glycine, may further be included into the carboxyl terminus of the amino acid sequence of the light chain variable region defined in accordance with the definition of IMGT. Such antibody or the binding fragment thereof comprising such light chain variable region is within the scope of the antibody according to the present invention or the binding fragment thereof.

[0578] The antibody and the like of the present invention may be prepared by introducing a mutation into a binding fragment of the anti-HLA/NY-ESO antibody according to the present invention and optimizing the ability to bind to HLA/NY-ESO, and, in particular, to human and/or cynomolgus monkey HLA/NY-ESO. Specific examples of methods for introduction of a mutation include a random mutagenesis using error-prone PCR, site-directed amino acid introduction using NHK libraries, site-directed mutagenesis using structural information, and a combination of any thereof.

3-2. Mutant of the Anti-HLA/NY-ESO Antibody (Mutant Antibody)

[0579] A mutant antibody of the anti-HLA/NY-ESO antibody according to the present invention can be preferably provided with, for example, lowered susceptibility to protein degradation or oxidation, maintained or improved biological activity or functions, suppression of lowering or change in such activity or functions, improved or regulated antigen-binding ability, physicochemical properties, or functional properties. A protein is known to change its functions or activity upon alternation of a particular amino acid side chain on its surface, and examples include deamidation of an asparagine side chain and isomerization of an aspartic acid side chain. An antibody resulting from substitution of a particular amino acid with another amino acid so as to prevent the amino acid side chain from changing is within the scope of the mutant antibody of the present invention.

[0580] An example of the mutant antibody of the present invention is an antibody comprising an amino acid sequence derived from the amino acid sequence of the original antibody by conservative amino acid substitution. Conservative amino acid substitution occurs within an amino acid group associated with the amino acid side chain.

[0581] Preferable amino acid groups are as follows: the acidic group: aspartic acid and glutamic acid; the basic group: lysine, arginine, and histidine; the non-polar group: alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, and tryptophan; and the uncharged polar family: glycine, asparagine, glutamine, cysteine, serine, threonine, and tyrosine. Other preferable amino acid groups are as follows: the aliphatic hydroxy group: serine and threonine; the amide-containing group: asparagine and glutamine; the aliphatic group: alanine, valine, leucine, and isoleucine; and the aromatic group: phenylalanine, tryptophan, and tyrosine. In such mutant antibody, amino acid substitution is preferably carried out by refraining from lowering the antigen-binding activity of the original antibody.

[0582] The anti-HLA/NY-ESO antibody according to the present invention, an antigen-binding fragment thereof, a mutant thereof (a mutant antibody or an antigen-binding fragment thereof), or the molecule of the present invention encompasses: a mutant antibody comprising an amino acid sequence derived from the amino acid sequence of the antibody of the present invention, such as NYA-2023, by conservative amino acid substitution and/or other mutation and binding to HLA/NY-ESO, an antigen-binding fragment thereof, and molecules comprising the same; a chimerized antibody, a humanized antibody, or a human antibody comprising CDR of an amino acid sequence derived from the amino acid sequence of any of CDRH1 to CDRH3 and CDRL1 to CDRL3 derived from the antibody of the present invention, including NYA-2023, by conservative amino acid substitution and/or other mutation and binding to HLA/NY-ESO, an antigen-binding fragment thereof, or a molecule comprising the same.

3-3. Binding Fragment of the Anti-HLA/NY-ESO Antibody

[0583] An aspect of the present invention provides an antigen-binding fragment of the anti-HLA/NY-ESO antibody according to the present invention (hereafter, it is simply referred to as a "binding fragment"). Binding fragments of the anti-HLA/NY-ESO antibody according to the present invention include binding fragments of a chimerized antibody, a humanized antibody, and a human antibody. A binding fragment of an antibody is a fragment that retains at least antigen-binding ability of functions of the original antibody or a modified product thereof. In general, examples of such antibody functions include antigen-binding activity, the activity of regulating antigen activity (e.g., agonist activity), the activity of internalizing an antigen in a cell, and the activity of inhibiting or promoting interactions between an antigen and a substance interacting therewith.

[0584] A binding fragment of an antibody is not particularly limited, provided that it is a fragment of a fragment retaining at least antigen-binding ability of the activity of the original antibody. Examples of such binding fragment

of an antibody include, but are not limited to, Fab, Fab', F(ab').sub.2, single chain Fab (scFab) in which the carboxyl terminus of the Fab light chain is ligated to the amino terminus of the Fab heavy chain via an adequate linker, Fv, single chain Fv (scFv) comprising heavy chain Fv ligated to light chain Fv via an adequate linker, and a single domain antibody (sdAb) with a single heavy chain variable region but without a light chain sequence, which is also referred to as a nanobody (Muyldemans S. et al., Protein Eng., 1994, 7 (9), 1129-35, Hamers-Casterman C. et al., Nature, 1993, 363 (6428), 446-448). A molecule comprising regions other than the binding fragment of the antibody of the present invention, such as scFab or scFv comprising a linker region, is within the scope of the binding fragment of the antibody of the present invention.

3-4. Modified Anti-HLA/NY-ESO Antibody, Modified Binding Fragment Thereof, or Complex Thereof

[0585] The present invention provides a modified antibody or a modified binding fragment thereof. The modified antibody according to the present invention or the modified binding fragment thereof has been subjected to chemical or biological modification. Examples of chemical modification include a bond of a chemical portion to the amino acid skeleton and chemical modification of N-bound or O-bound carbohydrate chains. Examples of biological modification include post-translational modification (e.g., sugar chain addition to an N-bond or O-bond, processing of the amino terminal or carboxyl terminal region, deamidation, aspartic acid isomerization, and methionine oxidation), and methionine addition to the amino terminus via expression in a prokaryotic host cell. Also, labels that enable detection or isolation of the antibody or antigen according to the present invention, such as an enzyme label, a fluorescence label, and an affinity label, are within the scope of the modified antibody or antigen as described above. The modified antibody according to the present invention or the binding fragment thereof as described above is useful for improvement of stability and retentivity in blood of the original antibody according to the present invention or the binding fragment thereof, reduction of the antigenicity, detection or isolation of the antibody or antigen, and other purposes.

[0586] Examples of chemical portions included in the chemical modified antibody or fragment include water-soluble polymers, such as polyethylene glycol (PEG), ethylene glycol/propylene glycol polymer, carboxymethyl cellulose, dextran, and polyvinyl alcohol.

[0587] Examples of biological modification include enzyme treatment, cell treatment, addition of other peptides such as tags via genetic recombination, and use of host cells expressing endogenous or exogenous sugar chain modified enzymes.

[0588] Such modification may be provided at any desired position in the antibody or the binding fragment thereof, and the same or two or more different types of modification may be provided at one or more positions.

[0589] However, deletion of such a heavy chain sequence or modification of a heavy chain or light chain sequence would not significantly affect the antigen-binding ability and effector functions of the antibody (e.g., complement activation or antibody-dependent cytotoxicity).

[0590] Accordingly, the present invention encompasses the antibody subjected to such deletion or modification. Examples include a deletion mutant lacking 1 or 2 amino acids from the heavy chain carboxyl terminus (Journal of Chromatography A; 705; 129-134, 1995), a deletion mutant lacking 2 amino acids (glycine and lysine) from the heavy chain carboxyl terminus and subjected to amidation of proline at the carboxyl terminus (Analytical Biochemistry, 360: 75-83, 2007), and an antibody resulting from pyroglutamylation of an amino-terminal glutamine or glutamic acid of the heavy chain or the light chain (WO 2013/147153) (they are collectively referred to as "deletion mutants"). As long as the antigen-binding ability and effector functions are retained, the antibody of the present invention lacking the heavy chain terminus and the light chain carboxyl terminus is not limited to the deletion mutants described above. When the antibody of the present invention comprises 2 or more chains (e.g., heavy chains), either of or both of such 2 or more chains (e.g., heavy chains) may be the full-length heavy chain or a heavy chain selected from the group consisting of the deletion mutants described above. While the quantitative ratio or the number ratio of molecules of the deletion mutant would be influenced by the type and culture conditions of cultured cells of mammalian animals producing the antibody of the present invention, the main components of the antibody of the present invention can be both of the 2 heavy chains lacking an amino acid residue from the carboxyl terminus.

[0591] In addition, the antibody of the present invention or an antigen-binding fragment thereof (including those comprised in the molecule, the multispecific molecule, and the bispecific molecule of the present invention) comprising one to several amino acids derived from an expression vector and/or a signal sequence added to the amino terminus and/or carboxy terminus (and partially or entirely modified as described above) or the like are within the scope of the modified antibody of the present invention or the modified antigen-binding fragment thereof, as long as the antigen-binding activity of interest is maintained. A molecule comprising such modified antibody or modified antigen-binding fragment thereof is within the scope of the molecule of the present invention.

[0592] In the present invention, "the antibody or the binding fragment thereof" encompasses "the modified antibody or the modified antigen-binding fragment thereof" In addition, "the antibody or antigen-binding fragment thereof" included in the molecule, the multispecific molecule, and the bispecific molecule of the present invention encompasses "the modified antibody or the modified antigen-binding fragment thereof."

[0593] Antibody dependent cellular cytotoxicity can be potentiated by regulation (glycosylation, afucosylation, and the like) of sugar chain modification bound to the antibody of the present invention. Known techniques for regulation of antibody sugar chain modification are disclosed in, for example, WO 99/54342, WO 00/61739, and WO 02/31140, although the techniques are not limited thereto.

[0594] The present invention encompasses a complex of the antibody and other molecules ligated to each other with a linker (i.e., an immunoconjugate). An example of an antibody-drug complex comprising the antibody bound to a radioactive substance or a compound having pharmacological activity is an antibody-drug conjugate (ADC) (Methods Mol. Biol., 2013, 1045: 1-27; Nature Biotechnology, 2005, 23, pp. 1137-1146).

[0595] Further, the present invention encompasses a complex comprising such an antibody bound to another functional polypeptide. An example of such antibody-peptide complex is a complex of the antibody with an albumin-bound polypeptide (Protein Eng. Des. Sel., 2012, (2): 81-8).

[0596] The modified antibody, the antibody with regulated sugar chain modification, and the complex described above are within the scope of the antibody of the present invention, and the binding fragments of the modified antibody, the antibody with regulated sugar chain modification, and the complex are within the scope of the binding fragment of the antibody of the present invention.

4. Method for Producing Antibody

[0597] The antibody of the present invention can be produced in a cell as a recombinant antibody by, for example, inserting a DNA encoding a heavy chain variable region or a DNA encoding a light chain variable region into an expression vector, transforming a host cell for expression with the vector, and culturing the host cell.

[0598] Concerning DNAs encoding antibodies, a DNA encoding a heavy chain can be obtained by ligating a DNA encoding a heavy chain variable region to a DNA encoding a heavy chain constant region, and a DNA encoding a light chain can be obtained by ligating a DNA encoding a light chain variable region to a DNA encoding a light chain constant region.

[0599] The anti-HLA/NY-ESO antibody of the present invention can be produced by inserting the DNA encoding the heavy chain and the DNA encoding the light chain mentioned above into an expression vector, transforming a host cell with the vector, and culturing the host cell. In such a case, the DNA encoding the heavy chain and the DNA encoding the light chain mentioned above may be introduced into the same expression vector and the host cell may be transformed with the vector. Alternatively, the DNA encoding the heavy chain and the DNA encoding the light chain may be inserted into separate vectors and the host cell may be transformed with the two vectors. In this case, DNAs encoding the heavy chain variable region and the light chain variable region may be introduced into a vector into which the DNA encoding the heavy chain constant region and the DNA encoding a light chain constant region have been introduced in advance. Further, the vector may contain a DNA encoding a signal peptide, which promotes secretion of an antibody from a host cell. In this case, the DNA encoding the signal peptide and the DNA encoding the antibody are ligated in-frame in advance. An antibody can be obtained as a mature protein by removing the signal peptide after an antibody is produced.

[0600] In this case, the DNA encoding the heavy chain variable region, the DNA encoding the light chain variable region, the DNA comprising the DNA encoding the heavy chain variable region ligated to the DNA encoding the heavy chain constant region, or the DNA comprising the DNA encoding the light chain variable region ligated to the DNA encoding the light chain constant region may be operably ligated to an element, such as a promoter, an enhancer, or a polyadenylation signal. When DNA is "operably ligated" herein, DNA is ligated to an element, so that the element can exert their functions.

[0601] An expression vector is not particularly limited, as long as it can be replicated in an animal cell, a bacterial cell, a yeast cell, or other host, and examples thereof include known plasmids and phages. Examples of a vector used to construct an expression vector include pcDNA™ (Thermo Fisher Scientific), Flexi® vector (Promega), pUC19, pUEX2 (Amersham), pGEX-4T, pKK233-2 (Pharmacia), and pMAMneo (Clontech). As host cells, prokaryotic cells such as *Escherichia coli* and *Bacillus subtilis* and eukaryotic cells such as yeasts and animal cells can be used, with the use of eukaryotic cells being preferable. Examples of animal cells include the human embryonic kidney cell line HEK293 and the Chinese hamster ovary (CHO) cell. It is sufficient to introduce an expression vector into a host cell by a known method to transform the host cell. Examples of methods include an electroporation method, a calcium phosphate precipitation method, and a DEAE-dextran transfection method. The produced antibody can be purified by usual protein isolation or purification methods. For example, affinity chromatography or other chromatography techniques, filtration, ultrafiltration, salting out, dialysis, and the like can be suitably selected and combined.

5. Molecule that Binds to HLA/NY-ESO

[0602] The molecule of the present invention comprises the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof. The molecule of the present invention is preferably a multispecific molecule comprising 2 or more antigen-binding sites. Specifically, the molecule of the present invention can bind to 2 or more different epitopes on a molecule or different epitopes on 2 or more different molecules and such a molecule comprises a plurality of different antigen-binding fragments. Examples of such a multispecific molecule include, but are not limited to, an IgG-type multispecific molecule and a multispecific molecule comprising two or

more types of variable regions, for example, antibody fragments, such as tandem scFv (taFv), single chain diabody, a diabody, and a triabody, and an antibody fragment resulting from covalent or noncovalent binding. A multispecific molecule may comprise Fc.

[0603] The multispecific molecule of the present invention may comprise, in addition to the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof, one type or two or more types of antibodies or antigen-binding fragments thereof. Examples of such antigen-binding fragments of the antibody include Fab, F(ab)', Fv, scFv, and sdAb.

[0604] A preferable multispecific molecule according to the present invention further comprises an anti-CD3 antibody or an antigen-binding fragment thereof, and it binds specifically to CD3 as well.

[0605] The anti-CD3 antibody or an antigen-binding fragment thereof included in the multispecific molecule according to the present invention is not particularly limited, provided that it is a human CD3-binding antibody or a binding fragment thereof, and it preferably binds to CD3 of a non-human primate, such as a cynomolgus monkey. A more preferable example of an anti-CD3 antibody or an antigen-binding fragment thereof is the antibody or antigen-binding fragment thereof comprising the heavy chain variable region CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 141, the heavy chain variable region CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 142, the heavy chain variable region CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 143, the light chain variable region CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 144, the light chain variable region CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 145, and the light chain variable region CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 146 (FIG. 142).

[0606] An example of the more preferable antibody or antigen-binding fragment thereof comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 is the antibody or antigen-binding fragment thereof comprising the C3E-7034 heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136 (FIG. 137), the C3E-7036 heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137 (FIG. 138), the C3E-7085 heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of SEQ ID NO: 138 (FIG. 139), the C3E-7088 heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139 (FIG. 140), the C3E-7093 heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140 (FIG. 141), the C3E-7096 heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155 (FIG. 151), the C3E-7096 heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152), and the C3E-7097 heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157 (FIG. 153).

[0607] An example of the more preferable antibody or antigen-binding fragment thereof comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 is the antibody or antigen-binding fragment thereof comprising the C3E-7034 light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136 (FIG. 137), the C3E-7036 light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137 (FIG. 138), the C3E-7085 light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138 (FIG. 139), the C3E-7088 light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139 (FIG. 140), the C3E-7093 light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140 (FIG. 141), the C3E-7096 light chain variable region consisting of an amino acid sequence of amino acids 405 to 511 of the amino acid sequence as shown in SEQ ID NO: 155 (FIG. 151), the C3E-7096 light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152), and the C3E-7097 light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157 (FIG. 153).

[0608] An example of the more preferable antibody or antigen-binding fragment thereof comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 is the antibody or antigen-binding fragment thereof comprising a combination of the C3E-7034 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 243 of SEQ ID NO: 136 (FIG. 137), a combination of the C3E-7036 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 241 of SEQ ID NO: 137 (FIG. 138), a combination of the C3E-7078 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 243 of SEQ ID NO: 147 (FIG. 143), a combination of the C3E-7085 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 241 of SEQ ID NO: 138 (FIG. 139), a combination of the C3E-7088 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 243 of SEQ ID NO: 139 (FIG. 140), a combination

of the C3E-7093 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 243 of SEQ ID NO: 140 (FIG. 141), a combination of the C3E-7096 heavy chain variable region and light chain variable region consisting of amino acids 272 to 389 and amino acids 405 to 511 of SEQ ID NO: 155 (FIG. 151), a combination of the C3E-7096 heavy chain variable region and light chain variable region consisting of amino acids 277 to 394 and amino acids 410 to 516 of SEQ ID NO: 156 (FIG. 152), and a combination of the C3E-7097 heavy chain variable region and light chain variable region consisting of amino acids 277 to 394 and amino acids 410 to 516 of SEQ ID NO: 157 (FIG. 153).

[0609] Further, examples of the more preferable antibody or antigen-binding fragment thereof comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 include C3E-7034 scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 136 (FIG. 137), C3E-7036 scFv consisting of an amino acid sequence of amino acids 2 to 241 of the amino acid sequence as shown in SEQ ID NO: 137 (FIG. 138), C3E-7078 scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 147 (FIG. 143), C3E-7085 scFv consisting of an amino acid sequence of amino acids 2 to 241 of the amino acid sequence as shown in SEQ ID NO: 138 (FIG. 139), C3E-7088 scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 139 (FIG. 140), C3E-7093 scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 140 (FIG. 141), and an antibody or a binding fragment thereof comprising any one of such scFv constructs. A preferable embodiment of CD3-specific scFv (which may be referred to as “anti-CD3 scFv”) includes scFv comprising a FLAG-His tag added to its carboxyl terminal side (may be simply referred to as “tag adduct”). Examples of preferable tag adducts include C3E-7034 (SEQ ID NO: 136; FIG. 137), C3E-7036 (SEQ ID NO: 137; FIG. 138), C3E-7085 (SEQ ID NO: 138; FIG. 139), C3E-7088 (SEQ ID NO: 139; FIG. 140), and C3E-7093 (SEQ ID NO: 140; FIG. 141), with C3E-7085 being more preferable.

[0610] A preferable example of the multispecific molecule of the present invention is a bispecific molecule. A “bispecific” molecule is capable of binding to two different epitopes on a single molecule or different epitopes on two molecules, and it includes an antibody or antigen-binding fragment having such bispecificity. The bispecific molecule of the present invention binds to HLA/NY-ESO and it further binds to CD3.

[0611] Examples of the bispecific molecule of the present invention include those having the structures (formats) described below.

[0612] In the dual scFv-type bispecific molecule, two types of scFv constructs binding to different epitopes are each allowed to bind to one Fc of a dimer with a linker or allowed to directly bind thereto without a linker. Alternatively, two types of scFv constructs binding to different epitopes are each allowed to bind to CH and CL with linkers and are further allowed to bind to one Fc of the dimer with linkers. In such a bispecific molecule, Fc comprising a heterodimer-forming mutation downstream of one scFv is heterologously associated with another Fc comprising a heterodimer-forming mutation downstream of another scFv. The dual scFv-type bispecific molecule is referred to as a dual-type bispecific molecule or simply as a dual-type (FIG. 6A (b)).

[0613] In the present invention, for example, a dual-type bispecific molecule consisting of anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv may be used.

[0614] Alternatively, the bispecific molecule of the present invention may comprise Fab and scFv binding to different epitopes, Fab of the first antibody may be bound to one Fc of a dimer, and scFv of the second antibody may be bound to another Fc with linkers. In such bispecific molecule, Fc comprising a heterodimer-forming mutation downstream of Fab is heterologously associated with Fc comprising a heterodimer-forming mutation downstream of scFv. Such bispecific molecule is referred to as a “hybrid-type bispecific molecule” or a “hybrid-type” (FIG. 6A (a)). In the present invention, for example, a hybrid type consisting of anti-HLA-A2/NY-ESO Fab and anti-CD3 scFv can be used.

[0615] In addition, a bispecific molecule may allow Fab of the first antibody and scFv of the second antibody to attach to one Fc of a dimer with linkers. In such a case, Fab may be allowed to attach to Fc and scFv may be allowed to attach to such Fab. Alternatively, scFv may be allowed to attach to Fc and Fab may be allowed to attach to such scFv. Preferably, Fab may be allowed to attach to Fc and scFv may be allowed to attach to such Fab. scFv may be allowed to attach to a variable region of Fab with a linker. In such a bispecific molecule, Fc comprising a heterodimer-forming mutation is associated with a site downstream of where scFv is ligated to Fab. Such a bispecific molecule is referred to as an scFv-Fab-heterodimer Fc-type bispecific molecule or scFv-Fab-heterodimer Fc-type (FIG. 6A (c)).

[0616] In the present invention, for example, scFv-Fab-heterodimer Fc-type consisting of anti-CD3 scFv and anti-HLA-A2/NY-ESO Fab may be used.

[0617] In addition, taFv (FIG. 3 (c)) comprising two types of scFv constructs (of the first antibody and the second antibody) ligated to each other with a linker may be attached to one Fc of a dimer with a linker or directly without a linker. Such a bispecific molecule is referred to as a taFv-heterodimer Fc-type bispecific molecule or taFv-heterodimer Fc-type (FIG. 3 (d)). In such a bispecific molecule, Fc comprising a heterodimer-forming mutation is heterologously associated with another Fc comprising a heterodimer-forming mutation in a site downstream of taFv.

The order for ligating the first antibody and the second antibody to taFv is not limited. The previously mentioned bispecific molecule is referred to as taFv-heterodimer Fc-type but when the order for ligating the first antibody and the second antibody to taFv is inverted, the bispecific molecule is referred to as taFv (inversed)-heterodimer Fc-type (taFv (inversed)-Fc type).

[0618] FIG. 6 A (a) shows a structure of a hybrid-type bispecific molecule, FIG. 6 A (b) shows a structure of a dual-type bispecific molecule, and FIG. 6 A (c) shows a structure of an scFv-Fab-heterodimer Fc-type bispecific molecule. FIG. 3 (a) shows a structure of scFv, FIG. 3 (b) shows a structure of Fab, FIG. 3 (c) shows a structure of taFv, FIG. 3 (d) shows a structure of taFv-heterodimer Fc-type bispecific molecule, and FIG. 3 (e) shows a structure of a taFv-Fab-heterodimer Fc-type bispecific molecule. FIG. 6 B (a) shows a structure of a taFv-heterodimer Fc-type bispecific molecule (the same as FIG. 3 (d)), FIG. 6 B (b) shows a structure of a taFv (inversed)-heterodimer Fc-type bispecific molecule, FIG. 6 B (c) shows a structure of the first polypeptide included in the taFv (inversed)-heterodimer Fc-type bispecific molecule, and FIG. 6 (d) shows a structure of the second polypeptide included in the taFv (inversed)-heterodimer Fc-type bispecific molecule.

[0619] In the bispecific molecule of the present invention, a plurality of polypeptides are associated.

[0620] In the present invention, a taFv, for example, one composed of an anti-HLA-A2/NY-ESO scFv and an anti-CD3 scFv may be used. The taFv-heterodimer Fc-type bispecific molecule preferably comprises (a) a first polypeptide comprising scFv that binds specifically to HLA/NY-ESO, scFv that binds specifically to CD3, and the immunoglobulin Fc region (i) in that order from the N terminus toward the C terminus and the second polypeptide comprising the hinge region and the Fc region (ii) of immunoglobulin. It more preferably comprises (b) the first polypeptide associated with a second polypeptide at the Fc region (i) and the Fc region (ii). Fc regions of the first polypeptide and of the second polypeptide may comprise a heterodimer-forming mutation. FIG. 3 (d) shows an example of the taFv-heterodimer Fc-type bispecific molecule. As shown in FIG. 3 (d), the Fc region (i) of the first polypeptide binds to the Fc region (ii) of the second polypeptide (solid), and the first polypeptide is thus associated with the second polypeptide. FIG. 3 (f) shows the first polypeptide and FIG. 3 (g) shows the second polypeptide. In FIG. 3 (d), for example, scFv shown as blank is anti-HLA-A2/NY-ESO scFv, and scFv indicated with positive slopes is anti-CD3 scFv.

[0621] The first polypeptide included in a more preferable taFv-heterodimer Fc-type bispecific molecule of the present invention comprises an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 149, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 150, an amino acid sequence of amino acids 20 to 511 of the amino acid sequence as shown in SEQ ID NO: 155, an amino acid sequence of amino acids 20 to 516 of the amino acid sequence as shown in SEQ ID NO: 156, or an amino acid sequence of amino acids 20 to 516 of the amino acid sequence as shown in SEQ ID NO: 157. The first polypeptide included in a further preferable taFv-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 529 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 86, 149, 150, or 155 or an amino acid sequence of amino acids 534 to 750 of the amino acid sequence as shown in SEQ ID NO: 156 or 157. The first polypeptide included in a still further preferable taFv-heterodimer Fc-type bispecific molecule consists of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 86, an amino

acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 149, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 150, or an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 155. Alternatively, the first polypeptide consists of an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 156 or an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 157.

[0622] The second polypeptide included in the preferable taFv-heterodimer Fc-type bispecific molecule of the present invention comprises a hinge region and mutant Fc derived from a human antibody. The second polypeptide included in the more preferable taFv-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.

[illegible]

[0624] Further examples of the preferable taFv-heterodimer Fc-type bispecific molecules of the present invention include NYZ-0038 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 155 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYZ-0082 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 156 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, and NYZ-0083 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as

shown in SEQ ID NO: 157 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.

[0625] Of these, NYF-0023, NYF-0047, NYF-0048, NYF-0060, NYF-0061, NYZ-0038, NYZ-0082, and NYZ-0083 exert excellent biological activity, physical properties, and the like and thus are particularly preferable.

[0626] In addition, taFv of the first antibody may be allowed to attach to one Fc of a dimer with a linker or directly bound thereto without a linker, and Fab of the first antibody or the second antibody may be allowed to attach to another Fc with a linker or directly bound thereto without a linker. In such a bispecific molecule, Fab is added to a site upstream of the Fc region (ii) (solid) of the second polypeptide of the taFv-heterodimer Fc-type. Such a bispecific molecule is referred to as the taFv-Fab-heterodimer Fc-type bispecific molecule or taFv-Fab-heterodimer Fc-type (FIG. 3).

[0627] In the present invention, for example, taFv of anti-HLA-A2/NY-ESO scFv, taFv of anti-CD3 scFv, and Fab of HLA/NY-ESO may be used to be included in the taFv-Fab-heterodimer Fc-type bispecific molecule.

[0628] The taFv-Fab-heterodimer Fc-type bispecific molecule preferably comprises (a) a first polypeptide comprising scFv that binds specifically to human HLA/NY-ESO, scFv that binds specifically to CD3, and the immunoglobulin Fc region (i) in that order from the N terminus toward the C terminus, a second polypeptide consisting of an immunoglobulin heavy chain including the Fc region (ii), and a third polypeptide consisting of an immunoglobulin light chain. More preferably, (b) the second polypeptide is associated with the third polypeptide, and (c) the first polypeptide is associated with the second polypeptide at the Fc region (i) and the Fc region (ii). FIG. 3 (e) shows an example of the taFv-Fab-heterodimer Fc-type bispecific molecule, FIG. 3 (f) shows the first polypeptide, FIG. 3 (h) shows the second polypeptide, and FIG. 3 (i) shows the third polypeptide. As shown in FIG. 3 (e), the Fc region (ii) (solid) of the second polypeptide, consisting of an immunoglobulin heavy chain comprising the Fc region (ii), binds to the Fc region (i) of the first polypeptide, and an immunoglobulin light chain further attaches to the second polypeptide. Such preferable taFv-Fab-heterodimer Fc-type bispecific molecule can be composed of the taFv-heterodimer Fc-type bispecific molecule comprising taFv and the immunoglobulin Fc region and Fab bound thereto. Examples of amino acid sequences included in the first polypeptide of a preferable taFv-Fab-heterodimer Fc-type bispecific molecule include an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 150. The first polypeptide included in a more preferable taFv-Fab-heterodimer Fc-type bispecific molecule consists of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 149, or an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 150.

[0629] The second polypeptide included in the preferable taFv-Fab-heterodimer Fc-type bispecific molecule of the present invention comprises the human antibody or humanized antibody heavy chain variable region, the CH1 region, a hinge region, and mutant Fc. The second polypeptide included in the more preferable taFv-Fab-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 20 to 242 of the amino acid sequence as shown in SEQ ID NO: 99.

[0630] The third polypeptide included in the preferable taFv-Fab-heterodimer Fc-type bispecific molecule of the

present invention comprises the human antibody or humanized antibody light chain variable region and constant region. The third polypeptide included in the more preferable taFv-Fab-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 21 to 131 of the amino acid sequence as shown in SEQ ID NO: 100.

[0631] Of the second polypeptide included in such preferable taFv-Fab-heterodimer Fc-type bispecific molecule, the variable region, the CH1 region, and the third polypeptide constitute Fab, and Fab is preferably of the anti-HLA/NY-ESO antibody, such as Fab of NYA-0001.

[0632] In the present invention, for example, anti-HLA-A2/NY-ESO scFv, anti-CD3 scFv, HLA/NY-ESO Fab, or anti-CD3 Fab may be included in the scFv-Fab-heterodimer Fc-type bispecific molecule.

[0633] The scFv-Fab-heterodimer Fc-type bispecific molecule preferably comprises (a) a first polypeptide comprising scFv that binds specifically to human HLA/NY-ESO, the antibody heavy chain variable region and constant region CH1 binding specifically to CD3, and the immunoglobulin Fc region (i) in that order from the N terminus toward the C terminus, a second polypeptide comprising the immunoglobulin hinge region and Fc region (ii), and a third polypeptide consisting of an antibody light chain comprising a variable region and a constant region. More preferably, (b) the first polypeptide is associated with the second polypeptide via the Fc region (i) and the Fc region (ii), and the first polypeptide is associated with (the antibody light chain of) the third polypeptide via the antibody heavy chain variable region and constant region CH1 of the first polypeptide. The Fc regions of the first polypeptide and of the second polypeptide may be a wild-type or may contain a heterodimer-forming mutation. FIG. 6 A (c) shows an example of the scFv-Fab-heterodimer Fc-type bispecific molecule. In FIG. 6 A (c), the right half shows the first polypeptide and the third polypeptide and the left half shows the second polypeptide. As shown in FIG. 6 A (c), the Fc region (i) of the first polypeptide is associated with the Fc region (ii) of the second polypeptide (solid), and the first polypeptide is associated with the third polypeptide. In FIG. 6 A (c), scFv indicated with positive slopes is anti-HLA-A2/NY-ESO scFv, and Fab each indicated as blank, a checkered pattern, or horizontal lines is anti-CD3 Fab.

[0634] An example of an amino acid sequence included in the first polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule is an amino acid sequence of amino acids 21 to 394 of the amino acid sequence as shown in SEQ ID NO: 160. A more preferable example thereof is an amino acid sequence of amino acids 20 to 724 of the amino acid sequence as shown in SEQ ID NO: 160.

[0635] An example of another amino acid sequence included in the first polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule is an amino acid sequence of amino acids 21 to 389 of the amino acid sequence as shown in SEQ ID NO: 197. A more preferable example thereof is an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 197. An example of another amino acid sequence included in the first polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule is an amino acid sequence of amino acids 21 to 389 of the amino acid sequence as shown in SEQ ID NO: 198. A more preferable example thereof is an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 198.

[0636] The second polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention comprises a hinge region and mutant Fc derived from a human antibody. The second polypeptide included in the more preferable scFv-Fab-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.

[0637] The third polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention comprises a human antibody-derived light chain. The third polypeptide included in the more preferable scFv-Fab-heterodimer Fc-type bispecific molecule comprises, for example, an amino acid sequence of amino acids 21 to 127 of the amino acid sequence as shown in SEQ ID NO: 161. A further preferable example thereof is an amino acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161.

[0638] Of these, NYZ-1010 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 724 of the amino acid sequence as shown in SEQ ID NO: 160, the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, and the third polypeptide consisting of an amino acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161 are associated with each other can be exemplified as the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention.

[0639] In addition, NYZ-1007 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 197, the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, and the third polypeptide consisting of an amino acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161 are associated with each other can be exemplified as the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention.

[0640] Further, NYZ-1017 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to

719 of the amino acid sequence as shown in SEQ ID NO: 198, the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, and the third polypeptide consisting of an amino acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161 are associated with each other can be exemplified as the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention.

[0641] At least 1, 2, or 3 peptides included in the bispecific molecule of the present invention may be the “deletion mutant” described above. Specifically, 1 or 2 (or more) amino acids at its carboxyl terminus (at the carboxyl terminus derived from the antibody heavy chain, in particular) may be mutated (may also be deleted). For example, the carboxyl terminus of the amino acid sequence of the first polypeptide included in NYZ-1010, which is a preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention, may be any of Lys at position 724 of SEQ ID NO: 160, Gly at position 723 resulting from deletion of an amino acid, or a mixture of first polypeptides including Lys and Gly at the carboxyl terminus. Also, the carboxyl terminus of the amino acid sequence of the second polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention may be any of Lys at position 246 of SEQ ID NO: 84, Gly at position 245 resulting from deletion of an amino acid, or a mixture of second polypeptides including Lys and Gly at the carboxyl terminus.

[0642] scFv and Fab included in the bispecific molecule of the present invention are preferably derived from a humanized antibody or human antibody, and Fc is preferably derived from a human antibody.

[0643] In a variable region in the bispecific molecule of the present invention, a heavy chain variable region and a light chain variable region may be attached in that order from the amino terminus of the antibody. Alternatively, a light chain variable region and a heavy chain variable region may be attached in that order. A linker may optionally be present between two variable regions. A glycine residue may optionally be present at the amino terminus of the amino-terminal variable region. In a tandem scFv-type bispecific molecule, a linker, a FLAG tag, and/or a His tag may optionally be attached to the carboxyl terminus of the carboxyl-terminal variable region. In a preferable embodiment, for example, a heavy chain variable region, a first linker, a light chain variable region, a second linker, a FLAG tag, and, a His tag are attached in that order from the amino terminus.

[0644] Examples of linkers include a single chain polypeptide, a single chain oligopeptide, and synthetic products, such as PEG, nucleotide, sugar chain, and compounds. Other known linkers can also be used without particular limitation, as long as they can attach 2 polypeptides to each other.

[0645] A peptide linker can comprise, for example, 5 to 30 amino acids. When a bispecific molecule comprises a plurality linkers, all the peptide linkers may be of the same length, or peptide linkers of different lengths may be used.

[0646] An example of a peptide linker is a repetition of (Gly·Gly·Gly·Gly·Ser) (SEQ ID NO: 161). One to several amino acid residues other than Gly and Ser may be added thereto.

[0647] Among constructs (formats) of the multispecific antibody and, in particular, the bispecific antibody of the present invention as described above, taFv-heterodimer Fc-type, taFv-Fab-heterodimer Fc-type, and scFv-Fab-heterodimer Fc-type are preferable, and taFv-heterodimer Fc-type is more preferable. A format in which anti-HLA/NY-ESO scFv and anti-CD3 scFv are positioned in that order from the N terminus toward the C terminus (the taFv-heterodimer Fc-type) is more preferable than the format in which anti-CD3 scFv and anti-HLA/NY-ESO scFv are positioned in that order (the taFv (inversed)-heterodimer Fc-type) (e.g., Example 11). Another example of a more preferable format is the scFv-Fab-heterodimer Fc-type.

[0648] The present invention also encompasses a molecule that comprises an amino acid sequence encoded by a nucleotide sequence included in a polynucleotide which hybridizes, under stringent conditions, to a complementary strand of the polynucleotide comprising a nucleotide sequence encoding an amino acid sequence included in the molecule of the present invention, and that binds to HLA/NY-ESO, and further binds to CD3.

[0649] The present invention also encompasses a molecule that comprises an amino acid sequence exhibiting at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to the amino acid sequence included in the molecule of the present invention, and that binds to HLA/NY-ESO, and further binds to CD3.

[0650] The antibody of the present invention, the binding fragment thereof, and the multispecific antibody comprising the same have excellent biological activity, physicochemical properties (hereafter referred to as “physical properties”), safety, pharmacokinetics, and other properties. (a) Examples of biological activity or indicators thereof include antigen-binding activity, in vitro cytotoxicity, and in vivo antitumor activity. For example, the dissociation constant (KD value) to HLA/NY-ESO is 100 nM or lower or 50 nM or lower, preferably 20 nM or lower or 10 nM or lower, and more preferably 5 nM or lower. The EC₅₀ value of cytotoxicity exerted with the use of human peripheral blood mononuclear cells as effector cells to human U266B1 and/or NCI-H1703 cells endogenously expressing NY-ESO is 20 nM or lower, preferably 10 nM or lower, and more preferably 5 nM or lower (in vitro cytotoxicity can be assayed and determined in accordance with the method described in Example 8, although the method is not limited thereto). For example, 0.1 ml of a suspension of human squamous lung cancer cell lines NCI-H1703 (6×10⁵.sup.7 cells/ml) is injected hypodermically to NOG mice, 0.2 ml of a suspension of human peripheral blood mononuclear cells (3.75×10⁵.sup.7 cells/ml) is injected intravenously 4 days later, and antibody administration is initiated 14 days

later and performed once a week in 3 instances. Thereafter, the tumor volume is measured. Tumor growth inhibitory activity relative to the control group to which a solvent has been administered is 50% or higher, preferably 75% or higher, and more preferably 90% or higher (in vivo antitumor activity can be assayed and determined in accordance with the method described in Example 9, although the method is not limited thereto). (b) Impurities contained in biopharmaceutical products are associated with drug safety. Thus, it is necessary to establish the adequate standard and to control the amount of impurities at the time of production and during storage. In particular, HMWS (aggregate) is a major impurity and is associated with an immunogenic risk or reduced drug efficacy. Thus, HMWS should be more strictly controlled. For the purpose of impurity control, evaluation should also be performed in respect of stability (whether or not the amount of impurities was increased) with the elapse of time during and after production, as well as at the time of production. Since the effective period of a pharmaceutical product is determined based on the results of the long-term stability test, a longer effective period can be determined for an antibody exhibiting stability over time. Thus, long-term stability can be used as an indicator to select a preferable antibody for a biopharmaceutical product. Examples of physicochemical properties in the present invention include acid resistance (for inhibition of HMWS production) and solution stability (for inhibition of HMWS production). An example of another indicator is a high yield of a culture product of recombinant cells obtained by introducing a gene encoding the amino acid sequence of the antibody of the present invention, the binding fragment thereof, or the molecule comprising the same into a host cell suitable for production thereof, such as the Expi293F cell. The preferable antibody of the present invention, the antigen-binding fragment thereof, and the multispecific antibody comprising the same with such physicochemical properties are advantageous for the following reasons. That is, a solution containing the antibody, the binding fragment thereof, or the multispecific antibody can be exposed to acidic conditions, and production processes thereof, such as protein A or ion-exchange chromatography and virus inactivation, can be thus performed or facilitated. Also, HMWS production can be controlled at low levels even in the form of a solution of the antibody, the binding fragment thereof, or the multispecific antibody, and, thus, production thereof, drug preparation, distribution or storage of pharmaceutical products comprising the same can be performed and facilitated. Further, pharmaceutical products containing the same can be efficiently produced. Concerning acid resistance, for example, the HMWS content determined by heating at pH 3.5 at room temperature for 1 hour and then measuring HMWS by size exclusion chromatography is 5% or lower, preferably 2% or lower, and more preferably 1% or lower (the HMWS content can be measured and determined for acid resistance evaluation in accordance with the method described in Example 19-1, although the method is not limited thereto). Concerning solution stability, for example, the analyte is dissolved to a concentration of 25 mg/ml in a solution comprising 25 mM histidine and 5% sorbitol at pH 6.0, the resultant is stored at 25° C. for 6 days, and HMWS is then measured by size exclusion chromatography. The HMWS content determined thereby is 20% or lower, and preferably 10% or lower (the HMWS content can be measured and determined for evaluation of solution stability in accordance with the method described in Example 19-2, although the method is not limited thereto). Production efficiency or yield can be measured and determined in accordance with the method described in Examples 20 and 21, although the method is not limited thereto. (c) Examples of safety and indicators thereof include antigen recognition properties and observations at the time of administration. For example, an antibody of interest that recognizes a plurality of amino acids on a wild-type NY-ESO peptide but does not bind to a homologous peptide comprising an amino acid sequence similar to but not identical to the wild-type NY-ESO peptide would produce a low risk of side effects caused by off-target effects. In addition, such an antibody can be predicted to exhibit low immunogenicity in ISPRI web-based immunogenicity screening (EpiVax, Inc) and exhibit a low risk of side effects such as cytokine production. Also, NYF-0023, NYF-0045, NYF-0047, NYF-0048, NYF-0060, NYF-0061, NYZ-0082, or NYZ-1010 included in the bispecific antibody of the present invention were administered to Balb/c mice. As a result, no significant problems were observed in the half-life in blood, and body weight loss or other apparent toxicity problems were not observed. As a result of single administration of NYZ-0082 or NYZ-1010 to cynomolgus monkeys, no significant problems were observed in the half-life in blood, and no changes resulting from administration were observed in terms of general conditions, body weight, the amount of feed intake, body temperature, and plasma cytokine levels. (d) Examples of kinetics or indicators thereof include the half-life in blood. In the present invention, no significant problems were observed in the half-life in blood as a result of administration of several bispecific antibodies to Balb/c mice or cynomolgus monkeys. The antibody of the present invention, the binding fragment thereof, and molecules provided with such excellent biological activity, physicochemical properties, safety, and kinetics can be preferably integrated into a pharmaceutical composition. Preferable examples of the antibody of the present invention or the antigen-binding fragment thereof having the antigen-binding activity described in (a) and various properties described in (b) include, but are not limited to, NYA-1143, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, NYA-2031, NYA-2047, NYA-2061, NYA-2143, and NYA-3061. More preferable examples thereof include, but are not limited to, NYA-2047, NYA-2061, NYA-2143, and NYA-3061. Preferable examples of the multispecific antibody of the present invention having the properties (a) to (d) include, but are not limited to, NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, NYF-0045, NYF-0047, NYF-0048, NYF-00058, NYF-0060, NYF-0061,

NYZ-0038, NYZ-0082, NYZ-0088, and NYZ-1010. More preferable examples thereof include, but are not limited to, NYF-0061, NYZ-0038, NYZ-0082, NYZ-0088, NYZ-1007, NYZ-1010, and NYZ-1017.

[0651] In the present invention, a “site” to which an antibody binds; i.e., a “site” that is recognized by an antibody, is a partial peptide or a partial higher-order structure of an antigen to which an antibody binds or which is recognized by the antibody. In the present invention, such a site is also referred to as an epitope or an antibody binding site. Examples of sites to which the anti-HLA/NY-ESO antibody according to the present invention binds or sites that are recognized thereby on HLA/NY-ESO include a plurality of amino acids and partial higher-order structures in the HLA/NY-ESO peptide.

[0652] The “antibody or the binding fragment thereof that binds to the same site” as the antibody according to the present invention or the binding fragment thereof is also within the scope of the present invention. The “antibody that binds to the same site” as a given antibody is another antibody that binds to a site on an antigen molecule recognized by such given antibody. If a second antibody binds to a partial peptide or partial conformation on an antigen molecule to which a first antibody binds, the first antibody and the second antibody can be evaluated to bind to the same site. When the first antibody of the present invention has the antigen-binding activity described in (a) above, the second antibody that binds to the same site on HLA/NY-ESO is highly likely to have similar activity, and such second antibody is thus within the scope of the present invention. An antibody that binds to HLA/NY-ESO competitively with the first antibody of the present invention having the antigen-binding activity described in (a) is within the scope of the present invention. An antibody that binds to a site on HLA/NY-ESO recognized by the monoclonal antibody of the present invention, an antibody that binds to HLA/NY-ESO competitively with the monoclonal antibody of the present invention, and binding fragments thereof preferably have at least one of in vitro cytotoxicity and in vivo antitumor activity described in (a) and properties described in (b) to (d), more preferably 3 or more thereof, and optimally all thereof.

[0653] A binding site of the antibody can be determined by a method well known to a person skilled in the art, such as an immunoassay technique. For example, a series of peptides are prepared by removing an antigen amino acid sequence from the C or N terminus as appropriate, reactivity of an antibody thereto is examined, a recognition site is roughly determined, a shorter peptide sequence is synthesized, and reactivity of an antibody thereto is examined. Thus, the binding site can be determined. Alternatively, a particular site or region of an amino acid sequence of an antigen or an antigen fragment peptide is deleted or substituted with another amino acid sequence, or a mutation is introduced into such site or region. By examining reactivity of an antibody to such peptide, the binding site can be determined. An antigen fragment peptide can be prepared by, for example, genetic recombination or peptide synthesis.

[0654] When an antibody binds to or recognizes a partial higher-order structure of an antigen, such antigen-binding site of the antibody can be determined by identifying an amino acid residue on an antigen adjacent to the antibody via x-ray structural analysis. For example, an antibody or a fragment thereof and an antigen or a fragment thereof are bound to each other, crystallized, and subjected to structural analysis. Thus, an amino acid residue on an antigen being at an interaction distance with the antibody can be identified. The interaction distance is 8 Å or less, preferably 6 Å or less, and more preferably 4 Å or less. One or more amino acid residues being at such an interaction distance with the antibody can constitute the antigen-binding site of the antibody (epitope). When there are two or more such amino acid residues, such amino acid residues may not be adjacent to each other on a one-dimensional sequence.

[0655] The anti-HLA/NY-ESO antibody according to the present invention or the binding fragment thereof specifically recognizes a plurality of amino acids in the HLA/NY-ESO amino acid sequence. An antibody or a binding fragment thereof that recognizes a plurality of amino acids, an antibody or a binding fragment thereof that binds to HLA/NY-ESO competitively with the antibody according to the present invention or the binding fragment thereof, and an antibody or a binding fragment thereof that has an interaction distance with such a plurality of amino acids are within the scope of the present invention. A multispecific antibody comprising such antibody or binding fragment thereof is also within the scope of the present invention.

6. Pharmaceutical Composition

[0656] The present invention encompasses an anti-cancer agent comprising, as an active ingredient, the anti-HLA/NY-ESO antibody according to the present invention or a multispecific molecule that binds to HLA/NY-ESO.

[0657] The anti-cancer agent of the present invention can be used for one type or two or more types of cancer species selected from carcinoma, sarcoma, lymphoma, leukemia, myeloma, germinoma, brain tumor, carcinoid, neuroblastoma, retinoblastoma, and nephroblastoma. Specific examples of carcinoma include kidney cancer, melanoma, squamous cell cancer, basal cell cancer, conjunctival cancer, oral cavity cancer, laryngeal cancer, pharyngeal cancer, thyroid gland cancer, lung cancer (non-small cell lung cancer (adenocarcinoma, epidermoid cancer, large cell cancer), and small cell lung cancer), breast cancer, esophageal cancer, gastric cancer, duodenal cancer, small bowel cancer, large bowel cancer, rectal cancer, appendiceal cancer, anal cancer, liver cancer, gallbladder cancer, bile duct cancer, pancreatic cancer, adrenal cancer, bladder cancer, prostate cancer, uterine cancer, and vaginal cancer. Specific examples of sarcoma include liposarcoma, angiosarcoma, chondrosarcoma, rhabdomyosarcoma, Ewing's sarcoma, osteosarcoma, undifferentiated pleomorphic sarcoma, myxofibrosarcoma,

malignant peripheral neurilemmoma, retroperitoneal sarcoma, synoviosarcoma, uterine sarcoma, gastrointestinal stromal tumor, leiomyosarcoma, and epithelioid sarcoma. Specific examples of lymphoma include B-cell lymphoma, T/NK-cell lymphoma, and Hodgkin's lymphoma. Specific examples of leukemia include myelogenous leukemia, lymphatic leukemia, myeloproliferative disorder, and myelodysplastic syndrome. Specific examples of myeloma include multiple myeloma. Specific examples of germinoma include testicular cancer and ovarian cancer. Specific examples of brain tumor include neuroglioma and meningioma.

[0658] The anti-cancer agent of the present invention can contain the anti-HLA/NY-ESO antibody or a multispecific molecule that binds to HLA/NY-ESO in an amount effective for treatment, as well as pharmaceutically acceptable carriers, diluents, solubilizers, emulsifiers, preservatives, aids, and the like. The “pharmaceutically acceptable carriers” and the like can be suitably selected from a broad range according to the type of target disease and the dosage form of a drug. An administration method for the anti-cancer agent of the present invention can be suitably selected. For example, the anti-cancer agent can be injected, and local injection, intraperitoneal injection, selective intravenous infusion, intravenous injection, subcutaneous injection, organ perfusate infusion, and the like can be employed. Further, an injection solution can be formulated using a carrier comprising a salt solution, a glucose solution, or a mixture of salt water and a glucose solution, various types of buffer solutions, or the like. Further, a powder may be formulated and mixed with a liquid carrier to prepare an injection solution before use.

[0659] Other administration methods can be suitably selected along with development of a formulation. For example, oral solutions, powders, pills, capsules, tablets, and the like can be applied for oral administration. For oral solutions, oral liquid preparations such as suspensions and syrups can be produced using water, saccharides such as sucrose, sorbitol, and fructose, glycols such as PEG, oils such as sesame oil and soybean oil, preservatives such as alkyl parahydroxybenzoates, flavors such as strawberry flavor and peppermint, and the like. Powders, pills, capsules, and tablets can be formulated using excipients such as lactose, glucose, sucrose, and mannitol, disintegrating agents such as starch and alginate soda, lubricants such as magnesium stearate and talc, binders such as polyvinyl alcohol, hydroxypropyl cellulose, and gelatin, surfactants such as fatty acid esters, plasticizers such as glycerin, and the like. Tablets and capsules are preferred unit dosage forms for the composition of the present invention in that they are easily administered. Solid production carriers are used to produce tablets and capsules.

[0660] The effective dose of the anti-HLA/NY-ESO antibody or a multispecific molecule that binds to HLA/NY-ESO used for treatment can be changed according to the characteristics of the symptoms to be treated and the patient's age and condition and can be finally determined by a physician. For example, one dose is 0.0001 mg to 100 mg per kg of body weight. The predetermined dose may be administered once every one to 180 days, or the dose may be divided into two doses, three doses, four doses, or more doses per day at appropriate intervals.

[0661] The present invention also encompasses a polynucleotide encoding an amino acid sequence included in the anti-HLA/NY-ESO antibody according to the present invention or a multispecific molecule binding to HLA/NY-ESO, a vector comprising such a polynucleotide, a cell comprising such a polynucleotide or vector, and a pharmaceutical composition comprising, as an active ingredient, any of the polynucleotide, the vector and the cell. Such a pharmaceutical composition is preferably an anti-cancer agent.

[0662] In addition, the pharmaceutical composition of the present invention can be used in combination with other therapeutic agents or therapeutic techniques. Examples of other therapeutic agents or therapeutic techniques include, but are not particularly limited to, chemotherapeutic agents, radiation therapy, and biopharmaceutical products. The pharmaceutical composition of the present invention and other therapeutic agents can be administered simultaneously or according to a different schedule in the form of a single formulation or two or more different formulations containing the pharmaceutical composition of the present invention. Administration of the pharmaceutical composition can be performed in combination with other therapeutic techniques.

EXAMPLES

[0663] Hereafter, the present invention is described in greater detail with reference to the examples, although the present invention is not limited to these examples.

[0664] In the following examples, genetic engineering procedures were performed in accordance with the method described in Molecular Cloning (Sambrook, J., Fritsch, E. F. and Maniatis, T., Cold Spring Harbor Laboratory Press, 1989) and methods described in other experimental protocols employed by a person skilled in the art, unless otherwise specified. When a commercially available reagent or kit was to be used, the procedures in accordance with the instructions of such a commercial product were employed. Synthesis of primers required for gene synthesis or vector construction was outsourced, according to need (Fasmac Co., Ltd., Thermo Fisher Scientific, and Eurofins Genomics).

(Example 1) Acquisition of Human Antibody Phage Library-Derived Anti-HLA/NY-ESO Antibody

1)-1 Preparation of HLA/NY-ESO Antigen Protein

[0665] Inclusion bodies prepared from *E. coli* cells (BL21 (DE3), Agilent Technologies) expressing the HLA-A*0201 (GenBank: ASA47534.1) truncate (biotin ligase recognition sequence added: SEQ ID NO: 33) and β 2-microglobulin (UniProtKB-P61769: SEQ ID NO: 34) and the NY-ESO peptide: SLLMWITQC (SEQ ID NO: 1 of Sequence Listing) were subjected to refolding at a high dilution factor, and the HLA-A*0201/ β 2-microglobulin/NY-

ESO peptide complexed (hereafter, referred to as “HLA/NY-ESO”) was then prepared using gel filtration columns (Superdex 200 10/300, GE Healthcare).

[0666] As a negative control antigen, the HLA-A*0201/β2-microglobulin/MAGEC-1 peptide complex (hereafter, referred to as “HLA/MC1”) was prepared by refolding using the MAGEC-1 peptide: ILFGISLREV (SEQ ID NO: 2 of Sequence Listing). Further, the prepared HLA/NY-ESO and HLA/MC1 were biotinylated with *E. coli* biotin ligase and then fractionated using gel filtration columns (Superdex 200 10/300, GE Healthcare) to prepare biotinylated proteins.

1)-2 Isolation of scFv Capable of Binding to HLA/NY-ESO

[0667] scFv binding to HLA/NY-ESO was isolated from the human antibody phage library. At the outset, phages were added to the biotinylated HLA/MC1-immobilized Dynabeads Streptavidin M-280 (Thermo Fisher Scientific), and unbound phages were collected. Subsequently, phages were added to the biotinylated HLA/NY-ESO-immobilized Dynabeads Streptavidin M-280Ag, and unbound phages were removed via washing using a magnet stand (DynaMag-2, Thermo Fisher Scientific).

[0668] Thereafter, *E. coli* (XL-1 Blue, Agilent Technologies) was infected with phages bound to HLA/NY-ESO, and the phages were collected and amplified. After a panning procedure was performed 3 times in total, the phages were transferred from the polyclonal phagemid to the *E. coli* expression vector comprising FLAG tag and a His tag added to the carboxyl terminus of scFv, the *E. coli* was transformed, scFv was expressed in the presence of IPTG (isopropyl-β-D-thiogalactopyranoside) (Sigma-Aldrich), and the expressed scFv was subjected to screening via ELISA.

1)-3 Screening of HLA/NY-ESO-Bound scFv Via ELISA

[0669] NeutrAvidin (Life Technologies) diluted to 1 μg/ml in PBS (0.01 M phosphate buffered saline containing 0.138 M sodium chloride and 0.0027 M potassium chloride (pH7.4), Sigma-Aldrich) was added in an amount of 50 μl each to a 384-well Maxi-sorp plate (Black, Nunc), and the plate was allowed to stand at 4° C. overnight for immobilization. The plate was washed 3 times with PBS (ELISA buffer) containing 0.05% Tween-20 (BioRad), biotinylated HLA/NY-ESO, which was diluted to 1 μg/ml in PBS and used in Example 1)-2, was added, and the resultant was agitated at room temperature for 1 hour. The plate was washed 3 times with ELISA buffer, blocked with Blocker Casein (Thermo Fisher Scientific), and then washed 3 times with ELISA buffer. Thereafter, a culture solution of scFv-expressing *E. coli* was added, and the reaction was allowed to proceed at room temperature for 2 hours. After the resultant was washed 3 times with ELISA buffer, 50 μl of the horseradish peroxidase (HRP)-labeled anti-FLAG antibody (Sigma-Aldrich) diluted to 5,000-fold with ELISA buffer was added, and the reaction was allowed to proceed at room temperature for 1 hour. After the resultant was washed 5 times with ELISA buffer, the SuperSignal Pico ELISA Chemiluminescent substrate (Thermo Fisher Scientific) was added, chemiluminescence 10 minutes later was assayed using a plate reader (Envision 2104 Multilabel Reader, Perkin Elmer), and positive clones in HLA/NY-ESO-binding ELISA were then selected.

1)-4 Determination of Nucleotide Sequence and Amino Acid Sequence of ELISA-Positive Clone NY-R119

[0670] From among the ELISA-positive clones obtained in 1)-3, NY-R119 was selected as scFv exhibiting high binding affinity to HLA/NY-ESO and excellent recognition specificity. The nucleotide sequences of the heavy chain and light chain variable regions of NY-R119 were analyzed by the Dye Terminator method (BigDye®, Terminator v3.1, Thermo Fisher Scientific). Sequences of the primers used for sequence analysis are as demonstrated below.

[0671] Primer A: 5'-CTCTTCGCTATTACGCCAGCTGGCGA-3' (SEQ ID NO: 3 of Sequence Listing (FIG. 10))

[0672] Primer B: 5'-ATAACAATTTACACAGGAAACAGCTATGA-3' (SEQ ID NO: 4 of Sequence Listing (FIG. 11))

[0673] SEQ ID NO: 5 (FIG. 12) shows the determined nucleotide sequence of cDNA encoding the heavy chain variable region of NY-R119 and SEQ ID NO: 6 (FIG. 13) shows the determined amino acid sequence thereof.

[0674] SEQ ID NO: 7 (FIG. 14) shows the determined nucleotide sequence of cDNA encoding the light chain variable region of NY-R119 and SEQ ID NO: 8 (FIG. 15) shows the determined amino acid sequence thereof.

[0675] The CDR sequences of NY-R119 in accordance with the definition of CDR provided by IMGT are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 56 (FIG. 61); CDRL1: SEQ ID NO: 57 (FIG. 61); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 59 (FIG. 61).

1)-5 Preparation of NYA-0001

1)-5-1 Construction of NYA-0001 Expression Vector

[0676] In order to prepare various evaluation samples, an NY-R119 expression vector for mammalian cells was constructed. NY-R119 expressed in a cultured mammalian cell was designated as “NYA-0001.” The amino acid sequences constituting the scFv region, the heavy chain and light chain variable regions, and CDRH1 to 3 and CDRL1 to 3 of NY-R119 are identical to those of NYA-0001. A DNA fragment encoding NYA-0001 was inserted into an expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone using the In-Fusion HD cloning kit (CLONTECH) to construct the NYA-0001 expression vector.

[0677] The nucleotide sequence of the constructed NYA-0001 expression vector was reanalyzed, and the nucleotide

sequence of the full-length NYA-0001 was found to be the nucleotide sequence as shown in SEQ ID NO: 69 of Sequence Listing (FIG. 70). On the basis of the nucleotide sequence, in addition, the amino acid sequence of the entire NYA-0001 encoded thereby was found to be the amino acid sequence as shown in SEQ ID NO: 70 (FIG. 71). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-0001, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

1)-5-2 Expression and Purification of NYA-0001

[0678] Expi293F cells (Thermo Fisher Scientific) were subcultured in accordance with the instructions. A culture solution of the Expi293F cells in a logarithmic growth phase was diluted to 2.5×10^6 cells/ml in the Expi293 Expression medium (Thermo Fisher Scientific) and used for NYA-0001 production. The NYA-0001 expression vector (0.3 mg) and 0.9 mg of polyethyleneimine (Polyscience #24765) were added to 20 ml of Opti-Pro SFM medium (Thermo Fisher Scientific). The mixture was stirred gently, allowed to stand for five minutes, and then added to the Expi293F cells. The culture supernatant obtained by agitation culture in an incubator at 37° C. in the presence of 8% CO₂ at 135 rpm for 6 days was filtered through a 0.2 µm-filter (Millipore). Thus, the culture supernatant of NYA-0001 was obtained. Purification was carried out by elution and concentration using Ni Sepharose excel (GE Healthcare) and then filtration through a gel filtration column (Superdex 200 Increase, GE Healthcare) equilibrated with 25 mM histidine, 300 mM NaCl, and 5% Sorbitol, pH6.0. Purified protein samples were subjected to analytical size exclusion chromatography (SEC), the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

(Example 2) Preparation of NYA-0001 Mutant

2)-1 Acquisition of NYA-0001 Mutant

[0679] A phage library was constructed by a method in which the NYA-0001 gene was used as a template and a mutation was introduced via PCR (i.e., the error-prone-based library) or a method in which an oligomer was synthesized through random mutation of 20 types of amino acids for each of all CDR residues (i.e., the oligo-based library), clones with high binding affinity were screened for, NYA-0060, NYA-0068, and NYA-0082 were obtained as mutants with high binding affinity, and the nucleotide sequences thereof were then determined.

[0680] SEQ ID NO: 9 (FIG. 16) shows the nucleotide sequence of cDNA encoding the heavy chain variable region of NYA-0060 and SEQ ID NO: 10 (FIG. 17) shows the amino acid sequence thereof.

[0681] SEQ ID NO: 11 (FIG. 18) shows the nucleotide sequence of cDNA encoding the light chain variable region of NYA-0060 and SEQ ID NO: 12 (FIG. 19) shows the amino acid sequence thereof.

[0682] SEQ ID NO: 13 (FIG. 20) shows the nucleotide sequence of cDNA encoding the heavy chain variable region of NYA-0068 and SEQ ID NO: 14 (FIG. 21) shows the amino acid sequence thereof.

[0683] SEQ ID NO: 15 (FIG. 22) shows the nucleotide sequence of cDNA encoding the light chain variable region of NYA-0068 and SEQ ID NO: 16 (FIG. 23) shows the amino acid sequence thereof.

[0684] SEQ ID NO: 17 (FIG. 24) shows the nucleotide sequence of cDNA encoding the heavy chain variable region of NYA-0082 and SEQ ID NO: 18 (FIG. 25) shows the amino acid sequence thereof.

[0685] SEQ ID NO: 19 (FIG. 26) shows the nucleotide sequence of cDNA encoding the light chain variable region of NYA-0082 and SEQ ID NO: 20 (FIG. 27) shows the amino acid sequence thereof.

2)-2 Preparation of Mutants with High Binding Affinity Using the Sites of Mutation and Combination Identified in NYA-0060, NYA-0068, and NYA-0082

[0686] DNA fragments encoding NYA-1163 and NYA-2023 with sites of mutation of NYA-0060 and NYA-0068 were inserted into an expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone using the In-Fusion HD cloning kit (CLONTECH) to construct the scFv expression vectors for mammalian cell culture.

[0687] According to the sites of mutation and combination thereof identified in NYA-0060, NYA-0068, and NYA-0082, so NYA-2027, NYA-1143, and NYA-2143 were designed. DNA fragments encoding target scFv were inserted into the NYA-0001 expression vector constructed in 1)-5 via site-directed mutagenesis or an expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone using the In-Fusion HD cloning kit (CLONTECH) to construct the scFv expression vectors for mammal cell culture.

[0688] The nucleotide sequences of the constructed scFv expression vectors were reanalyzed, and the nucleotide sequence of the full-length NYA-1163 was found to be the nucleotide sequence as shown in SEQ ID NO: 21 of Sequence Listing (FIG. 28).

[0689] The nucleotide sequence of the full-length NYA-2023 was found to be the nucleotide sequence as shown in SEQ ID NO: 22 of Sequence Listing (FIG. 29).

[0690] The nucleotide sequence of the full-length NYA-2027 was found to be the nucleotide sequence as shown in SEQ ID NO: 23 of Sequence Listing (FIG. 30).

[0691] The nucleotide sequence of the full-length NYA-1143 was found to be the nucleotide sequence as shown in SEQ ID NO: 24 of Sequence Listing (FIG. 31).

[0692] The nucleotide sequence of the full-length NYA-2143 was found to be the nucleotide sequence as shown in SEQ ID NO: 25 of Sequence Listing (FIG. 32).

[0693] On the basis of the nucleotide sequences above, in addition, the amino acid sequences of the full-length NYA-1163, NYA-2023, NYA-2027, NYA-1143, and NYA-2143 encoded thereby were identified.

[0694] The amino acid sequence of the full-length NYA-1163 is the amino acid sequence as shown in SEQ ID NO: 26 of the Sequence Listing (FIG. 33). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-1163, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0695] The amino acid sequence of the full-length NYA-2023 is the amino acid sequence as shown in SEQ ID NO: 27 of the Sequence Listing (FIG. 34). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2023, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0696] The amino acid sequence of the full-length NYA-2027 is the amino acid sequence as shown in SEQ ID NO: 28 of the Sequence Listing (FIG. 35). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2027, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0697] The amino acid sequence of the full-length NYA-1143 is the amino acid sequence as shown in SEQ ID NO: 26 of the Sequence Listing (FIG. 36). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-1143, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0698] The amino acid sequence of the full-length NYA-2143 is the amino acid sequence as shown in SEQ ID NO: 30 of the Sequence Listing (FIG. 37). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2143, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0699] All the CDR sequences of NYA-1163 are identical to the CDR sequences of NYA-0001. The CDR sequences of NYA-1143, NYA-2143, and NYA-2023 are identical to each other, and these sequences are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 56 (FIG. 61); CDRL1: SEQ ID NO: 60 (FIG. 62); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 59 (FIG. 61). The CDR sequences of NYA-2027 are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 56 (FIG. 61); CDRL1: SEQ ID NO: 57 (FIG. 61); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 61 (FIG. 63).

[0700] Also, NYA-1154 comprising sites of binding and mutation observed at the time of screening of clones with high binding affinity in combination was designed and introduced into the NYA-0001 expression vector via site-directed mutagenesis to construct the NYA-1154 expression vector comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone. The nucleotide sequences of the constructed scFv expression vectors were reanalyzed and found to be the nucleotide sequence as shown in SEQ ID NO: 31 (FIG. 38). On the basis of the nucleotide sequences above, the amino acid sequence of the full-length NYA-1154 encoded thereby was identified (SEQ ID NO: 32 (FIG. 39)). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-1154, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0701] The CDR sequences of NYA-1154 are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 62 (FIG. 64); CDRL1: SEQ ID NO: 57 (FIG. 61); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 63 (FIG. 64).

[0702] NYA-1163, NYA-2023, NYA-2027, NYA-1143, NYA-2143, and NYA-1154 were expressed in the same manner as in 1)-5-2 to purify the target scFv constructs. Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various assays.

2)-3 Preparation of NYA-1143 Mutant

2)-3-1 Preparation of NYA-2035

[0703] NYA-2035 was designed as an NYA-1143 mutant and introduced into the NYA-1143 expression vector for mammal cell culture via site-directed mutagenesis to construct an NYA-2035 expression vector. The nucleotide sequence of the constructed scFv expression vector was reanalyzed and found to be the nucleotide sequence as shown in SEQ ID NO: 35 (FIG. 42). On the basis of the nucleotide sequences above, the nucleotide sequence of the full-length NYA-2025 encoded thereby was determined (SEQ ID NO: 36 (FIG. 43)). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2035, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3. The CDR sequences of NYA-2035 in accordance with the definition of CDR provided by IMGT are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 56 (FIG. 61); CDRL1: SEQ ID NO: 64 (FIG. 65); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 59 (FIG. 61).

2)-3-2 Preparation of NYA-1143 CDR Graft Mutant

[0704] In order to improve physical properties, the NYA-1143 CDR graft mutant was designed. The VH and VL framework sequences of NYA-1143 were compared with the framework sequences of the human subgroup consensus sequence or the germline sequence defined in KABAT et al. (Sequences of Proteins of Immunological Interest, 5th Ed., Public Health Service National Institutes of Health, Bethesda, MD., 1991). As a result, a consensus sequence between the human germline sequence IGHV3_30*15 and the human γ chain subgroup 3 was selected for VH and the human germline sequence IGLV1-44*01 was selected for VL as acceptors having a high identity in the framework regions. Subsequently, amino acid residues in the framework region of an acceptor were aligned with the amino acid residues of NYA-1143 to identify different amino acid residues.

[0705] Subsequently, a three-dimensional model of NYA-1143 was used to select framework residues to be transferred onto an acceptor with reference to the criteria provided by Queen et al. (Proc. Natl. Acad. Sci. U.S.A., 86, 10029-10033, 1989). In accordance with the technique described above, CDR graft mutant amino acid sequences of the VH regions of NYA-1143; i.e., NYA-1143-VH01, NYA-1143-VH02, and NYA-1143-VH03, were designed. Such amino acid sequences are as shown in SEQ ID NO: 37 (FIG. 44), SEQ ID NO:38 (FIG. 45), and SEQ ID NO:39 (FIG. 46). As a CDR graft mutant amino acid sequence of a VL region of NYA-1143, NYA-1143-VLO1 was designed. Such an amino acid sequence is as shown in SEQ ID NO: 40 (FIG. 47).

[0706] Various scFv constructs were designed using the VH and VL sequences in combination. scFv resulting from substitution of the VH region of NYA-1143 with the amino acid sequence of NYA-1143-VH01 was designated as NYA-2044. scFv resulting from substitution of the VL region of NYA-2044 with the amino acid sequence of NYA-1143-VLO1 was designated as NYA-2045.

[0707] scFv resulting from substitution of the VH region of NYA-1143 with the amino acid sequence of NYA-1143-VH02 was designated as NYA-2047. scFv resulting from substitution of the VL region of NYA-2047 with the amino acid sequence of NYA-1143-VLO1 was designated as NYA-2048.

[0708] scFv resulting from substitution of the VH region of NYA-1143 with the amino acid sequence of NYA-1143-VH03 was designated as NYA-2060. scFv resulting from substitution of the VL region of NYA-2060 with the amino acid sequence of NYA-1143-VLO1 was designated as NYA-2061.

[0709] The DNA fragments of the various designed NYA-1143 CDR graft mutants were fully synthesized (Fasmac Co., Ltd.) and bound to each other using the In-Fusion HD Cloning kit (CLONTECH) to construct the scFv expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone.

[0710] The nucleotide sequences of the constructed scFv expression vectors were reanalyzed, the nucleotide sequence of the full-length NYA-2044 was found to be the nucleotide sequence as shown in SEQ ID NO: 41 of the Sequence Listing (FIG. 48).

[0711] The nucleotide sequence of the full-length NYA-2045 was found to be the nucleotide sequence as shown in SEQ ID NO: 42 of the Sequence Listing (FIG. 49).

[0712] The nucleotide sequence of the full-length NYA-2047 was found to be the nucleotide sequence as shown in SEQ ID NO: 43 of the Sequence Listing (FIG. 50).

[0713] The nucleotide sequence of the full-length NYA-2048 was found to be the nucleotide sequence as shown in SEQ ID NO: 44 of the Sequence Listing (FIG. 51).

[0714] The nucleotide sequence of the full-length NYA-2060 was found to be the nucleotide sequence as shown in SEQ ID NO: 45 of the Sequence Listing (FIG. 52).

[0715] The nucleotide sequence of the full-length NYA-2061 was found to be the nucleotide sequence as shown in SEQ ID NO: 46 of the Sequence Listing (FIG. 53).

[0716] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, and NYA-2061 encoded thereby were identified. The CDR sequences of NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, and NYA-2061 are identical to those of NYA-1143.

[0717] The amino acid sequence of the full-length NYA-2044 is the amino acid sequence as shown in SEQ ID NO:

47 of the Sequence Listing (FIG. 54). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2044, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0718] The amino acid sequence of the full-length NYA-2045 is the amino acid sequence as shown in SEQ ID NO: 48 of the Sequence Listing (FIG. 55). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2045, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0719] The amino acid sequence of the full-length NYA-2047 is the amino acid sequence as shown in SEQ ID NO: 50 of the Sequence Listing (FIG. 57). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2047, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0720] The amino acid sequence of the full-length NYA-2048 is the amino acid sequence as shown in SEQ ID NO: 51 of the Sequence Listing (FIG. 58). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2048, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0721] The amino acid sequence of the full-length NYA-2060 is the amino acid sequence as shown in SEQ ID NO: 52 of the Sequence Listing (FIG. 59). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2060, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0722] The amino acid sequence of the full-length NYA-2061 is the amino acid sequence as shown in SEQ ID NO: 53 of the Sequence Listing (FIG. 60). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2061, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

2)-3-3 Expression and Purification of NYA-1143 Mutants

[0723] NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, and NYA-2061 prepared in 2)-3-1 and 2)-3-2 were expressed in the same manner as in 1)-5-2 to purify the target scFv constructs. Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various assays.

(Example 3) Preparation of scFv of Anti-HLA/NY-ESO Reference Antibody

[0724] The scFv constructs of the antibodies 3M4E5 and T1 with high binding affinity to HLA/NY-ESO (WO2010/106431) were designed; 3M4E5 scFv was designated as NYC-0003, and T1 scFv was designated as NYC-0004.

[0725] The DNA fragments of NYC-0003 and NYC-0004 were fully synthesized (Thermo Fisher Scientific), and the scFv expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone was constructed using the In-Fusion HD Cloning kit (CLONTECH).

[0726] The nucleotide sequences of the constructed scFv expression vectors were reanalyzed, and the nucleotide sequences of the full-length NYC-0003 and NYC-0004 were found to be the nucleotide sequences as shown in SEQ ID NO: 65 of the Sequence Listing (FIG. 66) and SEQ ID NO: 66 thereof (FIG. 67). On the basis of the nucleotide sequences, the amino acid sequence of the full-length NYC-0003 encoded thereby was found to be the amino acid sequence as shown in SEQ ID NO: 67 (FIG. 68), and the amino acid sequence of the full-length NYC-0004 was found to be the amino acid sequence as shown in SEQ ID NO: 68 (FIG. 69).

[0727] NYC-0003 and NYC-0004 were expressed in the same manner as in 1)-5-2 to purify the target scFv constructs. Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various assays.

(Example 4) Evaluation of Binding Affinity to HLA/NY-ESO Using Biacore

[0728] With the use of Biacore T200, anti-HLA/NY-ESO scFv was captured as a ligand by the immobilized anti-His antibody, and the antigen was assayed as an analyte. As the antigen, HLA/NY-ESO prepared in 1)-1 was used. The

anti-His antibody kit, GE Healthcare) was immobilized on Sensor Chip CM5 (GE Healthcare) in accordance with the instructions of the kit. The anti-HLA/NY-ESO scFv constructs diluted to 0.5 µg/ml in HBS-EP+ (GE Healthcare) to be evaluated were brought into contact therewith at 10 µl/min for 60 seconds for immobilization. Thereafter, the samples were added to the HLA/NY-ESO analytes diluted to various levels with HBS-EP+ at a flow rate of 30 µl/min for 120 seconds and dissociation was assayed for 600 seconds. The results of the calculation obtained by such single cycle kinetics analysis, K_{sub}D, are shown in Table 1. In comparison with the parent antibody NYA-0001, all the mutants thereof were found to exhibit higher binding affinity to HLA/NY-ESO. In comparison with NYC-0004 NYA-1143, NYA-2023, NYA-2143, NYA-2044, NYA-2045, NYA-2060, and NYA-2061 were, in addition, found to exhibit higher binding affinity, and, in particular, NYA-1143, NYA-2044, NYA-2045, and NYA-2143 exhibited K_{sub}D of 1 nM or lower.

TABLE-US-00001

Clone name	K _{sub} D (nM)
NYA-0001	33.8
NYA-1143	1.0
NYA-1163	13.0
NYA-2023	2.4
NYA-2027	11.2
NYA-2035	3.2
NYA-2044	0.8
NYA-2045	0.8
NYA-2047	5.3
NYA-2048	6.3
NYA-2060	1.1
NYA-2061	1.5
NYA-2143	0.9
NYC-0003	8.1
NYC-0004	2.8

(Example 5) Analysis of Recognized Amino Acid in NY-ESO Peptide of anti-HLA/NY-ESO scFv

[0729] The concentration of the T2 human lymphoblast fusion cells (ATCC) was adjusted to an adequate level in AIM-V medium (Thermo Fisher Scientific) containing 20% FBS, a solution of the NY-ESO peptide (SEQ ID NO: 1), the point-mutant NY-ESO peptides 1F, 2M, 3A, 4A, 5A, 6L, 7F, 8A, and 9A (SEQ ID NO: 121 (FIG. 122), SEQ ID NO: 122 (FIG. 123), SEQ ID NO: 123 (FIG. 124), SEQ ID NO: 124 (FIG. 125), SEQ ID NO: 125 (FIG. 126), SEQ ID NO: 126 (FIG. 127), SEQ ID NO: 127 (FIG. 128), SEQ ID NO: 128 (FIG. 129), and SEQ ID NO: 129 (FIG. 130)), or the gp100 peptide (SEQ ID NO: 130 (FIG. 131)) (Sigma Genosys) dissolved to 5 mM in DMSO was added to result in a final concentration of 50 µM, or DMSO was added in an amount of 1/100, and the resulting mixture was incubated at 37° C. for 4 hour. The resultant was washed two times in the AIM-V medium containing 20% FBS, the concentration thereof was adjusted to an adequate level with PBS containing 5% FBS, the LIVE/DEAD Fixable Dead Cell Stain Kit (Thermo Fisher Scientific) was added thereto, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, the washed cells were divided into two groups, the cells in PBS containing 5% FBS were seeded on a 96-well U-bottom microplate at 10^{sup}.5 cells/well, and the plate was subjected to centrifugation, followed by removal of the supernatant. Anti-HLA/NY-ESO scFv diluted to 100 nM with PBS containing 5% FBS was added at 25 µl/well to one of the divided cells, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, Penta-His Alexa Fluor488 (QIAGEN) diluted in PBS containing 5% FBS was added at 25 µl/well, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, Anti-mouse-IgG Alexa Fluor488 (Thermo Fisher Scientific) diluted in PBS containing 5% FBS was added at 25 µl/well, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, immobilized with Mildform 10 N (FUJIFILM Wako Pure Chemical Corporation) overnight, and resuspended in PBS containing 5% FBS. For standardization with the amount of the HLA/peptide complex, the HLA-A2 antibody BB7.2-Alexa Fluor 488 diluted to 10 µg/ml in PBS containing 5% FBS or mouse IgG2b-Alexa Fluor 488 was added at 25 µl/well to the other group of cells, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, immobilized with Mildform 10 N (FUJIFILM Wako Pure Chemical Corporation) overnight, and resuspended in PBS containing 5% FBS. The cell suspensions were subjected to detection using a flow cytometer (CantoII, Becton Dickinson). Data analysis was performed using Flowjo (Treestar), and the geometric mean fluorescence intensity (gMFI) of Alexa Fluor 488 in a T2 cell fraction from which dead cells were removed was measured. The standardized gMFI indicating the binding affinity of each scFv standardized with the amount of the HLA/peptide complex on the T2 cells was determined in accordance with the following equation.

[00001] Standardized gMFI = (A / B) / ((C / D) / (E / F)) [0730] A/B=Relative gMFI [0731] A: gMFI of T2 cells supplemented with DMSO or peptide in the presence of antibody [0732] B: gMFI of T2 cells supplemented with DMSO or peptide [0733] (C/D)/(E/F)=Corrected amount of HLA/peptide complex of T2 cells supplemented with DMSO or peptide [0734] C: gMFI of T2 cells supplemented with DMSO or peptide in the presence of HLA-A2 antibody [0735] D: gMFI of T2 cells supplemented with DMSO or peptide in the presence of mouse IgG2b antibody or peptide [0736] E: gMFI of T2 cells supplemented with DMSO containing HLA-A2 antibody [0737] F: gMFI of T2 cells supplemented with DMSO containing mouse IgG2b antibody

[0738] As shown in FIG. 1, anti-HLA/NY-ESO scFv: NYA-0001, 1143, 2044, 2045, 2047, 2048, 2060, and 2061, exhibited binding affinity to T2 cells supplemented with peptides comprising point mutations introduced into amino acids 1, 4, 5, and 7 decreased to a half or lower, compared with the wild-type NY-ESO peptide. This indicates that such scFv recognizes amino acids 1, 4, 5, and 7 of the NY-ESO peptide. This also indicates that NYA-1154 recognizes amino acids 1 and 5, NYA-1163 recognizes amino acids 1, 3, 4, 5, 6, and 7, NYA-2023, 2027, and 2035 recognize amino acids 1, 4, and 5, and NYA-2143 recognizes amino acids 1, 5, and 7. This further indicates that NYC-0003 and 0004 recognize amino acid 4 and 5.

(Example 6) Evaluation of Antigen-Binding Specificity of Anti-HLA/NY-ESO scFv

[0739] In order to search for a human peptide having an amino acid sequence similar but not identical to the amino

acid sequence of the NY-ESO peptide: SLLMWITQC (SEQ ID NO: 1) to which the antibody may bind (hereafter such peptide is referred to as a “homologous peptide”) from among human proteomes (Swiss-Prot), a 9-mer peptide comprising amino acids 1, 4, and 5 that are consistent with those of peptides recognized by the antibody was searched. The searched 9-mer peptide was analyzed using NetMHCpan2.8 concerning binding affinity to HLA-A0201, and IC50 thereof was predicted to be 500 nM or lower. The 9-mer peptide shown in FIG. 2A was selected as a homologous peptide for evaluation of binding specificity of anti-HLA/NY-ESO scFv. The concentration of the T2 cells was adjusted to an adequate level in AIM-V medium (Thermo Fisher Scientific) containing 20% FBS, a solution of the NY-ESO peptide (SEQ ID NO: 1), the homologous peptides DOLPP1, IL20RB, PRKD2, CD163, and P2RY8 (SEQ ID NO: 131 (FIG. 132), SEQ ID NO:132 (FIG. 133), SEQ ID NO:133 (FIG. 134), SEQ ID NO:134 (FIG. 135), and SEQ ID NO:135 (FIG. 136)), or the gp100 peptide (SEQ ID NO: 130 (FIG. 131)) (Sigma Genosys) dissolved to 5 mM in DMSO was added to result in a final concentration of 50 μM, or DMSO was added in an amount of 1/100, and binding affinity of the antibodies was evaluated in the same manner as in Example 5. The standardized gMFI indicating the binding affinity of each scFv standardized with the amount of the HLA/peptide complex on the T2 cells was determined in accordance with the following equation.

[00002]StandardizedgMFI = $(A / B) / ((C / D) / (E / F))$ [0740] A/B=Relative gMFI [0741] A: gMFI of T2 cells supplemented with DMSO or peptide in the presence of antibody [0742] B: gMFI of T2 cells supplemented with DMSO or peptide [0743] (C/D)/(E/F)=Corrected amount of HLA/peptide complex of T2 cells supplemented with DMSO or peptide [0744] C: gMFI of T2 cells supplemented with DMSO containing HLA-A2 antibody or peptide [0745] D: gMFI of T2 cells supplemented with DMSO or peptide in the presence of mouse IgG2b antibody [0746] E: gMFI of T2 cells supplemented with DMSO containing HLA-A2 antibody [0747] F: gMFI of T2 cells supplemented with DMSO containing mouse IgG2b antibody

[0748] As shown in FIG. 2B, anti-HLA/NY-ESO scFv: NYA-0001, 1143, 1163, 2023, 2027, 2035, 2044, 2045, 2047, 2048, 2060, 2061, and 2143, did not bind to T2 cells supplemented with any homologous peptides. This indicates that such scFv has high specificity. In contrast, NYA-1154, NYC-0003, and 0004 were observed to have bound to T2 cells supplemented with some homologous peptides.

(Example 7) Preparation of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule

7)-1 Preparation of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule Expression Vector

7)-1-1 Preparation of taFv-Heterodimer Fc-Type Bispecific Molecule Expression Vector

[0749] In order to evaluate taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules, expression vectors for various molecules were designed. The anti-HLA/NY-ESO antibodies used were NYA-1143, NYA-2143, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, and NYA-2061. As the anti-CD3 antibody, humanized anti-CD3 scFv; i.e., C3E-7085 (WO 2018/117237), was used. As the heterodimer Fc sequence, the Fc sequence comprising a mutation introduced thereinto to lower effector functions and form a heteropolymer (WO 2014/190441) was used.

[0750] A DNA fragment encoding Fc (HC1 or HC2) comprising a mutation introduced thereinto to lower effector functions and form a heteropolymer was synthesized (Fasmac Co., Ltd.), an expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone was prepared using the In-Fusion HD cloning kit (CLONTECH), and the resulting vector was designated as “p_HC1.”

[0751] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-1143 ligated to C3E-7085 with a GGGGS linker was prepared and designated as “p_NYF-0016-HC2.”

[0752] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2143 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0016-HC2 and designated as “p_NYF-0019-HC2.”

[0753] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-1163 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-HC2 with a DNA fragment encoding NYA-1163 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0022-HC2.”

[0754] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2023 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0016-HC2 and designated as “p_NYF-0023-HC2.”

[0755] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2027 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis of an expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-0001 ligated to C3E-7085 with a GGGGS linker and designated as “p_NYF-0027-HC2.”

[0756] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2035 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0016-HC2 and designated as “p_NYF-0035-HC2.”

[0757] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2044 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-HC2 with a DNA fragment encoding NYA-2044 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0044-HC2.”

[0758] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2045 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-HC2 with a DNA fragment encoding NYA-2045 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0045-HC2.”

[0759] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2047 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-HC2 with a DNA fragment encoding NYA-2047 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0047-HC2.”

[0760] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2048 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-HC2 with a DNA fragment encoding NYA-2048 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0048-HC2.”

[0761] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2060 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0044-HC2 and designated as “p_NYF-0060-HC2.”

[0762] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2061 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0045-HC2 and designated as “p_NYF-0061-HC2.”

[0763] The nucleotide sequence of p_HC1 was reanalyzed, and the nucleotide sequence of the full-length HC1 was found to be the nucleotide sequence as shown in SEQ ID NO: 71 of the Sequence Listing (FIG. 72).

[0764] The nucleotide sequence of p_NYF-0016-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0016-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 72 of the Sequence Listing (FIG. 73).

[0765] The nucleotide sequence of p_NYF-0019-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0019-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 73 of the Sequence Listing (FIG. 74).

[0766] The nucleotide sequence of p_NYF-0022-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0022-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 74 of the Sequence Listing (FIG. 75).

[0767] The nucleotide sequence of p_NYF-0023-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0023-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 75 of the Sequence Listing (FIG. 76).

[0768] The nucleotide sequence of p_NYF-0027-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0027-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 76 of the Sequence Listing (FIG. 77).

[0769] The nucleotide sequence of p_NYF-0035-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0035-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 77 of the Sequence Listing (FIG. 78).

[0770] The nucleotide sequence of p_NYF-0044-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0044-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 78 of the Sequence Listing (FIG. 79).

[0771] The nucleotide sequence of p_NYF-0045-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0045-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 79 of the Sequence Listing (FIG. 80).

[0772] The nucleotide sequence of p_NYF-0047-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0047-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 80 of the Sequence Listing (FIG. 81).

[0773] The nucleotide sequence of p_NYF-0048-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0048-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 81 of the Sequence Listing (FIG. 82).

[0774] The nucleotide sequence of p_NYF-0060-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0060-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 82 of the Sequence Listing (FIG. 83).

[0775] The nucleotide sequence of p_NYF-0061-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0061-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 83 of the Sequence Listing (FIG.

84).

[0776] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length HC1, NYF-0016-HC2, NYF-0019-HC2, NYF-0022-HC2, NYF-0023-HC2, NYF-0027-HC2, NYF-0035-HC2, NYF-0044-HC2, NYF-0045-HC2, NYF-0047-HC2, NYF-0048-HC2, NYF-0060-HC2, and NYF-0061-HC2 encoded thereby were identified.

[0777] The amino acid sequence of the full-length HC1 is the amino acid sequence as shown in SEQ ID NO: 84 of the Sequence Listing (FIG. 85). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, and amino acids 20 to 246 constitute HC.

[0778] The amino acid sequence of the full-length NYF-0016-HC2 is as shown in SEQ ID NO: 85 of the Sequence Listing (FIG. 86). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-1143-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0779] The amino acid sequence of the full-length NYF-0019-HC2 is as shown in SEQ ID NO: 86 of the Sequence Listing (FIG. 87). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2143-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0780] The amino acid sequence of the full-length NYF-0022-HC2 is as shown in SEQ ID NO: 87 of the Sequence Listing (FIG. 88). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-1163-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0781] The amino acid sequence of the full-length NYF-0023-HC2 is as shown in SEQ ID NO: 88 of the Sequence Listing (FIG. 89). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2023-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0782] The amino acid sequence of the full-length NYF-0027-HC2 is as shown in SEQ ID NO: 89 of the Sequence Listing (FIG. 90). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2027-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0783] The amino acid sequence of the full-length NYF-0035-HC2 is as shown in SEQ ID NO: 90 of the Sequence Listing (FIG. 91). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2035-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0784] The amino acid sequence of the full-length NYF-0044-HC2 is as shown in SEQ ID NO: 91 of the Sequence Listing (FIG. 92). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2044-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0785] The amino acid sequence of the full-length NYF-0045-HC2 is as shown in SEQ ID NO: 92 of the Sequence Listing (FIG. 93). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2045-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0786] The amino acid sequence of the full-length NYF-0047-HC2 is as shown in SEQ ID NO: 93 of the Sequence Listing (FIG. 94). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2047-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0787] The amino acid sequence of the full-length NYF-0048-HC2 is as shown in SEQ ID NO: 94 of the Sequence Listing (FIG. 95). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2048-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0788] The amino acid sequence of the full-length NYF-0060-HC2 is as shown in SEQ ID NO: 95 of Sequence Listing (FIG. 96). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2060-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0789] The amino acid sequence of the full-length NYF-0061-HC2 is as shown in SEQ ID NO: 96 of the Sequence Listing (FIG. 97). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2061-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

7)-1-2 Preparation of taFv-Fab-Heterodimer Fc-Type Bispecific Molecule Expression Vector

[0790] A taFv-Fab-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule expression vector was designed. As the anti-HLA/NY-ESO antibody, NYA-0001 was used. As the anti-CD3 antibody, humanized anti-CD3

scFv; i.e., C3E-7085 (WO 2018/117237), was used. As the heterodimer Fc sequence, the Fc sequence comprising a mutation introduced thereinto to lower effector functions and form a heteropolymer (WO 2014/190441) was used. [0791] An expression vector for mammal cells comprising DNA fragments encoding the heavy chain variable region of NYA-0001, the human IgG-derived CH1 region, and a region comprising a mutation that lowers effector functions and encodes HC1-k delete integrated thereinto was prepared and designated as “p_NYA-0001-Fab-HC1-k delete.” Also, an expression vector for mammal cells comprising DNA fragments encoding the NYA-0001 light chain variable region and the human IgG-derived CL region integrated thereinto was prepared and designated as “p_NYA-0001-LC.”

[0792] The nucleotide sequence of p_NYA-0001-Fab-HC1-k delete was reanalyzed, and the nucleotide sequence of the full-length NYA-0001-Fab-HC1-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 97 (FIG. 98) of the Sequence Listing.

[0793] The nucleotide sequence of p_NYA-0001-LC was reanalyzed, and the nucleotide sequence of the full-length NYA-0001-LC was found to be the nucleotide sequence as shown in SEQ ID NO: 98 (FIG. 99) of the Sequence Listing.

[0794] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYA-0001-Fab-HC1-k delete and NYA-0001-LC encoded thereby were identified.

[0795] The amino acid sequence of the full-length NYA-0001-Fab-HC1-k delete is the amino acid sequence as shown in SEQ ID NO: 99 of the Sequence Listing (FIG. 100). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 20 to 139 constitute a variable region, and amino acids 140 to 468 constitute a constant region. Also, amino acids 45 to 52 constitute CDRH1 (SEQ ID NO: 54 (FIG. 61)), amino acids 70 to 77 constitute CDRH2 (SEQ ID NO: 55 (FIG. 61)), and amino acids 116 to 128 constitute CDRH3 (SEQ ID NO: 56 (FIG. 61)).

[0796] The amino acid sequence of the full-length NYA-0001-LC is as shown in SEQ ID NO: 100 of the Sequence Listing (FIG. 101). In such an amino acid sequence, amino acids 1 to 20 constitute a signal sequence, amino acids 21 to 131 constitute a variable region, and amino acids 132 to 237 constitute a constant region. Also, amino acids 46 to 53 constitute CDRL1 (SEQ ID NO: 57 (FIG. 61)), amino acids 71 to 73 constitute CDRL2 (SEQ ID NO: 58 (FIG. 61)), and amino acids 110 to 121 constitute CDRL3 (SEQ ID NO: 59 (FIG. 61)).

7)-2 Expression of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule

7)-2-1 Expression of taFv-Heterodimer Fc-Type Bispecific Molecule

[0797] Expi293F cells (Thermo Fisher Scientific) were subcultured in accordance with the instructions. A culture solution of the Expi293F cells in a logarithmic growth phase was diluted to 2.5×10^6 cells/ml in the Expi293 Expression medium (Thermo Fisher Scientific) and used for production of various bispecific molecules. A mixture (0.3 mg) comprising the p_NYF-0016-HC2 vector and p_HC1 at 1:1.5 and 0.9 mg of polyethyleneimine (Polyscience #24765) were added to 20 ml of Opti-Pro SFM medium (Thermo Fisher Scientific). The mixture was stirred gently, allowed to stand for five minutes, and then added to the Expi293F cells. The culture supernatant obtained by agitation culture in an incubator at 37° C. in the presence of 8% CO₂ at 135 rpm for 6 days was filtered through a 0.2 µm-filter (Millipore). Thus, the culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0016) was obtained. The amino acid sequences obtained via expression of vectors constituting NYF-0016 are shown in SEQ ID NO: 85 of the Sequence Listing (FIG. 86) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0798] In the same manner, the culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0019) was prepared using p_NYF-0019-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0019 are shown in SEQ ID NO: 86 of Sequence Listing (FIG. 87) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0799] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0022) was prepared using p_NYF-0022-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0022 are shown in SEQ ID NO: 87 of the Sequence Listing (FIG. 88) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0800] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0023) was prepared using p_NYF-0023-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0023 are shown in SEQ ID NO: 88 of the Sequence Listing (FIG. 89) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0801] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0027) was prepared using p_NYF-0027-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0027 are shown in SEQ ID NO: 89 of the Sequence Listing (FIG. 90) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0802] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0035) was prepared using p_NYF-0035-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0035 are shown in SEQ ID NO: 90 of the Sequence Listing (FIG. 91) and SEQ ID NO:

84 of the Sequence Listing (FIG. 85).

[0803] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0044) was prepared using p_NYF-0044-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0044 are shown in SEQ ID NO: 91 of the Sequence Listing (FIG. 92) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0804] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0045) was prepared using p_NYF-0045-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0045 are shown in SEQ ID NO: 92 of the Sequence Listing (FIG. 93) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0805] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0047) was prepared using p_NYF-0047-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0047 are shown in SEQ ID NO: 93 of the Sequence Listing (FIG. 94) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0806] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0048) was prepared using p_NYF-0048-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0048 are shown in SEQ ID NO: 94 of the Sequence Listing (FIG. 95) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0807] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0060) was prepared using p_NYF-0060-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0060 are shown in SEQ ID NO: 95 of the Sequence Listing (FIG. 96) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0808] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0061) was prepared using p_NYF-0061-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0061 are shown in SEQ ID NO: 96 of the Sequence Listing (FIG. 97) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

7)-2-2 Expression of taFv-Fab-heterodimer Fc-type bispecific molecules

[0809] In the same manner as in 7)-2-1, the culture supernatant of the taFv-Fab-heterodimer Fc-type bispecific molecule (NYF-0058) was prepared using a vector mixture comprising p_NYF-0023-HC2, p_NYA-0001-Fab-HC1-k delete, and p_NYA-0001-LC at 1:1:1.5. The amino acid sequences obtained via expression of vectors constituting NYF-0058 are shown in amino acids 20 to 745 of SEQ ID NO: 88 of the Sequence Listing (FIG. 89), amino acids 20 to 468 of SEQ ID NO: 99 of the Sequence Listing (FIG. 100), and amino acids 21 to 237 of SEQ ID NO: 100 of the Sequence Listing (FIG. 101).

7)-3 Purification of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule

[0810] Various bispecific molecules were purified from the culture supernatants obtained in 7)-2 by 2 steps of protein A affinity chromatography and gel filtration chromatography.

[0811] The culture supernatant was applied to the MabSelectSuRe column equilibrated with PBS at pH 7.4 (GE Healthcare Bioscience, also referred to simply as "GE Healthcare") to allow the target bispecific molecules to adsorb thereto. After the non-adsorbed components were removed by PBS, the adsorbed components were eluted using acetate buffer (pH 3.5). The elution fraction was neutralized with the aid of Tris buffer (pH 9.5), concentrated, and then applied to the gel filtration column Superdex 200 10/300 (GE Healthcare Bioscience) equilibrated with 25 mM histidine, 300 mM NaCl, 5% Sorbitol at pH 5.5 in advance. From the peak fraction obtained via gel filtration chromatography, fractions equivalent to the target heterodimer were collected, and generation of the target anti-HLA/NY-ESO-anti-CD3 bispecific molecule was confirmed via SDS-polyacrylamide electrophoresis (SDS-PAGE). Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

(Example 8) Evaluation of Cytotoxicity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecule

8)-1 Preparation of Target Cells

[0812] The endogenous human NY-ESO-expressing cells (U266B1 and NCI-H1703) and the endogenous human NY-ESO-non-expressing cells (AGS and CFPAC-1) were adjusted to 1×10^6 cells/ml in RPMI 1640 medium containing 10% FBS (FUJIFILM Wako Pure Chemical Corporation), 100 μ l of Chromium-51 Radionuclide (PerkinElmer) was added per 1 ml of each cell suspension, and the resultants were cultured at 37° C. in the presence of 5% CO₂ for 2 hours. The cells were washed 2 times in RPMI 1640 medium containing 10% FBS, resuspended to 1×10^5 cells/ml in RPMI 1640 medium containing 10% FBS, and then used as the target cells.

8)-2 Preparation of Effector Cells

[0813] Commercially available frozen human PBMCs (Cellular Technology Limited) were thawed at 37° C., transferred to a solution comprising RPMI 1640 medium containing 10% FBS supplemented with the Anti-aggregate Wash reagent (Cellular Technology Limited), washed 2 times, and adjusted to 1×10^6 cells/ml in RPMI 1640 medium containing 10% FBS to prepare effector cells.

8)-3 Cytotoxicity Assay

[0814] The target cells obtained in 8)-1 were seeded on a 96-well U-bottom microplate at 50 µl/well. Various Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules prepared in Example 7 adjusted to various concentrations were added thereto at 50 µl/well, the effector cells prepared in Example 8)-2 were added thereto at 100 l/well, centrifugation was carried out at room temperature and 1,000 rpm for 1 minute, and then they were incubated at 37° C. in the presence of 5% CO.sub.2 for 20 to 24 hours. The supernatant (50 µl) was collected on a LumaPlate (PerkinElmer), dried at 50° C. for approximately 2 hours, and then assayed using a plate reader (TopCount, PerkinElmer). The test was performed in triplicate, and the rate of cell lysis was determined in accordance with the following equation.

[00003]Rateofcelllysis(%) = (A - B) / (C - B) × 100 [0815] A: Sample well count [0816] B: Mean of background (wells not supplemented with antibody) count (n=3).

[0817] When the antibody was added, 50 µl of an assay medium was added. Other procedures were the same as those for the sample wells. [0818] C: Mean of maximal release (wells in which target cells were lysed with a surfactant) count (n=3).

[0819] When the antibody was added, 50 µl of an assay medium was added. A surfactant was added in an amount of 100 µl, a fraction of 50 µl was transferred to the LumaPlate as in the case of the sample wells, and assays were then performed.

[0820] As shown in FIG. 4A to FIG. 4F, various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules exerted cytotoxicity on the endogenous human NY-ESO-expressing cells (U266B1 and NCI-H1703). The results of calculation of EC.sub.50 obtained using the Four Parameter Logistic Curve equation of analytical software (Sigmaplot, version 12.0) concerning the U266B1 cells are shown in Table 2, and the results concerning the NCI-H1703 cells are shown in Table 3. "Not Applicable (NA)" indicates that it is impossible to perform curve fitting because the correlation coefficient (R) was not calculated. As shown in FIG. 4G to FIG. 4L, in contrast, no cytotoxicity was observed on the endogenous human NY-ESO-non-expressing cells (AGS and CFPAC-1).

TABLE-US-00002 TABLE 2 Name EC.sub.50 (nM) 1 NYF-0016 0.30 2 NYF-0019 0.20 3 NYF-0022 1.30 4 NYF-0023 0.90 5 NYF-0027 2.68 6 NYF-0035 0.97 7 NYF-0044 0.10 8 NYF-0045 0.18 9 NYF-0047 0.93 10 NYF-0048 1.10 11 NYF-0058 0.95 12 NYF-0060 0.39 13 NYF-0061 0.40

TABLE-US-00003 TABLE 3 Name EC.sub.50 (nM) 1 NYF-0016 0.90 2 NYF-0019 4.28 3 NYF-0022 NA* 4 NYF-0023 1.16 5 NYF-0027 6.40 6 NYF-0035 2.10 7 NYF-0044 1.11 8 NYF-0045 5.76 9 NYF-0047 7.63 10 NYF-0048 4.21 11 NYF-0058 1.03 12 NYF-0060 0.74 13 NYF-0061 1.04 *NA = Not Applicable

(Example 9) Evaluation of In Vivo Activity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules in Human PBMC-Transfected Models

[0821] The human squamous lung cancer cell lines NCI-H1703 (ATCC) were adjusted to 6×10^{sup.7} cells/ml in PBS containing 50% Matrigel (Corning) and 0.1 ml thereof was injected hypodermically into NOG mice (female, 6- to 7-week-old) (Day 0). On Day 4, human PBMCs were adjusted to 3.75×10^{sup.7} cells/ml in PBS, and 0.2 ml thereof was injected intravenously. Approximately 1 week thereafter (Day 6 to Day 7), measurement of the major diameter (mm) and the minor diameter (mm) of the tumor was initiated, measurement was performed using an electronic digital caliper with the elapse of time, and the estimated tumor volume was calculated in accordance with the equation below.

Estimated tumor volume (mm^{sup.3})=mean of estimated tumor volume among individuals Estimated tumor volume of an individual=major diameter×[minor diameter]²/2

[0822] On Day 14, mice were divided into groups each consisting of 5 or 6 mice on the basis of tumor volumes, and various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules were administered intravenously (1 mg/kg, 1.5 mg/kg of NYF-0058 for comparison at the same molar weight). Administration was performed on Day 14, Day 21, and Day 28. Antitumor effects were observed in the treatment groups to which various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules had been administered (FIG. 5A to FIG. 5C). The tumor growth inhibition (%) on Day 31 to Day 32 was calculated in accordance with the following equation and shown in Table 4.

[00004]

Tumorgrowthinhibition(%) = 100 - (estimatedtumorvolumeoftreatmentgroup / estimatedtumorvolumeofvehiclecontrol × 100

TABLE-US-00004 TABLE 4 Name Tumor Growth Inhibition (%) 1 NYF-0016 100.0 2 NYF-0019 98.8 3 NYF-0022 90.3 4 NYF-0023 98.6 5 NYF-0027 99.1 6 NYF-0035 92.5 7 NYF-0044 100.0 8 NYF-0045 95.4 9 NYF-0047 78.9 10 NYF-0048 96.3 11 NYF-0058 88.5 12 NYF-0060 88.8 13 NYF-0061 100.0

(Example 10) Preparation of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules

10)-1 Preparation of Various Fc-Conjugated Anti-HLA/NY-ESO-anti-CD3 Bispecific Molecule Expression Vectors

10)-1-1 Preparation of Hybrid-Type Bispecific Molecule Expression Vector

[0823] In order to evaluate hybrid-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules, expression vectors for various molecules were designed. As the anti-HLA/NY-ESO antibody, NYA-1143 was used. As the anti-CD3 antibody, humanized anti-CD3 scFv; i.e., C3E-7085 (PCT/JP2017/046006), was used. As the heterodimer Fc sequence, the Fc sequence comprising a mutation introduced thereinto to lower effector functions and form a

heteropolymer (WO 2014/190441) was used.

[0824] An expression vector for mammal cells comprising DNA fragments encoding the heavy chain variable region of NYA-1143, the human JgG-derived CH1 region, and the Fc region into which a mutation has been introduced therein to lower effector functions and form a heteropolymer was prepared and designated as “p_NYA-1143-Fab-HC1-k delete.” Also, an expression vector for mammal cells comprising DNA fragments encoding the NYA-1143 light chain variable region and the human IgG-derived CL region was prepared and designated as “p_NYA-1143-LC.” Further, an expression vector for mammal cells comprising a DNA fragment encoding the Fc region comprising a mutation to lower effector functions and form a heteropolymer introduced into the carboxyl terminus of humanized anti-CD3 scFv (C3E-7085) was prepared and designated as “p_C3E-7085-HC2-k delete.”

[0825] The nucleotide sequence of p_NYA-1143-Fab-HC1-k delete was reanalyzed, and the nucleotide sequence of the full-length NYA-1143-Fab-HC1-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 101 of the Sequence Listing (FIG. 102).

[0826] The nucleotide sequence of p_NYA-1143-LC was reanalyzed, and the nucleotide sequence of the full-length NYA-1143-LC was found to be the nucleotide sequence as shown in SEQ ID NO: 102 of the Sequence Listing (FIG. 103).

[0827] The nucleotide sequence of p_C3E-7085-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length C3E-7085-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 103 of the Sequence Listing (FIG. 104).

[0828] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYA-1143-Fab-HC1-k delete, NYA-1143-LC, and C3E-7085-HC2-k delete encoded thereby were identified.

[0829] The amino acid sequence of the full-length NYA-1143-Fab-HC1-k delete is the amino acid sequence as shown in SEQ ID NO: 104 of the Sequence Listing (FIG. 105). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 139 constitute a variable region, and amino acids 140 to 468 constitute a constant region. Also, amino acids 45 to 52 constitute CDRH1 (SEQ ID NO: 54 (FIG. 61)), amino acids 70 to 77 constitute CDRH2 (SEQ ID NO: 55 (FIG. 61)), amino acids 116 to 128 constitute CDRH3 (SEQ ID NO: 56 (FIG. 61)).

[0830] The amino acid sequence of the full-length NYA-1143-LC is the amino acid sequence as shown in SEQ ID NO: 105 of the Sequence Listing (FIG. 106). In such an amino acid sequence, amino acids 1 to 20 constitute a signal sequence, amino acids 21 to 131 constitute a variable region, and amino acids 132 to 237 constitute a constant region. Also, amino acids 46 to 53 constitute CDRL1 (SEQ ID NO: 60 (FIG. 62)), amino acids 71 to 73 constitute CDRL2 (SEQ ID NO: 58 (FIG. 61)), and amino acids 110 to 121 constitute CDRL3 (SEQ ID NO: 59 (FIG. 61)).

[0831] The amino acid sequence of the full-length C3E-7085-HC2-k delete is the amino acid sequence as shown in SEQ ID NO: 106 of the Sequence Listing (FIG. 107). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 260 constitute C3E-7085, amino acids 261 to 262 constitute a linker, and amino acids 263 to 493 constitute Fc comprising a mutation introduced thereinto to lower effector functions and form a heterodimer.

10)-1-2 Preparation of Dual-Type Bispecific Molecule Expression Vector

[0832] In order to evaluate the dual-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules, the expression vector described below was designed. An expression vector for mammal cells comprising a DNA fragment encoding the Fc region comprising a mutation to lower effector functions and form a heteropolymer integrated into the carboxyl terminus of NYA-1143 was prepared and designated as “p_NYA-1143-HC1-k delete.”

[0833] The nucleotide sequence of p_NYA-1143-HC1-k delete was reanalyzed, and the nucleotide sequence of the full-length NYA-1143-HC1-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 107 of the Sequence Listing (FIG. 108). On the basis of the nucleotide sequences above, the amino acid sequence of the full-length NYA-1143-HC1-k delete encoded was identified. The amino acid sequence of the full-length NYA-1143-HC1-k delete is the amino acid sequence as shown in SEQ ID NO: 108 (FIG. 109) of the Sequence Listing. In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-1143, amino acids 267 to 268 constitute a linker, amino acids 269 to 499 constitute Fc comprising a mutation introduced thereinto to lower effector functions and form a heterodimer.

10)-1-3 Preparation of scFv-Fab-Heterodimer Fc-Type Bispecific Molecule Expression Vector

[0834] In order to evaluate scFv-Fab-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules, expression vectors for various molecules were designed. As the anti-HLA/NY-ESO antibody, NYA-1154 was used, and, as the anti-CD3 antibody, C3E-7085 was used. As the heterodimer Fc sequence, the Fc sequence comprising a mutation introduced thereinto to lower effector functions and form a heteropolymer was used.

[0835] An expression vector for mammal cells comprising DNA fragments encoding the heavy chain variable region of NYA-1154, the human JgG-derived CH1 region, and the Fc region comprising a mutation to lower effector functions and form a heteropolymer integrated into the carboxyl terminus of C3E-7085 was prepared and designated as “p_C3E-7085-NYA-1154-Fab-HC2-k delete.” Also, an expression vector for mammal cells comprising DNA fragments encoding the light chain variable region of NYA-1154 and the human IgG-derived CL region integrated

therein was prepared and designated as “p_NYA-1154-LC.” Further, an expression vector for mammal cells comprising a DNA fragment encoding HC1-k delete integrated therein was prepared and designated as “p_OAA-HC1-k delete.”

[0836] The nucleotide sequence of p_C3E-7085-NYA-1154-Fab-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length C3E-7085-NYA-1154-Fab-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 109 of the Sequence Listing (FIG. 110).

[0837] The nucleotide sequence of p_NYA-1154-LC was reanalyzed, and the nucleotide sequence of the full-length NYA-1154-LC was found to be the nucleotide sequence as shown in SEQ ID NO: 110 (FIG. 111) of the Sequence Listing.

[0838] The nucleotide sequence of p_OAA-HC1-k delete was reanalyzed, and the nucleotide sequence of the full-length OAA-HC1-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 111 of the Sequence Listing (FIG. 112).

[0839] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length C3E-7085-NYA-1154-Fab-HC2-k delete, NYA-1154-LC, and OAA-HC1-k delete encoded thereby were identified.

[0840] The amino acid sequence of the full-length C3E-7085-NYA-1154-Fab-HC2-k delete is the amino acid sequence as shown in SEQ ID NO: 112 of the Sequence Listing (FIG. 113). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 260 constitute C3E-7085, amino acids 261 to 385 constitute a linker, and amino acids 266 to 385 constitute the heavy chain variable region of NYA-1154. Also, amino acids 386 to 714 constitute HC2-k delete.

[0841] The amino acid sequence of the full-length NYA-1154-LC is the amino acid sequence as shown in SEQ ID NO: 113 of the Sequence Listing (FIG. 114). In such an amino acid sequence, amino acids 1 to 20 constitute a signal sequence, amino acids 21 to 131 constitute a variable region, and amino acids 132 to 237 constitute a constant region. Also, amino acids 46 to 53 constitute CDRL1 (SEQ ID NO: 57 (FIG. 61)), amino acids 71 to 73 constitute CDRL2 (SEQ ID NO: 58 (FIG. 61)), and amino acids 110 to 121 constitute CDRL3 (SEQ ID NO: 63 (FIG. 64)).

[0842] The amino acid sequence of the full-length OAA-HC1-k delete is the amino acid sequence as shown in SEQ ID NO: 114 of the Sequence Listing (FIG. 115). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, and amino acids 20 to 245 constitute OAA-HC1-k delete.

10)-1-4 Preparation of taFv-Heterodimer Fc-Type Bispecific Molecule Expression Vectors for Evaluation of Various Formats

[0843] In order to evaluate taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules for evaluation of various formats, expression vectors for various molecules were designed. As the anti-HLA/NY-ESO antibodies, NYA-1143 and NYA-1154 were used. As the anti-CD3 antibody, humanized anti-CD3 scFv; i.e., C3E-7085, was used. As the heterodimer Fc sequence, the Fc sequence comprising a mutation introduced therein to lower effector functions and form a heteropolymer was used.

[0844] An expression vector for mammal cells comprising a DNA fragment encoding HC2-k delete integrated into the carboxyl terminus of taFv comprising NYA-1154 ligated to C3E-7085 with a GGGGS linker was prepared and designated as “p_NYF-0010-HC2-k delete.”

[0845] An expression vector for mammal cells comprising a DNA fragment encoding HC2-k delete integrated into the carboxyl terminus of taFv comprising C3E-7085 ligated to NYA-1154 with a GGGGS linker was prepared and designated as “p_NYF-0004-HC2-k delete.”

[0846] An expression vector for mammal cells comprising a DNA fragment encoding HC2-k delete integrated into the carboxyl terminus of taFv comprising NYA-1143 ligated to C3E-7085 with a GGGGS linker was prepared and designated as “p_NYF-0011-HC2-k delete.”

[0847] The nucleotide sequence of p_NYF-0010-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length NYF-0010-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 115 of the Sequence Listing (FIG. 116).

[0848] The nucleotide sequence of p_NYF-0004-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length NYF-0004-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 116 of the Sequence Listing (FIG. 117).

[0849] The nucleotide sequence of p_NYF-0011-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length NYF-0011-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 117 of the Sequence Listing (FIG. 118).

[0850] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYF-0010-HC2-k delete, NYF-0004-HC2-k delete, and NYF-0011-HC2-k delete encoded thereby were identified.

[0851] The amino acid sequence of the full-length NYF-0010-HC2-k delete is the amino acid sequence as shown in SEQ ID NO: 118 (FIG. 119) of the Sequence Listing. In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-1154-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 744 constitute HC2-k delete.

[0852] The amino acid sequence of the full-length NYF-0004-HC2-k delete is the amino acid sequence as shown in

SEQ ID NO: 119 (FIG. **120**) of the Sequence Listing. In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute C3E-7085-NYA-1154 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 744 constitute HC2-k delete.

[0853] The amino acid sequence of the full-length NYF-0011-HC2-k delete is the amino acid sequence as shown in SEQ ID NO: 120 of the Sequence Listing (FIG. **121**). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-1143-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2-k delete.

10)-2 Expression of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules of Various Formats

10)-2-1 Expression of Hybrid-Type Bispecific Molecules

[0854] In the same manner as in 7)-2-1, the culture supernatant of the hybrid-type bispecific molecule (NYG-3143) was prepared using a vector mixture comprising p_NYA-1143-Fab-HC1-k delete, p_C3E-7085-HC2-k delete, and p_NYA-1143-LC at 1:1:1.5. The amino acid sequences obtained via expression of vectors constituting NYG-3143 are shown in SEQ ID NO: 104 of the Sequence Listing (FIG. **105**), SEQ ID NO: 105 of the Sequence Listing (FIG. **106**), and SEQ ID NO: 106 of the Sequence Listing (FIG. **107**).

10)-2-2 Expression of Dual-Type Bispecific Molecules

[0855] In the same manner as in 7)-2-1, the culture supernatant of the dual-type bispecific molecule (NYG-2143) was prepared using a vector mixture comprising p_NYA-1143-HC1-k delete and p_C3E-7085-HC2-k delete at 2:1. The amino acid sequences obtained via expression of vectors constituting NYG-2143 are shown in SEQ ID NO: 108 of the Sequence Listing (FIG. **109**) and SEQ ID NO: 106 of the Sequence Listing (FIG. **107**).

10)-2-3 Expression of scFv-Fab-Heterodimer Fc-Type Bispecific Molecules

[0856] In the same manner as in 7)-2-1, the culture supernatant of the scFv-Fab-heterodimer Fc-type (scFv-Fab-Fc-type) bispecific molecule (NYF-0003) was prepared using a vector mixture comprising p_C3E-7085-NYA-1154-Fab-HC2-k delete, p_OAA-HC1-k delete, and p_NYA-1154-LC at 1:1:1.5. The amino acid sequences obtained via expression of vectors constituting NYF-0003 are shown in SEQ ID NO: 112 of the Sequence Listing (FIG. **113**), SEQ ID NO: 113 of the Sequence Listing (FIG. **114**), and SEQ ID NO: 114 of the Sequence Listing (FIG. **115**).

10)-2-4 Expression of taFv-Heterodimer Fc-Type Bispecific Molecules for Evaluation of Various Formats

[0857] In the same manner as in 7)-2-1, the culture supernatant of the taFv-heterodimer Fc-type (taFv-Fc-type) anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0010) was prepared using p_NYF-0010-HC2-k delete and p_OAA-HC1-k delete. The amino acid sequences obtained via expression of vectors constituting NYF-0010 are shown in SEQ ID NO: 118 of the Sequence Listing (FIG. **119**) and SEQ ID NO: 114 of the Sequence Listing (FIG. **115**).

[0858] The culture supernatant of the taFv (inversed)-heterodimer Fc-type (taFv (inversed)-Fc-type) anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0004) was prepared using p_NYF-0004-HC2-k delete and p_OAA-HC1-k delete. The amino acid sequences obtained via expression of vectors constituting NYF-0004 are shown in SEQ ID NO: 119 of the Sequence Listing (FIG. **120**) and SEQ ID NO: 114 of the Sequence Listing (FIG. **115**).

[0859] The culture supernatant of the taFv-Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0011) was prepared using p_NYF-0011-HC2-k delete and p_OAA-HC1-k delete. The amino acid sequences obtained via expression of vectors constituting NYF-0011 are shown in SEQ ID NO: 120 of the Sequence Listing (FIG. **121**) and SEQ ID NO: 114 of the Sequence Listing (FIG. **115**).

10)-3 Purification of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules of Various Formats

[0860] Various bispecific molecules were purified from the culture supernatants obtained in 10)-2-1 and 10)-2-2 by 2 steps of protein A affinity chromatography and ceramic hydroxyapatite chromatography.

[0861] The culture supernatant was applied to the MabSelectSuRe column equilibrated with PBS at pH 7.4 (GE Healthcare Bioscience) to allow the target bispecific molecules to adsorb thereto. After the non-adsorbed components were removed by PBS, the adsorbed components were eluted using acetate buffer (pH 3.6). The elution fraction was neutralized with the aid of Tris buffer (pH 9.5), and a buffer was exchanged with 25 mM histidine, 150 mM NaCl, 5% Sorbitol, at pH 5.5. The solution of the target fraction diluted to 5-fold with a buffer comprising 10 mM potassium phosphate and 50 mM MES (pH 6.5) was applied to the ceramic hydroxyapatite column (Bio-Scale CHT Type-1 Hydroxyapatite Column, BioRad Japan) equilibrated with 10 mM potassium phosphate and 50 mM MES (pH 6.5). Linear concentration gradient elution was performed using sodium chloride, and fractions equivalent to the target heterodimer were collected. The fractions were applied to the gel filtration column Superdex 200 10/300 (GE Healthcare Bioscience) equilibrated with 25 mM histidine, 300 mM NaCl, 5% Sorbitol at pH 6.0 in advance. From the peak fraction obtained via gel filtration chromatography, fractions equivalent to the target heterodimer were collected, and the buffer was exchanged with 25 mM histidine, 300 mM NaCl, 5% Sorbitol at pH 5.5. Generation of the target anti-HLA/NY-ESO-anti-CD3 bispecific molecule was confirmed via SDS-polyacrylamide electrophoresis (SDS-PAGE). Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

[0862] Various bispecific molecules were purified from the culture supernatants obtained in 10)-2-3 and 10)-2-4 by 2 steps of protein A affinity chromatography and gel filtration chromatography in the same manner as in 7)-3.

Generation of the target anti-HLA/NY-ESO-anti-CD3 bispecific molecule was confirmed via SDS-polyacrylamide electrophoresis (SDS-PAGE). Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation. (Example 11) Evaluation of In Vitro Activity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules of Various Formats

11)-1 Evaluation of In Vitro Activity of Hybrid-Type, Dual-Type, and taFv-Fc-Type Anti-HLA-A2/NY-ESO-anti-CD3 Bispecific Molecules

11)-1-1 Preparation of Target Cells

[0863] U266B1 cells prepared in the same manner as in Example 8)-1 were used as the target cells.

11)-1-2 Preparation of Effector Cells

[0864] Commercially available frozen human PBMCs (Cellular Technology Limited) prepared in the same manner as in Example 8)-2 were used as the effector cells.

11)-1-3 Cytotoxicity Assay

[0865] The target cells obtained in Example 11)-1-1 were seeded on a 96-well U-bottom microplate at 50 μ l/well. The anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules of various formats prepared in Example 10 and adjusted to various concentration levels were added thereto at 50 μ l/well, the effector cells prepared in Example 11)-1-2 were added thereto at 100 l/well, centrifugation was carried out at room temperature and 1,000 rpm for 1 minute, and then they were incubated at 37° C. in the presence of 5% CO.sub.2 for 20 to 24 hours. The supernatant (50 μ l) was collected on a LumaPlate (PerkinElmer), dried at 50° C. for approximately 2 hours, and then assayed using a plate reader (TopCount, PerkinElmer). The test was performed in triplicate, and the rate of cell lysis was determined in accordance with the following equation.

[00005]Rateofcelllysis(%) = $(A - B) / (C - B) \times 100$ [0866] A: Sample well count [0867] B: Mean of background (wells not supplemented with antibody) count (n=3).

[0868] When the antibody was added, 50 μ l of an assay medium was added. Other procedures were the same as those for the sample wells. [0869] C: Mean of maximal release (wells in which target cells were lysed with a surfactant) count (n=3).

[0870] When the antibody was added, 50 μ l of an assay medium was added. A surfactant was added in an amount of 100 μ l, a fraction of 50 μ l was transferred to the LumaPlate as in the case of the sample wells, and assays were then performed. As shown in FIG. 7A, the taFv-Fc type exerted cytotoxicity. The results of calculations of EC₅₀ obtained using the Four Parameter Logistic Curve equation of analytical software (Sigmaplot, version 12.0) are shown in Table 5. "Not Applicable (NA)" indicates that it is impossible to perform curve fitting because the correlation coefficient (R) was not calculated. Neither Hybrid-type nor Dual-type exerted cytotoxicity.

TABLE-US-00005 TABLE 5 Format Name EC.sub.50 (nM) 1 Hybrid NYG-3143 NA* 2 Dual NYG-2143 NA* 3 taFv-Fc NYF-0011 0.27 *NA = Not Applicable

11)-2 Evaluation of In Vitro Activity of scFv-Fab-Fc-Type, taFv-Fc-Type, and taFv (Inversed)-Fc-Type Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules

11)-2-1 Preparation of Target Cells

[0871] U266B1 cells prepared in the same manner as in Example 8)-1 were used as the target cells.

11)-2-2 Preparation of Effector Cells

[0872] Commercially available frozen human PBMCs (Cellular Technology Limited) prepared in the same manner as in Example 8)-2 were used as the effector cells.

11)-2-3 Cytotoxicity Assay

[0873] The target cells obtained in Example 11)-2-1 were seeded on a 96-well U-bottom microplate at 50 μ l/well. The anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules of various formats prepared in Example 10 and adjusted to various concentration levels were added thereto at 50 μ l/well, the effector cells prepared in Example 11)-1-2 were added thereto at 100 l/well, centrifugation was carried out at room temperature and 1,000 rpm for 1 minute, and then they were incubated at 37° C. in the presence of 5% CO.sub.2 for 20 to 24 hours. The supernatant (50 μ l) was collected on a LumaPlate (PerkinElmer), dried at 50° C. for approximately 2 hours, and then assayed using a plate reader (TopCount, PerkinElmer). The test was performed in triplicate, and the rate of cell lysis was determined in accordance with the following equation.

[00006]Rateofcelllysis(%) = $(A - B) / (C - B) \times 100$ [0874] A: Sample well count [0875] B: Mean of background (wells not supplemented with antibody) count (n=3).

[0876] When the antibody was added, 50 μ l of an assay medium was added. Other procedures were the same as those for the sample wells. [0877] C: Mean of maximal release (wells in which target cells were lysed with a surfactant) count (n=3).

[0878] When the antibody was added, 50 μ l of an assay medium was added. A surfactant was added in an amount of 100 μ l, a fraction of 50 μ l was transferred to the LumaPlate as in the case of the sample wells, and assays were then performed. As shown in FIG. 7B, the taFv-Fc type exerted cytotoxicity. The results of calculations of EC₅₀ obtained using the Four Parameter Logistic Curve equation of analytical software (Sigmaplot, version 12.0) are

shown in Table 6. “Not Applicable (NA)” indicates that it is impossible to perform curve fitting because the correlation coefficient (R) was not calculated. In comparison with the taFv-heterodimer Fc-type, the scFv-Fab-heterodimer Fc-type and the taFv (inversed)-heterodimer Fc-type exhibited lower activity.

TABLE-US-00006 TABLE 6 Format Name EC.sub.50 (nM) 1 scFv-Fab-heterodimer Fc NYF-0003 NA* 2 taFv-heterodimer Fc NYF-0010 0.13 3 taFv (inversed)-heterodimer Fc NYF-0004 0.88 *NA = Not Applicable (Example 12) Preparation of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules Via Mutagenesis and Format Modification of NYF-0061

12)-1 Preparation of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule Expression Vectors Via Mutagenesis and Format Modification of NYF-0061

[0879] In order to improve physical properties, NYF-0061 mutants were designed and the vectors constituting the molecules described below were prepared.

[0880] A mutation was introduced into the C3E-7085 sequence of the p_NYF-0061-HC2 vector containing the nucleotide sequence encoding the amino acid sequence of NYF-0061 and the resultant was designated as “p_NYZ-0038-HC2.”

[0881] A mutation was introduced into the NYA-2061 sequence and the C3E-7085 sequence of the p_NYF-0061-HC2 vector containing the nucleotide sequence encoding the amino acid sequence of NYF-0061 and the resultants were designated as “p_NYZ-0082-HC2” and “p_NYZ-0083-HC2,” respectively.

[0882] The nucleotide sequence of p_NYZ-0038-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYZ-0038-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 152 of the Sequence Listing (FIG. 148).

[0883] The nucleotide sequence of p_NYZ-0082-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYZ-0082-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 153 of the Sequence Listing (FIG. 149).

[0884] The nucleotide sequence of p_NYZ-0083-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYZ-0083-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 154 of the Sequence Listing (FIG. 150).

[0885] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYZ-0038-HC2, NYZ-0082-HC2, and NYZ-0083-HC2 encoded thereby were identified.

[0886] The amino acid sequence of the full-length NYZ-0038-HC2 is the amino acid sequence as shown in SEQ ID NO: 155 of the Sequence Listing (FIG. 151). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2061-C3E-7096 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0887] The amino acid sequence of the full-length NYZ-0082-HC2 is the amino acid sequence as shown in SEQ ID NO: 156 of the Sequence Listing (FIG. 152). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 516 constitute NYA-3061-C3E-7096 taFv, amino acids 517 to 518 constitute a linker, and amino acids 519 to 750 constitute HC2.

[0888] The amino acid sequence of the full-length NYZ-0083-HC2 is the amino acid sequence as shown in SEQ ID NO: 157 of the Sequence Listing (FIG. 153). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 516 constitute NYA-3061-C3E-7097 taFv, amino acids 517 to 518 constitute a linker, and amino acids 519 to 750 constitute HC2.

[0889] In order to examine format modification of NYF-0061 into the scFv-Fab-heterodimer Fc-type, the vectors constituting the molecules described below were designed.

[0890] An expression vector for mammal cells comprising DNA fragments encoding an amino acid sequence included in a polypeptide comprising (i) the heavy chain variable region of C3E-7085, (ii) the human IgG-derived CH1 region, and (iii) the Fc region comprising a mutation to lower effector functions and form a heteropolymer integrated in that order into the carboxyl terminus of NYA-3061 was prepared and designated as “p_NYZ-1010-HC2.” Also, an expression vector for mammal cells comprising a DNA fragment encoding an amino acid sequence included in a polypeptide comprising the human IgG-derived CL region added to the carboxyl terminus of the light chain variable region of C3E-7085 was prepared and designated as “p_C3E-7085-LC.”

[0891] The nucleotide sequence of p_NYZ-1010-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYZ-1010-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 158 of the Sequence Listing (FIG. 154).

[0892] The nucleotide sequence of p_C3E-7085-LC was reanalyzed, and the nucleotide sequence of the full-length C3E-7085-LC was found to be the nucleotide sequence as shown in SEQ ID NO: 159 of the Sequence Listing (FIG. 155).

[0893] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYZ-1010-HC2 and C3E-7085-LC encoded thereby were identified.

[0894] The amino acid sequence of the full-length NYZ-1010-HC2 is the amino acid sequence as shown in SEQ ID NO: 160 of the Sequence Listing (FIG. 156). In such an amino acid sequence, amino acids 1 to 19 constitute a signal

sequence, amino acids 21 to 271 constitute NYA-3061, amino acids 272 to 276 constitute a linker, and amino acids 277 to 394 constitute the heavy chain variable region of C3E-7085. Also, amino acids 395 to 724 constitute a constant region.

[0895] The amino acid sequence of the full-length C3E-7085-LC is the amino acid sequence as shown in SEQ ID NO: 161 of the Sequence Listing (FIG. 157). In such an amino acid sequence, amino acids 1 to 20 constitute a signal sequence, amino acids 21 to 127 constitute a variable region, and amino acids 128 to 233 constitute a constant region. Also, amino acids 46 to 53 constitute CDRL1 (SEQ ID NO: 144 (FIG. 142)), amino acids 71 to 73 constitute CDRL2 (SEQ ID NO: 145 (FIG. 142)), and amino acids 110 to 117 constitute CDRL3 (SEQ ID NO: 146 (FIG. 142)).

12)-2 Expression and Purification of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules Via Mutagenesis and Format Modification or NYF-0061

[0896] In the same manner as in 7)-2-1, the culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYZ-0038) was prepared using p_NYZ-0038-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYZ-0038 are shown in SEQ ID NO: 155 of the Sequence Listing (FIG. 151) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0897] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYZ-0082) was prepared using p_NYZ-0082-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYZ-0082 are shown in SEQ ID NO: 156 of the Sequence Listing (FIG. 152) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0898] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYZ-0083) was prepared using p_NYZ-0083-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYZ-0083 are shown in SEQ ID NO: 157 of the Sequence Listing (FIG. 153) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0899] The culture supernatant of the scFv-Fab-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYZ-1010) was prepared using p_NYZ-1010-HC2, p_C3E-7085-LC, and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYZ-1010 are shown in SEQ ID NO: 160 of the Sequence Listing (FIG. 156), SEQ ID NO: 161 of the Sequence Listing (FIG. 157), and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0900] The culture supernatants prepared above were purified in the same manner as in 7)-3. Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

(Example 13) Evaluation of Binding Affinity to HLA/NY-ESO Using Biacore

[0901] With the use of Biacore T200, various Fc-conjugated anti-HLA/NY-ESO-anti-CD3 bispecific molecules were captured as ligands to the immobilized anti-human IgG (Fc) antibody, and the antigen was assayed as an analyte. As the antigen, HLA/NY-ESO prepared in 1)-1 was used. The anti-human IgG (Fc) antibody (Human antibody Capture kit, GE Healthcare) was immobilized on Sensor Chip CM5(GE Healthcare) in accordance with the instructions of the kit. The Fc-conjugated anti-HLA/NY-ESO-anti-CD3 bispecific molecules diluted to 0.5 µg/ml in HBS-EP+(GE Healthcare) to be evaluated were brought into contact therewith at 10 µl/min for 60 seconds for immobilization. Thereafter, the samples were added to the HLA/NY-ESO analytes diluted to various levels with HBS-EP+ at a flow rate of 30 µl/min for 120 seconds and dissociation was assayed for 600 seconds. The results of calculations obtained by such single cycle kinetics analysis, K.sub.D, are shown in Table 7. NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010 retained binding affinity equivalent to that of NYF-0061.

TABLE-US-00007 TABLE 7 Clone name K.sub.D (nM) NYF-0061 1.5 NYZ-0038 1.5 NYZ-0082 1.8 NYZ-0083 1.7 NYZ-1010 1.9

(Example 14) Preparation of CD3e Knockout T2 Human Lymphoblast Fusion Cells

[0902] In order to knockout CD3e in the genome sequence of the T2 human lymphoblast fusion cells (ATCC) by the technique of CRISPR, the Cas9 expression plasmid (GE Healthcare) and the sgRNA expression plasmid were introduced via electroporation (LONZA). Thereafter, cell cloning was performed by limited dilution. The cells in which deletion of the target gene fragment from the introduced cells and CD3e gene expression were not observed as a result of genome analysis and RT-PCR analysis of total RNA, respectively, were subjected to the subsequent experiments.

(Example 15) Analysis of Recognition Amino Acid in NY-ESO Peptide of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecule

[0903] The concentration of the CD3e knockout T2 cells prepared in Example 14 was adjusted to an adequate level in AIM-V medium (Thermo Fisher Scientific) containing 20% FBS, the NY-ESO peptide (SEQ ID NO: 1), the point-mutant NY-ESO peptides 1F, 2M, 3A, 4A, 5A, 6L, 7F, 8A, and 9A (SEQ ID NO: 121 (FIG. 122), 122 (FIG. 123), 123 (FIG. 124), 124 (FIG. 125), 125 (FIG. 126), 126 (FIG. 127), 127 (FIG. 128), 128 (FIG. 129), and 129 (FIG. 130)), and the gp100 peptide (SEQ ID NO: 130 (FIG. 131)) (Sigma Genosys) were added in the same manner as in Example 5, cells stained with the use of the LIVE/DEAD Fixable Dead Cell Stain Kit (Thermo Fisher Scientific)

divided into two groups, the cells in PBS containing 5% FBS were seeded on a 96-well U-bottom microplate at 10.5 cells/well, and the plate was subjected to centrifugation, followed by removal of the supernatant. The various Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules prepared in Example 12 and diluted to 100 nM with PBS containing 5% FBS were added at 25 µl/well to one of the two groups, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, PE conjugated F(ab')₂ Fragment Anti-Human IgG (Jackson Immuno Research Laboratories) diluted with PBS containing 5% FBS was added at 25 l/well, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, immobilized with Mildform 10 N (FUJIFILM Wako Pure Chemical Corporation) overnight, and resuspended in PBS containing 5% FBS. For standardization with the amount of the HLA/peptide complex, the HLA-A2 antibody BB7.2-Alexa Fluor 488 diluted to 10 µg/ml in PBS containing 5% FBS or mouse IgG2b-Alexa Fluor 488 (BioRAD) was added at 25 µl/well, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, immobilized with Mildform 10 N (FUJIFILM Wako Pure Chemical Corporation) overnight, and resuspended in PBS containing 5% FBS. The cell suspensions were subjected to detection using a flow cytometer (CantoII, Becton Dickinson). Data analysis was performed using Flowjo (Treestar), and the geometric mean fluorescence intensity (gMFI) of PE or Alexa Fluor 488 in the CD3e knockout T2 cells from which dead cells were removed was measured. The standardized gMFI indicating the binding affinity of each Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule standardized with the amount of the HLA/peptide complex on the CD3e knockout T2 cells was determined in accordance with the following equation.

[00007]StandardizedgMFI = $(A / B) / ((C / D) / (E / F))$ [0904] A/B=Relative gMFI [0905] A: gMFI of PE in CD3e knockout T2 cells supplemented with DMSO or peptide in the presence of antibody [0906] B: gMFI of PE in CD3e knockout T2 cells supplemented with DMSO or peptide [0907] (C/D)/(E/F)=Corrected amount of HLA/peptide complex in CD3e knockout T2 cells supplemented with DMSO or peptide [0908] C: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO or peptide in the presence of HLA-A2 antibody [0909] D: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO or peptide in the presence of mouse IgG2b antibody [0910] E: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO containing HLA-A2 antibody [0911] F: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO containing mouse IgG2b antibody

[0912] As shown in FIG. 158A, the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010, exhibited binding affinity to the CD3e knockout T2 cells, supplemented with peptides comprising point mutations introduced into amino acids 1, 4, 5, and 7, that had been lowered to a quarter of the binding affinity for the wild-type NY-ESO peptide or lower. This indicates that such constructs strongly recognize amino acids 1, 4, 5, and 7 of the NY-ESO peptide.

(Example 16) Evaluation of Antigen-Binding Specificity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules

[0913] In the same manner as in Example 6), the 9-mer peptide shown in FIG. 2A exhibiting identity to the sequence of the NY-ESO peptide: SLLMWITQC (SEQ ID NO: 1) at amino acids 1, 4, and 5, which are indicated to be strongly recognized by the four types of antibodies, was selected as a homologous peptide, and binding specificity thereto was evaluated. The concentration of the CD3e knockout T2 cells was adjusted to an adequate level in AIM-V medium (Thermo Fisher Scientific) containing 20% FBS, a solution of the NY-ESO peptide (SEQ ID NO: 1), the homologous peptides DOLPP1, IL20RB, PRKD2, CD163, and P2RY8 (SEQ ID NO: 131 (FIG. 132), SEQ ID NO: 132 (FIG. 133), SEQ ID NO: 133 (FIG. 134), SEQ ID NO: 134 (FIG. 135), and SEQ ID NO: 135 (FIG. 136)), or the gp100 peptide (SEQ ID NO: 130 (FIG. 131)) (Sigma Genosys) dissolved to 5 mM in DMSO was added to result in a final concentration of 50 µM, or DMSO was added in an amount of 1/100, and binding affinity of the antibodies was evaluated in the same manner as in Example 15. The standardized gMFI indicating the binding affinity of each Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule standardized with the amount of the HLA/peptide complex on the CD3e knockout T2 cells was determined in accordance with the following equation.

[00008]StandardizedgMFI = $(A / B) / ((C / D) / (E / F))$ [0914] A: gMFI of PE in CD3e knockout T2 cells supplemented with DMSO or peptide in the presence of antibody [0915] B: gMFI of PE in CD3e knockout T2 cells supplemented with DMSO or peptide [0916] (C/D)/(E/F)=Corrected amount of HLA/peptide complex in CD3e knockout T2 cells supplemented with DMSO or peptide [0917] C: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO or peptide of the presence of HLA-A2 antibody [0918] D: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO or peptide containing mouse IgG2b antibody [0919] E: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO containing HLA-A2 antibody [0920] F: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO containing mouse IgG2b antibody

[0921] As shown in FIG. 158B, none of the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010, had bound to cells supplemented with homologous peptides, and these molecules exhibited very high specificity.

(Example 17) Evaluation of Cytotoxicity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules

17)-1 Preparation of Target Cells

[0922] In the same manner as in Example 8)-1, suspensions of the endogenous human NY-ESO-expressing cells (U266B1 and NCI-H1703) and the endogenous human NY-ESO-non-expressing cells (AGS and CFPAC-1) were prepared and used as the target cells.

17)-2 Preparation of Effector Cells

[0923] In the same manner as in Example 8)-2, a suspension of commercially available frozen human PBMCs (Cellular Technology Limited) was prepared and used as the effector cells.

17)-3 Cytotoxicity Assay

[0924] The target cells obtained in Example 17)-1 were seeded on a 96-well U-bottom microplate at 50 μ l/well. The various Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules prepared in Example 12 and adjusted to various concentration levels were added thereto at 50 μ l/well, the effector cells prepared in Example 17)-2 were added thereto at 100 l/well, centrifugation was carried out at room temperature and 1,000 rpm for 1 minute, and then they were incubated at 37° C. in the presence of 5% CO.sub.2 for 20 to 24 hours. The supernatant (50 μ l) was collected on a LumaPlate (PerkinElmer), dried at 50° C. for approximately 2 hours, and then assayed using a plate reader (TopCount, PerkinElmer). The test was performed in triplicate, and the rate of cell lysis was determined in accordance with the following equation.

$$[00009] \text{Rate of cell lysis (\%)} = (A - B) / (C - B) \times 100$$

[0925] A: Sample well count [0926] B: Mean of background (wells not supplemented with antibody) count (n=3).

[0927] When the antibody was added, 50 μ l of an assay medium was added. Other procedures were the same as those for the sample wells. [0928] C: Mean of maximal release (wells in which target cells were lysed with a surfactant) count (n=3). [0929] When the antibody was added, 50 μ l of an assay medium was added. A surfactant was added in an amount of 100 μ l, a fraction of 50 μ l was transferred to the LumaPlate as in the case of the sample wells, and assays were then performed.

[0930] As shown in FIG. 159A to FIG. 159D, various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules exerted cytotoxicity on the endogenous human NY-ESO-expressing cells (U266B1 and NCI-H1703). The results of calculations of EC.sub.50 obtained using the Four Parameter Logistic Curve equation of analytical software (Sigmaplot, version 12.0) concerning the U266B1 cells are shown in Table 8, and the results concerning the NCI-H1703 cells are shown in Table 9. As shown in FIG. 159E to FIG. 159H, in contrast, no cytotoxicity was observed on the endogenous human NY-ESO-non-expressing cells (AGS and CFPAC-1).

TABLE-US-00008 TABLE 8 Name EC.sub.50 (nM) 1 NYZ-0038 1.01 2 NYZ-0082 0.45 3 NYZ-0083 0.25 4 NYZ-1010 0.96

TABLE-US-00009 TABLE 9 Name EC.sub.50 (nM) 1 NYZ-0038 0.61 2 NYZ-0082 1.00 3 NYZ-0083 0.71 4 NYZ-1010 2.56

(Example 18) Evaluation of In Vivo Activity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules in Human PBMC-Transfected Models

[0931] The human squamous lung cancer cell lines NCI-H1703 (ATCC) were adjusted to 6 $\times$ 10⁷ cells/ml in PBS containing 50% Matrigel (Corning) and 0.1 ml thereof was injected hypodermically to NOG mice (female, 6- to 7-week-old) (Day 0). On Day 4, human PBMCs were adjusted to 3.75 $\times$ 10⁷ to 5 $\times$ 10⁷ cells/ml in PBS, and 0.2 ml thereof was injected intravenously. Approximately 1 week thereafter (Day 6 to Day 7), measurement of the major diameter (mm) and the minor diameter (mm) of the tumor was initiated, measurement was performed using an electronic digital caliper with the elapse of time, and the estimated tumor volume was calculated in accordance with the equation below.

Estimated tumor volume (mm³)=mean of estimated tumor volume among individuals Estimated tumor volume of an individual=major diameter $\times$ [minor diameter]²/2

[0932] On Day 14, mice were divided into groups each consisting of 5 or 6 mice on the basis of tumor volumes, and various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules were administered intravenously (NYZ-1010 at 1 mg/kg, NYZ-1010 at 1.2 mg/kg for comparison at the same molar weight). Administration was performed on Day 14 and Day 21. Antitumor effects were observed in the treatment groups to which various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules had been administered (FIG. 160A to FIG. 160D). The tumor growth inhibition (%) on Day 28 to Day 29 was calculated in accordance with the following equation and shown in Table 10.

[00010]

$$\text{Tumor growth inhibition (\%)} = 100 - (\text{estimated tumor volume of treatment group} / \text{estimated tumor volume of vehicle control} \times 100)$$

TABLE-US-00010 TABLE 10 Name Tumor growth inhibition (%) 1 NYZ-0038 100.0 2 NYZ-0082 100.0 3 NYZ-0083 99.8 4 NYZ-1010 99.9

(Example 19) Comparison of Physicochemical Properties of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules

19)-1 Acid Treatment Evaluation

[0933] NYF-0016 and various NYA-1143 CDR graft mutants (NYF-0044, NYF-0045, NYF-0047, NYF-0048, NYF-

0060, and NYF-0061) prepared in 7)-2-1 were concentrated by centrifugation using Amicon (Millipore), the buffer was exchanged with 25 mM sodium acetate, 5% sorbitol (pH 5.5), and the concentration was adjusted to 15 mg/ml (NYF-0016 at 10 mg/ml). Subsequently, the samples were dialyzed using Xpress Micro Dialyzer (Scienova) against 100 mM sodium acetate (pH 3.5) or 25 mM sodium acetate, 5% sorbitol (pH 5.5) as a control, and the samples were collected and allowed to stand at room temperature for 1 hour. Subsequently, pH levels of the samples were adjusted to 5.0 using 500 mM Tris-HCl (pH 9.0). The samples were analyzed via size exclusion chromatography (SEC) using ACQUITY UPLC BEH200 SEC 1.7 m 4.6*150 mm (Waters). As the mobile phase, 0.2 M Ki/200 mM KCl/pH 7.0 was employed, and analysis was performed at a flow rate of 0.2 ml/min (detection wavelength: 280 nm). The peak contents (%) of polymers contained in the samples were analyzed and calculated by the area percentage method. The results are shown in FIG. 162. As a result of acid treatment, the polymer content in NYF-0016 was increased to 96%. In various NYA-1143 CDR graft mutants, in contrast, the amount of polymer formation upon acid treatment was decreased. In NYF-0047, NYF-0048, NYF-0060, and NYF-0061, in particular, the amount of polymer formation upon acid treatment was decreased to 1% or lower. This indicates that acid resistance was improved to a significant extent.

19)-2 Evaluation of Solution Stability

[0934] NYF-0061 prepared in 7)-2-1 and NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010 prepared in 12)-2 were concentrated by centrifugation using Amicon Ultra-4 (Millipore), the buffer was exchanged with 25 mM histidine, 5% sorbitol (pH 6.0), 25 mM histidine and 5% sorbitol (pH 6.0) were added thereto, and the sample concentration was adjusted to 25 mg/ml to prepare evaluation samples. In the beginning, the high molecular weight species (HMWS) of each evaluation sample was determined by the area percentage method via size exclusion chromatography using AdvanceBio SEC 300A 2.7 m 4.6*300 mm (Agilent). As the mobile phase, 0.2 M Ki/200 mM KCl/pH 7.0 was employed, and analysis was performed at a flow rate of 0.2 ml/min (detection wavelength: 280 nm). The solution stability test of each sample was performed by storage at 25° C. for 6 days (NYF-0061: 7 days), size exclusion chromatography under the conditions as described above, and calculation of HMWS (%) of each sample by the area percentage method. The results are shown in Table 11. HMWS (%) of NYF-0061 was increased to approximately 10.6% with the elapse of storage time. In contrast, an increase in HMWS (%) of NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010 was approximately 2% or lower. This demonstrates that solution stability of NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010 is superior to that of NYF-0061.

TABLE-US-00011 TABLE 11 Clone name HMWS (%) NYF-0061 10.6 NYZ-0038 2.1 NYZ-0082 1.7 NYZ-0083 1.9 NYZ-1010 0.8

(Example 20) Preparation of Anti-HLA/NY-ESO scFv

20)-1 Construction of Anti-HLA/NY-ESO scFv Expression Vector

[0935] scFv constructs of antibodies that may have high binding affinity to HLA-A2/NY-ESO-1; i.e., mAb24955N, mAb24956N, mAb28075P, mAb28105P, mAb28113P, and mAb29822P2 (WO2021/003357), were designed, and scFv of mAb24955N was designated as NYC-0005, scFv of mAb24956N was designated as NYC-0006, scFv of mAb28075P was designated as NYC-0007, scFv of mAb28105P was designated as NYC-0008, scFv of mAb28113P was designated as NYC-0009, and scFv of mAb29822P2 was designated as NYC-0010.

[0936] Also, scFv expression vectors for NYC-0005, NYC-0006, NYC-0007, NYC-0008, NYC-0009, and NYC-0010 each comprising pcDNA3.4 (ThermoFisher Scientific) as the backbone in mammalian cells were designed. In order to compare expression levels of the same vector backbones and formats in mammalian cells, in addition, scFv expression vectors for NYA-2047, NYA-2061, and NYA-3061 in mammalian cells were designed.

[0937] The nucleotide sequences of the constructed scFv expression vectors were reanalyzed, and the nucleotide sequences of the full-length NYA-2047, NYA-2061, NYA-3061, NYC-0005, NYC-0006, NYC-0007, NYC-0008, NYC-0009, and NYC-0010 were found to be the nucleotide sequences as shown in SEQ ID NO: 43 (FIG. 50), SEQ ID NO: 46 (FIG. 53), SEQ ID NO: 163 (FIG. 164), SEQ ID NO: 165 (FIG. 166), SEQ ID NO: 167 (FIG. 168), SEQ ID NO: 169 (FIG. 170), SEQ ID NO: 171 (FIG. 172), SEQ ID NO: 173 (FIG. 174), and SEQ ID NO: 175 (FIG. 176) of Sequence Listing. On the basis of the nucleotide sequences indicated above, the amino acid sequences encoded thereby are as follows: the amino acid sequence of the full-length NYA-2047 is as shown in SEQ ID NO: 50 (FIG. 57); the amino acid sequence of the full-length NYA-2061 is as shown in SEQ ID NO: 53 (FIG. 60); the amino acid sequence of the full-length NYA-3061 is as shown in SEQ ID NO: 164 (FIG. 165); the amino acid sequence of the full-length NYC-0005 is as shown in SEQ ID NO: 166 (FIG. 167); the amino acid sequence of the full-length NYC-0006 is as shown in SEQ ID NO: 168 (FIG. 169); the amino acid sequence of the full-length NYC-0007 is as shown in SEQ ID NO: 170 (FIG. 171); the amino acid sequence of the full-length NYC-0008 is as shown in SEQ ID NO: 172 (FIG. 173); the amino acid sequence of the full-length NYC-0009 is as shown in SEQ ID NO: 174 (FIG. 175); and the amino acid sequence of the full-length NYC-0010 is as shown in SEQ ID NO: 176 (FIG. 177).

20)-2 Expression and Purification of Anti-HLA/NY-ESO scFv

[0938] Culture of and gene transfection into the Expi293F cells (ThermoFisher Scientific) were performed in the same manner as described in 1)-5-2. For evaluation of the amount of production, the cells were subjected to agitation culture in an incubator at 37° C. in the presence of 8% CO₂ at 135 rpm for 4 days after gene introduction, the

culture supernatant was filtered through a 0.2-µm filter (ThermoFisher Scientific), and the culture supernatant of scFv of anti-HLA/NY-ESO was obtained. Purification was performed using Ni Sepharose excel (Cytiva), the eluate was subjected to analytical size exclusion chromatography (SEC), and the concentration was determined. Table 12 shows the amount of production. The amounts of production of NYA-2047, NYA-2061, and NYA-3061 determined with the use of Ni Sepharose excel relative to 1 liter of the culture supernatant in the purified eluate were higher than those of NYC-0005, NYC-0006, NYC-0007, NYC-0008, NYC-0009, and NYC-0010.

TABLE-US-00012 TABLE 12 Clone name Amount of production per 1 liter (mg) NYA-2047 53.3 NYA-2061 45.7 NYA-3061 52.5 NYC-0005 2.5 NYC-0006 1.1 NYC-0007 10.8 NYC-0008 0.5 NYC-0009 2.4 NYC-0010 1.8

[0939] The eluate of each scFv obtained with the use of Ni Sepharose excel was concentrated and purified through a gel filtration column (Superdex 200 Increase, Cytiva) equilibrated with 25 mM histidine, 300 mM NaCl, 5% Sorbitol (pH 5.5). Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

(Example 21) Expression and Purification of scFv-Heterodimer-Fc of Anti-HLA/NY-ESO

21)-1 Construction of scFv-Heterodimer-Fc Expression Vector of Anti-HLA/NY-ESO

[0940] For comparison of productivity resulting from different formats, an expression vector was constructed by format conversion of scFv of anti-HLA/NY-ESO into an Fc fusion type. As the heterodimer Fc sequences (hereafter, referred to as HC-h and HC-k), those reported in Nat. Biotechnol., July 1998, 16 (7), 677-81 were used.

[0941] scFv-heterodimer-Fc constructs of antibodies that may have high binding affinity to HLA-A2/NY-ESO-1; i.e., mAb24955N, mAb24956N, mAb28075P, mAb28105P, mAb28113P, and mAb29822P2 (WO2021/003357), were designed, and scFv-heterodimer-Fc of mAb24955N was designated as NYC-0011, scFv-heterodimer-Fc of mAb24956N was designated as NYC-0012, scFv-heterodimer-Fc of mAb28075P was designated as NYC-0013, scFv-heterodimer-Fc of mAb28105P was designated as NYC-0014, scFv-heterodimer-Fc of mAb28113P was designated as NYC-0015, and scFv-heterodimer-Fc of mAb29822P2 was designated as NYC-0016. Also, scFv-heterodimer-Fc of NYA-2047 was designated as NYD-2047, scFv-heterodimer-Fc of NYA-2061 was designated as NYD-2061, and scFv-heterodimer-Fc of NYA-3061 was designated as NYD-3061.

[0942] Expression vectors for HC-h, NYC-0011-HC-k, NYC-0012-HC-k, NYC-0013-HC-k, NYC-0014-HC-k, NYC-0015-HC-k, and NYC-0016-HC-k each comprising pcDNA3.3 or pcDNA3.4 (ThermoFisher Scientific) as the backbone in mammalian cells were designed. Further, expression vectors for NYD-2047-HC-k, NYD-2061-HC-k, and NYD-3061-HC-k in mammalian cells were designed to compare expression levels in mammalian cells with the same vector backbones and formats.

[0943] The nucleotide sequences of the constructed scFv-heterodimer-Fc expression vectors were reanalyzed, and the nucleotide sequences of the full-length HC-h, NYD-2047-HC-k, NYD-2061-HC-k, NYD-3061-HC-k, NYC-0011-HC-k, NYC-0012-HC-k, NYC-0013-HC-k, NYC-0014-HC-k, NYC-0015-HC-k, and NYC-0016-HC-k were the nucleotide sequences as shown in SEQ ID NO: 177 (FIG. 178), SEQ ID NO: 179 (FIG. 180), SEQ ID NO: 181 (FIG. 182), SEQ ID NO: 183 (FIG. 184), SEQ ID NO: 185 (FIG. 186), SEQ ID NO: 187 (FIG. 188), SEQ ID NO: 189 (FIG. 190), SEQ ID NO: 191 (FIG. 192), SEQ ID NO: 193 (FIG. 194), and SEQ ID NO: 195 (FIG. 196) of Sequence Listing. On the basis of the nucleotide sequences indicated above, the full-length amino acid sequences encoded thereby are as follows: the amino acid sequence of the entire HC-h is as shown in SEQ ID NO: 178 (FIG. 179); the amino acid sequence of the full-length NYD-2047-HC-k is as shown in SEQ ID NO: 180 (FIG. 181); the amino acid sequence of the full-length NYD-2061-HC-k is as shown in SEQ ID NO: 182 (FIG. 183); the amino acid sequence of the full-length NYD-3061-HC-k is as shown in SEQ ID NO: 184 (FIG. 185); the amino acid sequence of the full-length NYC-0011-HC-k is as shown in SEQ ID NO: 186 (FIG. 187); the amino acid sequence of the full-length NYC-0012-HC-k is as shown in SEQ ID NO: 188 (FIG. 189); the amino acid sequence of the full-length NYC-0013-HC-k is as shown in SEQ ID NO: 190 (FIG. 191); the amino acid sequence of the full-length NYC-0014-HC-k is as shown in SEQ ID NO: 192 (FIG. 193); the amino acid sequence of the full-length NYC-0015-HC-k is as shown in SEQ ID NO: 194 (FIG. 195); and the amino acid sequence of the full-length NYC-0016-HC-k is as shown in SEQ ID NO: 196 (FIG. 197).

21)-2 Expression and Purification of scFv-Heterodimer-Fc of Anti-HLA/NY-ESO

[0944] Culture of and gene introduction into the Expi293F cells (ThermoFisher Scientific) were performed in the same manner as described in 1)-5-2. Plasmids used for preparation of clones are shown in Table 13 below.

TABLE-US-00013 TABLE 13 Clone name HC-k HC-h NYD-2047 NYD-2047-HC-k HC-h NYD-2061 NYD-2061-HC-k HC-h NYD-3061 NYD-3061-HC-k HC-h NYC-0011 NYC-0011-HC-k HC-h NYC-0012 NYC-0012-HC-k HC-h NYC-0013 NYC-0013-HC-k HC-h NYC-0014 NYC-0014-HC-k HC-h NYC-0015 NYC-0015-HC-k HC-h NYC-0016 NYC-0016-HC-k HC-h

[0945] For evaluation of the amount of production, the cells were subjected to agitation culture in an incubator at 37° C. in the presence of 8% CO₂ at 135 rpm for 4 days after gene introduction, the culture supernatant was filtered through a 0.2-µm filter (ThermoFisher Scientific), and the culture supernatant of scFv-heterodimer-Fc of anti-HLA/NY-ESO was obtained. For purification, the culture supernatant was applied to MabSelectSuRe resin (Cytiva) equilibrated with PBS (pH 7.4) to allow the target scFv-heterodimer-Fc of anti-HLA/NY-ESO to adsorb thereto.

After the non-adsorbed components were removed by PBS, the adsorbed components were eluted using acetate buffer, the eluate was neutralized with the aid of Tris buffer, and the eluate was subjected to analytical size exclusion chromatography (SEC) to determine the purity and the concentration. Table 14 shows the amount of production relative to 1 liter of the culture supernatant. The amounts of production of NYA-2047, NYA-2061, and NYA-3061 determined with the use of MabSelectSuRe resin relative to 1 liter of the culture supernatant in the purified eluate were higher than those of NYC-0011, NYC-0012, NYC-0013, NYC-0014, NYC-0015, and NYC-0016.

TABLE-US-00014 TABLE 14 Clone name Amount of production per 1 liter (mg) NYD-2047 93.3 NYD-2061 112.2 NYD-3061 127.5 NYC-0011 17.2 NYC-0012 4.2 NYC-0013 54.1 NYC-0014 1.8 NYC-0015 12.2 NYC-0016 16.6 [0946] The eluate of each scFv-heterodimer-Fc obtained with the use of MabSelectSuRe resin was concentrated and purified through a gel filtration column (Superdex 200 Increase, Cytiva) equilibrated with 25 mM histidine, 300 mM NaCl, 5% Sorbitol (pH 5.5). Purified protein samples were subjected to analytical size exclusion chromatography (SEC), and NYD-2047, NYD-2061, NYD-3061, NYC-0011, NYC-0013, and NYC-0015 with sufficient degrees of purity and amounts were used as samples for evaluation of solution stability. A target object was not detected in NYC-0016 within the elution time deduced by SEC. It was thus determined that a target object was not expressed in the culture supernatant.

(Example 22) Evaluation of Solution Stability of scFv-Heterodimer-Fc of Anti-HLA/NY-ESO

[0947] NYD-2047, NYD-2061, NYD-3061, NYC-0011, NYC-0013, and NYC-0015 prepared in Example 21 were concentrated by centrifugation using Amicon Ultra-4 (Millipore), the buffer was exchanged with PBS, PBS was added thereto, and the sample concentration was adjusted to 5 mg/ml to prepare evaluation samples. In the beginning, HMWS (%) of each evaluation sample was determined by the area percentage method via size exclusion chromatography using AdvanceBio SEC 300A 2.7 m 4.6×150 mm (Agilent). As the mobile phase, 0.2 M Ki/200 mM KCl/pH 7.0 was employed, and analysis was performed at a flow rate of 0.2 ml/min (detection wavelength: 280 nm). The accelerated aging test of each sample was performed by storage at 40° C. for 1 day and 7 days, size exclusion chromatography under the conditions as described above, and calculation of HMWS (%) of each sample by the area percentage method. FIG. 163 shows the results demonstrating changes occurred in each sample upon storage at 40° C. with the elapse of time.

[0948] HMWS (%) of NYD-2047, NYD-2061, and NYD-3061 was increased to 8.7%, 8.1%, and 4.4%, respectively, with the elapse of storage time. In contrast, HMWS (%) of NYC-0011, NYC-0013, and NYC-0015 was increased to a significant extent; i.e., 71.8%, 42.4%, and 97.3%, respectively, with the elapse of storage time.

[0949] The results demonstrate that NYD-2047, NYD-2061 and NYD-3061 are superior to NYC-0011, NYC-0013, and NYC-0015 in terms of solution stability. In addition, HMWS (%) of NYD-3061 was the lowest among the evaluation samples. That is, NYD-3061 was excellent in terms of solution stability.

(Example 23) Evaluation of Binding Affinity to HLA/NY-ESO Using Biacore

[0950] NYC-0005, NYC-0007, and NYC-0008 equivalent to scFv constituting anti-HLA/NY-ESO of NYC-0011, NYC-0013, and NYC-0015 were used to evaluate binding affinity to HLA/NY-ESO

[0951] With the use of Biacore T200, the anti-HLA/NY-ESO scFv was captured as a ligand to the immobilized anti-His antibody, and the antigen was assayed as an analyte. As the antigen, HLA/NY-ESO prepared in 1)-1 was used. The anti-His antibody (His Capture kit, Cytiva) was immobilized on Sensor Chip CM5 (Cytiva) in accordance with the instructions of the kit. The anti-HLA/NY-ESO scFv constructs diluted to 0.5 pg/ml in HBS-EP+(Cytiva) to be evaluated were brought into contact therewith at 10 µl/min for 60 seconds for immobilization. Thereafter, the samples were added to the HLA/NY-ESO analytes diluted to various levels with HBS-EP+ at a flow rate of 30 µl/min for 120 seconds and dissociation was assayed for 600 seconds. The results of calculations obtained by such single cycle kinetics analysis, K.sub.D, are shown in Table 15. It was confirmed that NYC-0005, NYC-0007, and NYC-0008 would bind to HLA/NY-ESO under the conditions in which the binding affinity of NYA-2047 used as a positive control to HLA/NY-ESO was stably confirmed.

TABLE-US-00015 TABLE 15 Clone name K.sub.D NYC-0005 9.3 nM NYC-0007 1.0 µM NYC-0008 7.9 nM INDUSTRIAL APPLICABILITY

[0952] The bispecific antibody of the present invention can be used as a therapeutic or preventive agent for cancer and the like.

[0953] All publications, patents, and patent applications cited herein are incorporated herein by reference in their entirety.

SEQUENCE LISTING FREE TEXT

[0954] SEQ ID NO 1: The amino acid sequence of the peptide in NY-ESO (FIG. 8) [0955] SEQ ID NO 2: The amino acid sequence of the peptide in MAGEC-1 (FIG. 9) [0956] SEQ ID NO 3: The scFv sequence analysis primer 1 (FIG. 10) [0957] SEQ ID NO 4: The scFv sequence analysis primer 2 (FIG. 11) [0958] SEQ ID NO 5: The nucleotide sequence of the heavy chain variable region of NYA-0001 (FIG. 12) [0959] SEQ ID NO 6: The amino acid sequence of the heavy chain variable region of NYA-0001 (FIG. 13) [0960] SEQ ID NO 7: The nucleotide sequence of the light chain variable region of NYA-0001 (FIG. 14) [0961] SEQ ID NO 8: The amino acid sequence of the light chain variable region of NYA-0001 (FIG. 15) [0962] SEQ ID NO 9: The nucleotide sequence of the

heavy chain variable region of NYA-0060 (FIG. 16) [0963] SEQ ID NO 10: The amino acid sequence of the heavy chain variable region of NYA-0060 (FIG. 17) [0964] SEQ ID NO 11: The nucleotide sequence of the light chain variable region of NYA-0060 (FIG. 18) [0965] SEQ ID NO 12: The amino acid sequence of the light chain variable region of NYA-0060 (FIG. 19) [0966] SEQ ID NO 13: The nucleotide sequence of the heavy chain variable region of NYA-0068 (FIG. 20) [0967] SEQ ID NO 14: The amino acid sequence of the heavy chain variable region of NYA-0068 (FIG. 21) [0968] SEQ ID NO 15: The nucleotide sequence of the light chain variable region of NYA-0068 (FIG. 22) [0969] SEQ ID NO 16: The amino acid sequence of the light chain variable region of NYA-0068 (FIG. 23) [0970] SEQ ID NO 17: The nucleotide sequence of the heavy chain variable region of NYA-0082 (FIG. 24) [0971] SEQ ID NO 18: The amino acid sequence of the heavy chain variable region of NYA-0082 (FIG. 25) [0972] SEQ ID NO 19: The nucleotide sequence of the light chain variable region of NYA-0082 (FIG. 26) [0973] SEQ ID NO 20: The amino acid sequence of the light chain variable region of NYA-0082 (FIG. 27) [0974] SEQ ID NO 21: The nucleotide sequence of the NYA-1163 tag adduct (FIG. 28) [0975] SEQ ID NO 22: The nucleotide sequence of the NYA-2023 tag adduct (FIG. 29) [0976] SEQ ID NO 23: The nucleotide sequence of the NYA-2027 tag adduct (FIG. 30) [0977] SEQ ID NO 24: The nucleotide sequence of the NYA-1143 tag adduct (FIG. 31) [0978] SEQ ID NO 25: The nucleotide sequence of the NYA-2143 tag adduct (FIG. 32) [0979] SEQ ID NO 26: The amino acid sequence of the NYA-1163 tag adduct (FIG. 33); NYA-1163: amino acids 21 to 266 [0980] SEQ ID NO 27: The amino acid sequence of the NYA-2023 tag adduct (FIG. 34); NYA-2023: amino acids 21 to 266 [0981] SEQ ID NO 28: The amino acid sequence of the NYA-2027 tag adduct (FIG. 35); NYA-2027: amino acids 21 to 266 [0982] SEQ ID NO 29: The amino acid sequence of the NYA-1143 tag adduct (FIG. 36); NYA-1143: amino acids 21 to 266 [0983] SEQ ID NO 30: The amino acid sequence of the NYA-2143 tag adduct (FIG. 37); NYA-2143: amino acids 21 to 266 [0984] SEQ ID NO 31: The nucleotide sequence of the NYA-1154 tag adduct (FIG. 38) [0985] SEQ ID NO 32: The amino acid sequence of the NYA-1154 tag adduct (FIG. 39); NYA-1154: amino acids 21 to 266 [0986] SEQ ID NO 33: The amino acid sequence of the HLA-A*0201 (GenBank: ASA47534.1) truncate (FIG. 40) [0987] SEQ ID NO 34: The amino acid sequence of 02-microglobulin (FIG. 41) [0988] SEQ ID NO 35: The nucleotide sequence of the NYA-2035 tag adduct (FIG. 42) [0989] SEQ ID NO 36: The amino acid sequence of the NYA-2035 tag adduct (FIG. 43); NYA-2035: amino acids 21 to 266 [0990] SEQ ID NO 37: The amino acid sequence of NYA-1143-VH01 (FIG. 44) [0991] SEQ ID NO 38: The amino acid sequence of NYA-1143-VH02 (FIG. 45) [0992] SEQ ID NO 39: The amino acid sequence of NYA-1143-VH03 (FIG. 46) [0993] SEQ ID NO 40: The amino acid sequence of NYA-1143-VL01 (FIG. 47) [0994] SEQ ID NO 41: The nucleotide sequence of the NYA-2044 tag adduct (FIG. 48) [0995] SEQ ID NO 42: The nucleotide sequence of the NYA-2045 tag adduct (FIG. 49) [0996] SEQ ID NO 43: The nucleotide sequence of the NYA-2047 tag adduct (FIG. 50) [0997] SEQ ID NO 44: The nucleotide sequence of the NYA-2048 tag adduct (FIG. 51) [0998] SEQ ID NO 45: The nucleotide sequence of the NYA-2060 tag adduct (FIG. 52) [0999] SEQ ID NO 46: The nucleotide sequence of the NYA-2061 tag adduct (FIG. 53) [1000] SEQ ID NO 47: The amino acid sequence of the NYA-2044 tag adduct (FIG. 54); NYA-2044: amino acids 21 to 266 [1001] SEQ ID NO 48: The amino acid sequence of the NYA-2045 tag adduct (FIG. 55); NYA-2045: amino acids 21 to 266 [1002] SEQ ID NO 49: The amino acid sequence of NYA-0082 (FIG. 56) [1003] SEQ ID NO 50: The amino acid sequence of the NYA-2047 tag adduct (FIG. 57); NYA-2047: amino acids 21 to 266 [1004] SEQ ID NO 51: The amino acid sequence of the NYA-2048 tag adduct (FIG. 58); NYA-2048: amino acids 21 to 266 [1005] SEQ ID NO 52: The amino acid sequence of the NYA-2060 tag adduct (FIG. 59); NYA-2060: amino acids 21 to 266 [1006] SEQ ID NO 53: The amino acid sequence of the NYA-2061 tag adduct (FIG. 60); NYA-2061: amino acids 21 to 266 [1007] SEQ ID NO 54: The heavy chain CDRH1 of NYA-0001 (FIG. 61) [1008] SEQ ID NO 55: The heavy chain CDRH2 of NYA-0001 (FIG. 61) [1009] SEQ ID NO 56: The heavy chain CDRH3 of NYA-0001 (FIG. 61) [1010] SEQ ID NO 57: The light chain CDRL1 of NYA-0001 (FIG. 61) [1011] SEQ ID NO 58: The light chain CDRL2 of NYA-0001 (FIG. 61) [1012] SEQ ID NO 59: The light chain CDRL3 of NYA-0001 (FIG. 61) [1013] SEQ ID NO 60: The amino acid sequence of CDRL1 of NYA-2023 (FIG. 62) [1014] SEQ ID NO 61: The amino acid sequence of CDRL3 of NYA-2027 (FIG. 63) [1015] SEQ ID NO 62: The heavy chain CDRH3 of NYA-1154 (FIG. 64) [1016] SEQ ID NO 63: The light chain CDRL3 of NYA-1154 (FIG. 64) [1017] SEQ ID NO 64: The amino acid sequence of CDRL1 of NYA-0035 (FIG. 65) [1018] SEQ ID NO 65: The nucleotide sequence of the NYC-0003 tag adduct (FIG. 66) [1019] SEQ ID NO 66: The nucleotide sequence of the NYC-0004 tag adduct (FIG. 67) [1020] SEQ ID NO 67: The amino acid sequence of NYC-0003 tag adduct (FIG. 68); NYC-0003: amino acids 21 to 263 [1021] SEQ ID NO 68: The amino acid sequence of NYC-0004 tag adduct (FIG. 69); NYC-0004: amino acids 21 to 263 [1022] SEQ ID NO 69: The nucleotide sequence of the NYA-0001 tag adduct (FIG. 70) [1023] SEQ ID NO 70: The amino acid sequence of the NYA-0001 tag adduct (FIG. 71); NYA-0001: amino acids 21 to 266 [1024] SEQ ID NO 71: The nucleotide sequence of HC1 (FIG. 72) [1025] SEQ ID NO 72: The nucleotide sequence of NYF-0016-HC2 (FIG. 73) [1026] SEQ ID NO 73: The nucleotide sequence of NYF-0019-HC2 (FIG. 74) [1027] SEQ ID NO 74: The nucleotide sequence of NYF-0022-HC2 (FIG. 75) [1028] SEQ ID NO 75: The nucleotide sequence of NYF-0023-HC2 (FIG. 76) [1029] SEQ ID NO 76: The nucleotide sequence of NYF-0027-HC2 (FIG. 77) [1030] SEQ ID NO 77: The nucleotide sequence of NYF-0035-HC2 (FIG. 78) [1031] SEQ ID NO 78: The nucleotide sequence of NYF-0044-HC2 (FIG. 79) [1032] SEQ ID NO 79: The nucleotide sequence of NYF-0045-HC2 (FIG. 80) [1033] SEQ ID NO

80: The nucleotide sequence of NYF-0047-HC2 (FIG. **81**) [1034] SEQ ID NO 81: The nucleotide sequence of NYF-0048-HC2 (FIG. **82**) [1035] SEQ ID NO 82: The nucleotide sequence of NYF-0060-HC2 (FIG. **83**) [1036] SEQ ID NO 83: The nucleotide sequence of NYF-0061-HC2 (FIG. **84**) [1037] SEQ ID NO 84: The amino acid sequence of HC1 (FIG. **85**) [1038] SEQ ID NO 85: The amino acid sequence of NYF-0016-HC2 (FIG. **86**); NYA-1143: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1039] SEQ ID NO 86: The amino acid sequence of NYF-0019-HC2 (FIG. **87**); NYA-2143: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1040] SEQ ID NO 87: The amino acid sequence of NYF-0022-HC2 (FIG. **88**); NYA-1163: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1041] SEQ ID NO 88: The amino acid sequence of NYF-0023-HC2 (FIG. **89**); NYA-2023: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1042] SEQ ID NO 89: The amino acid sequence of NYF-0027-HC2 (FIG. **90**); NYA-2027: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1043] SEQ ID NO 90: The amino acid sequence of NYF-0035-HC2 (FIG. **91**); NYA-2035: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1044] SEQ ID NO 91: The amino acid sequence of NYF-0044-HC2 (FIG. **92**); NYA-2044: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1045] SEQ ID NO 92: The amino acid sequence of NYF-0045-HC2 (FIG. **93**); NYA-2045: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1046] SEQ ID NO 93: The amino acid sequence of NYF-0047-HC2 (FIG. **94**); NYA-2047: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1047] SEQ ID NO 94: The amino acid sequence of NYF-0048-HC2 (FIG. **95**); NYA-2048: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1048] SEQ ID NO 95: The amino acid sequence of NYF-0060-HC2 (FIG. **96**); NYA-2060: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1049] SEQ ID NO 96: The amino acid sequence of NYF-0061-HC2 (FIG. **97**); NYA-2061: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1050] SEQ ID NO 97: The nucleotide sequence of NYA-0001-Fab-HC1-k delete (FIG. **98**) [1051] SEQ ID NO 98: The nucleotide sequence of NYA-0001-LC (FIG. **99**) [1052] SEQ ID NO 99: The amino acid sequence of NYA-0001-Fab-HC1-k delete (FIG. **100**); the NYA-0001 heavy chain variable region: amino acids 20 to 139 [1053] SEQ ID NO 100: The amino acid sequence of NYA-0001-LC (FIG. **101**); the NYA-0001 light chain variable region: amino acids 21 to 131 [1054] SEQ ID NO 101: The nucleotide sequence of NYA-1143-Fab-HC1-k delete (FIG. **102**) [1055] SEQ ID NO 102: The nucleotide sequence of NYA-1143-LC (FIG. **103**) [1056] SEQ ID NO 103: The nucleotide sequence of C3E-7085-HC2-k deleteC (FIG. **104**) [1057] SEQ ID NO 104: The amino acid sequence of NYA-1143-Fab-HC1-k delete (FIG. **105**); the NYA-1143 heavy chain variable region: amino acids 20 to 139 [1058] SEQ ID NO 105: The amino acid sequence of NYA-1143-LC (FIG. **106**); the NYA-1143 light chain variable region: amino acids 21 to 131 [1059] SEQ ID NO 106: The amino acid sequence of C3E-7085-HC2-k delete (FIG. **107**); C3E-7085: amino acids 21 to 260 [1060] SEQ ID NO 107: The nucleotide sequence of NYA-1143-HC1-k delete (FIG. **108**) [1061] SEQ ID NO 108: The amino acid sequence of NYA-1143-HC1-k delete (FIG. **109**); NYA-1143: amino acids 21 to 266 [1062] SEQ ID NO 109: The nucleotide sequence of C3E-7085-NYA-1154-Fab-HC2-k delete (FIG. **110**) [1063] SEQ ID NO 110: The nucleotide sequence of NYA-1154-LC (FIG. **111**) [1064] SEQ ID NO 111: The nucleotide sequence of OAA-HC1-k delete (FIG. **112**) [1065] SEQ ID NO 112: The amino acid sequence of C3E-7085-NYA-1154-Fab-HC2-k delete (FIG. **113**); C3E-7085: amino acids 21 to 260; the NYA-1154 heavy chain variable region: amino acids 266 to 285 [1066] SEQ ID NO 113: The amino acid sequence of NYA-1154-LC (FIG. **114**); the NYA-1154 light chain variable region: amino acids 21 to 131 [1067] SEQ ID NO 114: The amino acid sequence of OAA-HC1-k delete (FIG. **115**) [1068] SEQ ID NO 115: The nucleotide sequence of NYF-0010-HC2-k delete (FIG. **116**) [1069] SEQ ID NO 116: The nucleotide sequence of NYF-0004-HC2-k delete (FIG. **117**) [1070] SEQ ID NO 117: The nucleotide sequence of NYF-0011-HC2-k delete (FIG. **118**) [1071] SEQ ID NO 118: The amino acid sequence of NYF-0010-HC2-k delete (FIG. **119**); NYA-1154: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1072] SEQ ID NO 119: The amino acid sequence of NYF-0004-HC2-k delete (FIG. **120**); C3E-7085: amino acids 21 to 260; and NYA-1154: amino acids 272 to 511 [1073] SEQ ID NO 120: The amino acid sequence of NYF-0011-HC2-k delete (FIG. **121**); NYA-1143: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1074] SEQ ID NO 121: The amino acid sequence of the point-mutant NY-ESO peptide 1F (FIG. **122**) [1075] SEQ ID NO 122: The amino acid sequence of the point-mutant NY-ESO peptide 2M (FIG. **123**) [1076] SEQ ID NO 123: The amino acid sequence of the point-mutant NY-ESO peptide 3A (FIG. **124**) [1077] SEQ ID NO 124: The amino acid sequence of the point-mutant NY-ESO peptide 4A (FIG. **125**) [1078] SEQ ID NO 125: The amino acid sequence of the point-mutant NY-ESO peptide 5A (FIG. **126**) [1079] SEQ ID NO 126: The amino acid sequence of the point-mutant NY-ESO peptide 6L (FIG. **127**) [1080] SEQ ID NO 127: The amino acid sequence of the point-mutant NY-ESO peptide 7F (FIG. **128**) [1081] SEQ ID NO 128: The amino acid sequence of the point-mutant NY-ESO peptide 8A (FIG. **129**) [1082] SEQ ID NO 129: The amino acid sequence of the point-mutant NY-ESO peptide 9A (FIG. **130**) [1083] SEQ ID NO 130: The amino acid sequence of the gp100 peptide (FIG. **131**) [1084] SEQ ID NO 131: The amino acid sequence of the homologous peptide DOLPP1 (FIG. **132**) [1085] SEQ ID NO 132: The amino acid sequence of the homologous peptide IL20RB (FIG. **133**) [1086] SEQ ID NO 133: The amino acid sequence of the homologous peptide PRKD2 (FIG. **134**) [1087] SEQ ID NO 134: The amino acid sequence of the homologous peptide CD163 (FIG. **135**) [1088] SEQ ID NO 135: The amino acid sequence of the homologous peptide P2RY8 (FIG. **136**) [1089] SEQ ID NO 136: The amino acid sequence of C3E-7034 (FIG. **137**) [1090] SEQ ID NO 137: The amino acid sequence of C3E-7036 (FIG. **138**) [1091] SEQ ID NO

138: The amino acid sequence of C3E-7085 (FIG. **139**) [1092] SEQ ID NO 139: The amino acid sequence of C3E-7088 (FIG. **140**) [1093] SEQ ID NO 140: The amino acid sequence of C3E-7093 (FIG. **141**) [1094] SEQ ID NO 141: The heavy chain CDRH1 of C3E-7085 (FIG. **142**) [1095] SEQ ID NO 142: The heavy chain CDRH2 of C3E-7085 (FIG. **142**) [1096] SEQ ID NO 143: The heavy chain CDRH3 of C3E-7085 (FIG. **142**) [1097] SEQ ID NO 144: The light chain CDRL1 of C3E-7085 (FIG. **142**) [1098] SEQ ID NO 145: The light chain CDRL2 of C3E-7085 (FIG. **142**) [1099] SEQ ID NO 146: The light chain CDRL3 of C3E-7085 (FIG. **142**) [1100] SEQ ID NO 147: The amino acid sequence of C3E-7078 (FIG. **143**) [1101] SEQ ID NO 148: The nucleotide sequence of NYF-0014-HC2 (FIG. **144**) [1102] SEQ ID NO 149: The amino acid sequence of NYF-0014-HC2 (FIG. **145**); NYA-0001: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1103] SEQ ID NO 150: The amino acid sequence of NYF-0082-HC2 (FIG. **146**); NYA-0082: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511 [1104] SEQ ID NO 151: The amino acid sequence of human CD3 P (FIG. **147**) [1105] SEQ ID NO 152: The nucleotide sequence of the full-length NYZ-0038-HC2 (FIG. **148**) [1106] SEQ ID NO 153: The nucleotide sequence of the full-length NYZ-0082-HC2 (FIG. **149**) [1107] SEQ ID NO 154: The nucleotide sequence of the full-length NYZ-0083-HC2 (FIG. **150**) [1108] SEQ ID NO 155: The amino acid sequence of the full-length NYZ-0038-HC2 (FIG. **151**) [1109] SEQ ID NO 156: The amino acid sequence of the full-length NYZ-0082-HC2 (FIG. **152**) [1110] SEQ ID NO 157: The amino acid sequence of the full-length NYZ-0083-HC2 (FIG. **153**) [1111] SEQ ID NO 158: The nucleotide sequence of the full-length NYZ-1010-HC2 (FIG. **154**) [1112] SEQ ID NO 159: The nucleotide sequence of the full-length C3E-7085-LC (FIG. **155**) [1113] SEQ ID NO 160: The amino acid sequence of the full-length NYZ-1010-HC2 (FIG. **156**) [1114] SEQ ID NO 161: The amino acid sequence of the full-length C3E-7085-LC (FIG. **157**) [1115] SEQ ID NO 162: The amino acid sequence of the peptide linker (FIG. **161**) [1116] SEQ ID NO 163: The nucleotide sequence of the full-length NYA-3061 (FIG. **164**) [1117] SEQ ID NO 164: The amino acid sequence of the full-length NYA-3061 (FIG. **165**) [1118] SEQ ID NO 165: The nucleotide sequence of the full-length NYC-0005 (FIG. **166**) [1119] SEQ ID NO 166: The amino acid sequence of the full-length NYC-0005 (FIG. **167**) [1120] SEQ ID NO 167: The nucleotide sequence of the full-length NYC-0006 (FIG. **168**) [1121] SEQ ID NO 168: The amino acid sequence of the full-length NYC-0006 (FIG. **169**) [1122] SEQ ID NO 169: The nucleotide sequence of the full-length NYC-0007 (FIG. **170**) [1123] SEQ ID NO 170: The amino acid sequence of the full-length NYC-0007 (FIG. **171**) [1124] SEQ ID NO 171: The nucleotide sequence of the full-length NYC-0008 (FIG. **172**) [1125] SEQ ID NO 172: The amino acid sequence of the full-length NYC-0008 (FIG. **173**) [1126] SEQ ID NO 173: The nucleotide sequence of the full-length NYC-0009 (FIG. **174**) [1127] SEQ ID NO 174: The amino acid sequence of the full-length NYC-0009 (FIG. **175**) [1128] SEQ ID NO 175: The nucleotide sequence of the full-length NYC-0010 (FIG. **176**) [1129] SEQ ID NO 176: The amino acid sequence of the full-length NYC-0010 (FIG. **177**) [1130] SEQ ID NO 177: The nucleotide sequence of the full-length HC-h (FIG. **178**) [1131] SEQ ID NO 178: The amino acid sequence of the full-length HC-h (FIG. **179**) [1132] SEQ ID NO 179: The nucleotide sequence of the full-length NYD-2047-HC-k (FIG. **180**) [1133] SEQ ID NO 180: The amino acid sequence of the full-length NYD-2047-HC-k (FIG. **181**) [1134] SEQ ID NO 181: The nucleotide sequence of the full-length NYD-2061-HC-k (FIG. **182**) [1135] SEQ ID NO 182: The amino acid sequence of the full-length NYD-2061-HC-k (FIG. **183**) [1136] SEQ ID NO 183: The nucleotide sequence of the full-length NYD-3061-HC-k (FIG. **184**) [1137] SEQ ID NO 184: The amino acid sequence of the full-length NYD-3061-HC-k (FIG. **185**) [1138] SEQ ID NO 185: The nucleotide sequence of the full-length NYC-0011-HC-k (FIG. **186**) [1139] SEQ ID NO 186: The amino acid sequence of the full-length NYC-0011-HC-k (FIG. **187**) [1140] SEQ ID NO 187: The nucleotide sequence of the full-length NYC-0012-HC-k (FIG. **188**) [1141] SEQ ID NO 188: The amino acid sequence of the full-length NYC-0012-HC-k (FIG. **189**) [1142] SEQ ID NO 189: The nucleotide sequence of the full-length NYC-0013-HC-k (FIG. **190**) [1143] SEQ ID NO 190: The amino acid sequence of the full-length NYC-0013-HC-k (FIG. **191**) [1144] SEQ ID NO 191: The nucleotide sequence of the full-length NYC-0014-HC-k (FIG. **192**) [1145] SEQ ID NO 192: The amino acid sequence of the full-length NYC-0014-HC-k (FIG. **193**) [1146] SEQ ID NO 193: The nucleotide sequence of the full-length NYC-0015-HC-k (FIG. **194**) [1147] SEQ ID NO 194: The amino acid sequence of the full-length NYC-0015-HC-k (FIG. **195**) [1148] SEQ ID NO 195: The nucleotide sequence of the full-length NYC-0016-HC-k (FIG. **196**) [1149] SEQ ID NO 196: The amino acid sequence of the full-length NYC-0016-HC-k (FIG. **197**) [1150] SEQ ID NO 197: The amino acid sequence of the full-length NYZ-1007-HC (FIG. **198**) [1151] SEQ ID NO 198: The amino acid sequence of the full-length NYZ-1017-HC (FIG. **199**)

Claims

1. A polynucleotide comprising a nucleotide sequence encoding a bispecific antibody that comprises a site which binds specifically to human HLA/NY-ESO and a site which binds specifically to human CD3, wherein the site which binds specifically to human HLA/NY-ESO comprises: a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54, a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55, a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56, a light chain CDRL1

consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 in which amino acid 7 is W, a light chain CDRL2 consisting of the amino acid sequence represented by DNN, and a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59.

2. The polynucleotide according to claim 1, wherein the site which binds specifically to human HLA/NY-ESO comprises: a heavy chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 53 and a light chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53, or a heavy chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156 and a light chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156.

3. The polynucleotide according to claim 1, wherein the site which binds specifically to human HLA/NY-ESO comprises: a heavy chain variable region consisting of an amino acid sequence having 98% or higher sequence identity to an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 53 and a light chain variable region consisting of an amino acid sequence having 98% or higher sequence identity to an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53, or a heavy chain variable region consisting of an amino acid sequence having 98% or higher sequence identity to an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156 and a light chain variable region consisting of an amino acid sequence having 98% or higher sequence identity to an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156.

4. The polynucleotide according to claim 1 wherein the bispecific antibody comprises: a heavy chain variable region consisting of an amino acid sequence derived from an amino acid sequence as shown in SEQ ID NO: 53 by substitution, deletion, addition or insertion of one or more amino acids and a light chain variable region consisting of an amino acid sequence derived from an amino acid sequence as shown in SEQ ID NO: 53 by substitution, deletion, addition or insertion of one or more amino acids, or a heavy chain variable region consisting of an amino acid sequence derived from an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156 by substitution, deletion, addition or insertion of one or more amino acids and a light chain variable region consisting of an amino acid sequence derived from an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156 by substitution, deletion, addition or insertion of one or more amino acids.

5. The polynucleotide according to claim 1, wherein the site which binds specifically to human HLA/NY-ESO is scFv.

6. The polynucleotide according to claim 1, wherein the site which binds specifically to human CD3 comprises: (CCH1) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 141; (CCH2) a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 142 or a heavy chain CDRH2 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 142 in which amino acid 3 is N or S; (CCH3) a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 143; (CCL1) a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 144; (CCL2) a light chain CDRL2 consisting of the amino acid sequence represented by RDD or a light chain CDRL2 consisting of an amino acid sequence derived from the amino acid sequence represented by RDD in which amino acid 2 is N; and (CCL3) a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 146.

7. The polynucleotide according to claim 1, wherein the site which binds specifically to human CD3 comprises: (CH1) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136, (CH2) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137, (CH3) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 147, (CH4) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 138, (CH5) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139, (CH6) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140, (CH7) a heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155, (CH8) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156; or (CH9) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157; and (CL1) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136, (CL2) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137, (CL3) a light chain variable region consisting of an amino acid sequence of amino acids 135 to

243 of the amino acid sequence as shown in SEQ ID NO: 147, (CL4) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138, (CL5) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139, (CL6) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140, (CL7) a light chain variable region consisting of an amino acid sequence of amino acids 405 to 511 of the amino acid sequence as shown in SEQ ID NO: 155, (CL8) a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156, or (CL9) a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157.

8. The polynucleotide according to claim 1, wherein the site which binds specifically to human CD3 comprises: (CH1CL1) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136; (CH2CL2) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137; (CH3CL3) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 147 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 147; (CH4CL4) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 138 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138; (CH5CL5) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139; (CH6CL6) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140; (CH7CL7) a heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155 and a light chain variable region consisting of an amino acid sequence of amino acids 405 to 511 of the amino acid sequence as shown in SEQ ID NO: 155; (CH8CL8) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156 and a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156; or (CH9CL9) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157 and a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157.

9. The polynucleotide according to claim 1, wherein the site which binds specifically to human CD3 is Fab.

10. The polynucleotide according to claim 1, wherein the bispecific antibody comprises: a first polypeptide comprising scFv that binds specifically to human HLA/NY-ESO, a variable region of an antibody that binds specifically to human CD3, and an Fc region (i) in that order from the N terminus toward the C terminus, a second polypeptide comprising a hinge region and an Fc region (ii) of the immunoglobulin, and a third polypeptide comprising an antibody light chain consisting of a variable region and a constant region, wherein the first polypeptide is associated with the second polypeptide at the Fc region (i) and the Fc region (ii), and the first polypeptide is associated with the third polypeptide in the heavy chain variable region and the constant region CH1 of the antibody.

11. The polynucleotide according to claim 10, wherein the first polypeptide comprises an amino acid sequence of amino acids 21 to 394 of the amino acid sequence as shown in SEQ ID NO: 160.

12. The polynucleotide according to claim 10, wherein the first polypeptide comprises the amino acid sequence according to (i) or (ii) below: (i) an amino acid sequence of amino acids 20 to 724 of the amino acid sequence as shown in SEQ ID NO: 160; and (ii) an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 197.

13. The polynucleotide according to claim 10, wherein the second polypeptide comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.

14. The polynucleotide according to claim 10, wherein the third polypeptide comprises an amino acid sequence of amino acids 21 to 127 of the amino acid sequence as shown in SEQ ID NO: 161.

15. A vector comprising the polynucleotide according to claim 1.

16. A host cell comprising the polynucleotide according to claim 1 or a vector comprising the polynucleotide according to claim 1.

17. A vector comprising the polynucleotide according to claim 10.

18. A host cell comprising the polynucleotide according to claim 10 or a vector comprising the polynucleotide

according to claim 10.

19. A method for producing a bispecific antibody that binds specifically to human HLA/NY-ESO and human CD3 comprising: (i) a step of culturing the host cell according to claim 16; and (ii) a step of purifying an antibody or a binding fragment thereof from the culture product obtained in the step (i).

20. A bispecific antibody that binds specifically to human HLA/NY-ESO and human CD3, which is obtained by the method according to claim 19.
